

MEMORY SQUARE

Database and event report

Complete snapshot of events.json - generated 2026-07-18T11:25:47.596Z

EVENTS

210

verified memory nodes

MAP

15 x 15

225 cells / 15 vacant

SNAPSHOT SCOPE

This report contains every top-level database property, every memory-map coordinate, and every field stored on every event. Values are never shortened. Non-ASCII symbols are represented as explicit U+ code points so the source data remains unambiguous in every PDF reader.

SOURCE FILE

events.json

SCHEMA

memory-event-node/2.0

DATABASE UPDATED AT

2026-07-18T11:25:00.044Z

REPORT GENERATED AT

2026-07-18T11:25:47.596Z

Database details

Top-level fields and complete square-map placement

TOP-LEVEL PROPERTIES

SCHEMA

memory-event-node/2.0

UPDATED-AT

2026-07-18T11:25:00.044Z

COMPILER.NAME

gpilc

COMPILER.VERSION

1.0.0

COMPILER.LANGUAGE

GPIL/1.0

COMPILER.SOURCE

examples/language.gpil

COMPILER.LINE-CONTRACT

one nonblank physical source line equals one event

INSTRUCTION-SET.ARRAY-1-CONSTRUCTION[0]

add

INSTRUCTION-SET.ARRAY-1-CONSTRUCTION[1]

apply_directional_polarity

INSTRUCTION-SET.ARRAY-1-CONSTRUCTION[2]

containerize

INSTRUCTION-SET.ARRAY-1-CONSTRUCTION[3]

decontainerize

INSTRUCTION-SET.ARRAY-2-CONTAINER-TRANSFORMATION[0]

transform_container

INSTRUCTION-SET.ARRAY-3-ARITHMETIC[0]

subtract

INSTRUCTION-SET.ARRAY-3-ARITHMETIC[1]

multiply

INSTRUCTION-SET.ARRAY-3-ARITHMETIC[2]

divide

INSTRUCTION-SET.ARRAY-3-ARITHMETIC[3]

modulo

INSTRUCTION-SET.ARRAY-4-TEXT[0]

literal

INSTRUCTION-SET.ARRAY-4-TEXT[1]

identity

INSTRUCTION-SET.ARRAY-4-TEXT[2]

concat

INSTRUCTION-SET.ARRAY-4-TEXT[3]	symbol_length
INSTRUCTION-SET.ARRAY-4-TEXT[4]	reverse
INSTRUCTION-SET.ARRAY-4-TEXT[5]	replace
INSTRUCTION-SET.ARRAY-5-LOGIC[0]	equal
INSTRUCTION-SET.ARRAY-5-LOGIC[1]	not
INSTRUCTION-SET.ARRAY-5-LOGIC[2]	and
INSTRUCTION-SET.ARRAY-5-LOGIC[3]	or
INSTRUCTION-SET.ARRAY-5-LOGIC[4]	select
INSTRUCTION-SET.ARRAY-5-LOGIC[5]	assert_equal
INSTRUCTION-SET.ARRAY-6-SEMANTICS[0]	semantic_triple
INSTRUCTION-SET.ARRAY-7-SEQUENTIAL-ID[0]	sequential_encode
INSTRUCTION-SET.ARRAY-7-SEQUENTIAL-ID[1]	sequential_decode
INSTRUCTION-SET.ARRAY-7-SEQUENTIAL-ID[2]	sequential_verify
INSTRUCTION-SET.ARRAY-8-MEMORY-MAP[0]	reference
INSTRUCTION-SET.ARRAY-8-MEMORY-MAP[1]	map_coordinate
INSTRUCTION-SET.ARRAY-8-MEMORY-MAP[2]	automatic_event_capture
INSTRUCTION-SET.ARRAY-8-MEMORY-MAP[3]	automatic_square_map

MAP SUMMARY

MAP.DIMENSIONS.ROWS	15
MAP.DIMENSIONS.COLUMNS	15
MAP.CELLS.TOTAL	225
MAP.CELLS.OCCUPIED	210

MAP.CELLS.VACANT

15

MAP CELLS

MAP.CELLS[0][0]

gpil_0001_7fb5c4c3e1f62579

MAP.CELLS[0][1]

gpil_0002_bb60bd992cc5229c

MAP.CELLS[0][2]

gpil_0003_4d610d745dda33e1

MAP.CELLS[0][3]

gpil_0004_1bd9b0567747f5e0

MAP.CELLS[0][4]

gpil_0005_acf482484385991e

MAP.CELLS[0][5]

gpil_0006_d40f12701dbcdcb1

MAP.CELLS[0][6]

gpil_0007_fba64ff6395fb867

MAP.CELLS[0][7]

gpil_0008_4e0deac514dec5e5

MAP.CELLS[0][8]

gpil_0009_2987cc2491361c98

MAP.CELLS[0][9]

gpil_0010_8fda7f31708deb9b

MAP.CELLS[0][10]

gpil_0011_1e722f196156c83e

MAP.CELLS[0][11]

gpil_0012_f17fc7855d08e562

MAP.CELLS[0][12]

gpil_0013_4831d16cec7c30bc

MAP.CELLS[0][13]

gpil_0014_5aac4b604708e1f9

MAP.CELLS[0][14]

gpil_0015_952700ad4a411aa9

MAP.CELLS[1][0]

gpil_0016_78c79597d3362ca7

MAP.CELLS[1][1]

gpil_0017_cfd7520aace50ae6

MAP.CELLS[1][2]

gpil_0018_fa7e528d79edf0fd

MAP.CELLS[1][3]

gpil_0019_c4b50d8590a0220f

MAP.CELLS[1][4]

gpil_0020_f7554fa1d1153eb1

MAP.CELLS[1][5]
gpil_0021_8cc55a3c0396b095

MAP.CELLS[1][6]
gpil_0022_1f389ebc3a176377

MAP.CELLS[1][7]
gpil_0023_a5b02eb2fbecb62d

MAP.CELLS[1][8]
gpil_0024_27bcd996ec2d8020

MAP.CELLS[1][9]
gpil_0025_2069e01947081282

MAP.CELLS[1][10]
gpil_0026_9ed529d8a3edf3c2

MAP.CELLS[1][11]
gpil_0027_aa39b9096908f6be

MAP.CELLS[1][12]
gpil_0028_9ebe406732926a6a

MAP.CELLS[1][13]
gpil_0029_5ff306e67ec276d5

MAP.CELLS[1][14]
gpil_0030_4e4eed285eb7b434

MAP.CELLS[2][0]
gpil_0031_5f5e0aa9e89a3456

MAP.CELLS[2][1]
gpil_0032_541a7102313e8cb7

MAP.CELLS[2][2]
gpil_0033_e0e30fe0922faffd

MAP.CELLS[2][3]
gpil_0034_0a2f76a3516bf171

MAP.CELLS[2][4]
gpil_0035_326087f9c682cdfb

MAP.CELLS[2][5]
gpil_0036_f7b87461332e4fa1

MAP.CELLS[2][6]
gpil_0037_e63120160c3ddb1a

MAP.CELLS[2][7]
gpil_0038_83e991b7dbbc6e21

MAP.CELLS[2][8]
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MAP.CELLS[2][9]
gpil_0040_2e3c7cc8b7e24811

MAP.CELLS[2][10]
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MAP.CELLS[2][11]
gpil_0042_aa045656d1297f62

MAP.CELLS[2][12]
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MAP.CELLS[2][13]
gpil_0044_873cb26b232528fb

MAP.CELLS[2][14]
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MAP.CELLS[3][0]
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MAP.CELLS[3][1]
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MAP.CELLS[3][2]
gpil_0048_4b33e2b33d0708ea

MAP.CELLS[3][3]
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MAP.CELLS[3][4]
gpil_0050_2f258ecf9c102099

MAP.CELLS[3][5]
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MAP.CELLS[3][6]
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MAP.CELLS[3][8]
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MAP.CELLS[3][9]
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MAP.CELLS[3][10]
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MAP.CELLS[4][2]
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MAP.CELLS[4][3]
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MAP.CELLS[4][6]
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MAP.CELLS[4][7]
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MAP.CELLS[4][8]
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MAP.CELLS[4][9]
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MAP.CELLS[4][10]
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MAP.CELLS[4][11]
gpil_0072_1cc1c7b4e91ed3ee

MAP.CELLS[4][12]
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MAP.CELLS[4][14]
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MAP.CELLS[5][3]
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MAP.CELLS[5][12]
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MAP.CELLS[5][14]
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MAP.CELLS[8][2]
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MAP.CELLS[8][11]
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MAP.CELLS[8][14]
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MAP.CELLS[9][6]
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MAP.CELLS[9][8]
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MAP.CELLS[9][9]
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MAP.CELLS[9][12]
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MAP.CELLS[9][14]
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MAP.CELLS[10][1]
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MAP.CELLS[10][2]
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MAP.CELLS[10][7]
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MAP.CELLS[10][8]
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MAP.CELLS[10][9]
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MAP.CELLS[10][10]
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MAP.CELLS[10][11]
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MAP.CELLS[10][14]
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MAP.CELLS[11][0]
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MAP.CELLS[11][1]
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MAP.CELLS[11][2]
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MAP.CELLS[11][3]
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MAP.CELLS[11][4]
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MAP.CELLS[11][5]
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MAP.CELLS[11][6]
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MAP.CELLS[11][7]
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MAP.CELLS[11][8]
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MAP.CELLS[11][9]
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MAP.CELLS[11][10]
gpil_0176_d516d5a1999d6767

MAP.CELLS[11][11]
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MAP.CELLS[11][12]
gpil_0178_47ad40832165aa42

MAP.CELLS[11][13]
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MAP.CELLS[11][14]
gpil_0180_373d7f7dc466ea8b

MAP.CELLS[12][0]
gpil_0181_634a32a5626aaa51

MAP.CELLS[12][1]
gpil_0182_91620f7f6c6ed8b1

MAP.CELLS[12][2]
gpil_0183_5d9947b30ca5e072

MAP.CELLS[12][3]
gpil_0184_6837e313e038afc8

MAP.CELLS[12][4]
gpil_0185_e084e797e40f3712

MAP.CELLS[12][5]
gpil_0186_c36d130cbead6cda

MAP.CELLS[12][6]
gpil_0187_37c7b7944b466dbe

MAP.CELLS[12][7]
gpil_0188_5b55813c7958b5ab

MAP.CELLS[12][8]
gpil_0189_bc8f38c985144c9d

MAP.CELLS[12][9]
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MAP.CELLS[12][10]
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MAP.CELLS[12][11]
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MAP.CELLS[12][12]
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MAP.CELLS[12][13]
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MAP.CELLS[12][14]
gpil_0195_942fcb54bd2472bd

MAP.CELLS[13][0]
gpil_0196_a954d9ab1decdd54

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MAP.CELLS[13][1]
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MAP.CELLS[13][2]
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MAP.CELLS[13][3]
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MAP.CELLS[13][4]
gpil_0200_e1e431f70eb04177

MAP.CELLS[13][5]
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MAP.CELLS[13][6]
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MAP.CELLS[13][7]
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MAP.CELLS[13][8]
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MAP.CELLS[13][9]
gpil_0205_c0c26c7bbfb8fe63

MAP.CELLS[13][10]
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MAP.CELLS[13][11]
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MAP.CELLS[13][12]
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MAP.CELLS[13][13]
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MAP.CELLS[13][14]
gpil_0210_55fb754872de5788

MAP.CELLS[14][0]
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MAP.CELLS[14][1]
null

MAP.CELLS[14][2]
null

MAP.CELLS[14][3]
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MAP.CELLS[14][4]
null

MAP.CELLS[14][5]
null

MAP.CELLS[14][6]
null

MAP.CELLS[14][7]
null
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MAP.CELLS[14][8]

null

MAP.CELLS[14][9]

null

MAP.CELLS[14][10]

null

MAP.CELLS[14][11]

null

MAP.CELLS[14][12]

null

MAP.CELLS[14][13]

null

MAP.CELLS[14][14]

null

God

Complete event record 1 of 210

TIME-STAMP	2026-07-18T11:25:00.044Z
LABEL	God
VALUE	There exists one, and only one, point of origin. Where that origin gives rise to Heaven (The attainment of all Hope).
ID	gpil_0001_7fb5c4c3e1f62579
SEQUENTIAL-STRING ID	1401292924473084736124369145015115921891910476759968244545526280339227546566309284081167775893853534135043887234048191724336189881654530186612947027153636694909073472578238921237853623449111082616876856521414057511476732282183126965
BACKWARD-COMPUTED VALUE	There exists one, and only one, point of origin. Where that origin gives rise to Heaven (The attainment of all Hope).
BACKWARD-COMPUTATION.ALGORITHM	shortlex-unrank/1.0
BACKWARD-COMPUTATION.MATCHES-VALUE	true
ALPHABET	!"#\$%&'()*+,-./0123456789:;<=>?@ABCDEFGHIJKLMNOPQRSTUVWXYZ[\]^_`abcdefghijklmnopqrstuvwxyz{ }~
METRICS.ALPHABET-SYMBOLS	95
METRICS.VALUE-SYMBOLS	117
METRICS.SEQUENTIAL-ID-DIGITS	232
METRICS.NATIVE-INTEGER	false
INSTRUCTION.LANGUAGE	GPIL/1.0
INSTRUCTION.SOURCE-LINE	1
INSTRUCTION.OPERATION	literal
INSTRUCTION.SOURCE	event "God" : literal("There exists one, and only one, point of origin. Where that origin gives rise to Heaven (The attainment of all Hope).")

Constructivism

Complete event record 2 of 210

TIME-STAMP	2026-07-18T11:25:00.044Z
LABEL	Constructivism
VALUE	An approach to mathematics in which an assertion is justified by explicit evidence or a construction; in particular, existence requires a method or witness rather than only the denial of nonexistence.
ID	gpil_0002_bb60bd992cc5229c
SEQUENTIAL-STRING ID	128520666238732477672891879863742537930038120576985240199873260707608866433126834184013518307062292934757855666738756635340789579635358965236906040420652658611787158277831140311760697821161331314333436605124983406185899557432318311975646397789712677202776512322623026823278814316432671030163498884567968098244183635596266805096151991239379099831567729893682524480608904623226026774896233277193615
BACKWARD-COMPUTED VALUE	An approach to mathematics in which an assertion is justified by explicit evidence or a construction; in particular, existence requires a method or witness rather than only the denial of nonexistence.
BACKWARD-COMPUTATION.ALGORITHM	shortlex-unrank/1.0
BACKWARD-COMPUTATION.MATCHES-VALUE	true
ALPHABET	!"#\$%&'()*+,-./0123456789:;<=>?@ABCDEFGHIJKLMNOPQRSTUVWXYZ[\]^_`abcdefghijklmnopqrstuvwxyz{ }~
METRICS.ALPHABET-SYMBOLS	95
METRICS.VALUE-SYMBOLS	200
METRICS.SEQUENTIAL-ID-DIGITS	396
METRICS.NATIVE-INTEGER	false
INSTRUCTION.LANGUAGE	GPIL/1.0
INSTRUCTION.SOURCE-LINE	2
INSTRUCTION.OPERATION	literal
INSTRUCTION.SOURCE	event "Constructivism" : literal("An approach to mathematics in which an assertion is justified by explicit evidence or a construction; in particular, existence requires a method or witness rather than only the denial of nonexistence.")

Intuitionism

Complete event record 3 of 210

TIME-STAMP	2026-07-18T11:25:00.044Z
LABEL	Intuitionism
VALUE	The constructivist tradition initiated by Brouwer in which mathematical truth is grounded in construction and the accepted logic is intuitionistic rather than unrestricted classical logic.
ID	gpil_0003_4d610d745dda33e1
SEQUENTIAL-STRING ID	367201234115035413044329185327888707438817752987539747664019369386331955716314604446231097704674666633803875670201003411117271627096230909132920409462638994242003463602592658204196111862800566272593257813622320782348769679220358652000481191834896714909726094855212804935552221494012288922541164478603555258805256789298165073032184220807906330109308112978394978952890495950
BACKWARD-COMPUTED VALUE	The constructivist tradition initiated by Brouwer in which mathematical truth is grounded in construction and the accepted logic is intuitionistic rather than unrestricted classical logic.
BACKWARD-COMPUTATION.ALGORITHM	shortlex-unrank/1.0
BACKWARD-COMPUTATION.MATCHES-VALUE	true
ALPHABET	!"#\$%&'()*+,-./0123456789:;<=>?@ABCDEFGHIJKLMNOPQRSTUVWXYZ[\]^_`abcdefghijklmnopqrstuvwxyz{ }~
METRICS.ALPHABET-SYMBOLS	95
METRICS.VALUE-SYMBOLS	188
METRICS.SEQUENTIAL-ID-DIGITS	372
METRICS.NATIVE-INTEGER	false
INSTRUCTION.LANGUAGE	GPIL/1.0
INSTRUCTION.SOURCE-LINE	3
INSTRUCTION.OPERATION	literal
INSTRUCTION.SOURCE	event "Intuitionism" : literal("The constructivist tradition initiated by Brouwer in which mathematical truth is grounded in construction and the accepted logic is intuitionistic rather than unrestricted classical logic.")

Intuitionistic logic

Complete event record 4 of 210

TIME-STAMP	2026-07-18T11:25:00.044Z
LABEL	Intuitionistic logic
VALUE	A deductive logic that preserves constructive validity. It does not accept principles such as excluded middle or double-negation elimination for every proposition without further evidence.
ID	gpil_0004_1bd9b0567747f5e0
SEQUENTIAL-STRING ID	232287698859006001523104383714201795570817636612931497838724627367005143100770730751197001468183029245456057304018842349518997180891082054101119399275483890216672664973472946166197602725804939461231599903561487594867865040092565543490217381171289591028926830463923826618473614292228546767471264026336520486762244660180652666491902113738783528269866208244749963908733962365
BACKWARD-COMPUTED VALUE	A deductive logic that preserves constructive validity. It does not accept principles such as excluded middle or double-negation elimination for every proposition without further evidence.
BACKWARD-COMPUTATION.ALGORITHM	shortlex-unrank/1.0
BACKWARD-COMPUTATION.MATCHES-VALUE	true
ALPHABET	!"#\$%&'()*+,-./0123456789:;<=>?@ABCDEFGHIJKLMNOPQRSTUVWXYZ[\]^_`abcdefghijklmnopqrstuvwxyz{ }~
METRICS.ALPHABET-SYMBOLS	95
METRICS.VALUE-SYMBOLS	188
METRICS.SEQUENTIAL-ID-DIGITS	372
METRICS.NATIVE-INTEGER	false
INSTRUCTION.LANGUAGE	GPIL/1.0
INSTRUCTION.SOURCE-LINE	4
INSTRUCTION.OPERATION	literal
INSTRUCTION.SOURCE	event "Intuitionistic logic" : literal("A deductive logic that preserves constructive validity. It does not accept principles such as excluded middle or double-negation elimination for every proposition without further evidence.")

Constructive proof

Complete event record 5 of 210

TIME-STAMP	2026-07-18T11:25:00.044Z
LABEL	Constructive proof
VALUE	A derivation that supplies usable evidence for its conclusion, such as an algorithm, a witness for an existential claim, or a tagged choice for a disjunction.
ID	gpil_0005_acf482484385991e
SEQUENTIAL-STRING ID	1082226228641627964358480366401442077705569830929253184137928851765755486122167430673409934329412877422036061176876504154066906401163565275883935937157718847960343778296235916618806481096487657543386753893906251926975856522031221869496112047280029967805792784709939534190109829502003001409714127959271341459203395
BACKWARD-COMPUTED VALUE	A derivation that supplies usable evidence for its conclusion, such as an algorithm, a witness for an existential claim, or a tagged choice for a disjunction.
BACKWARD-COMPUTATION.ALGORITHM	shortlex-unrank/1.0
BACKWARD-COMPUTATION.MATCHES-VALUE	true
ALPHABET	!"#\$%&'()*+,-./0123456789:;<=>?@ABCDEFGHIJKLMNOPQRSTUVWXYZ[\]^_`abcdefghijklmnopqrstuvwxyz{ }~
METRICS.ALPHABET-SYMBOLS	95
METRICS.VALUE-SYMBOLS	158
METRICS.SEQUENTIAL-ID-DIGITS	313
METRICS.NATIVE-INTEGER	false
INSTRUCTION.LANGUAGE	GPIL/1.0
INSTRUCTION.SOURCE-LINE	5
INSTRUCTION.OPERATION	literal
INSTRUCTION.SOURCE	event "Constructive proof" : literal("A derivation that supplies usable evidence for its conclusion, such as an algorithm, a witness for an existential claim, or a tagged choice for a disjunction.")

BHK interpretation

Complete event record 6 of 210

TIME-STAMP	2026-07-18T11:25:00.044Z
LABEL	BHK interpretation
VALUE	The Brouwer-Heyting-Kolmogorov explanation of logical constants by their proofs: conjunction uses pairs, disjunction uses a tagged alternative, and implication uses a construction transforming proofs.
ID	gpil_0006_d40f12701dbcdbc1
SEQUENTIAL-STRING ID	198420889562748060043404951525626765577462803342689490920906701672476098347398128931850492726262727508574876287099898160865553586813743011525239645725070271286973579120223829294732899326263398068125544932891563951818647589528821843961532249179529281987460313754817592628118680430342224372683792611680641244756988999241498583840003046444314249458210505974637380944498239421315381705747030370045020
BACKWARD-COMPUTED VALUE	The Brouwer-Heyting-Kolmogorov explanation of logical constants by their proofs: conjunction uses pairs, disjunction uses a tagged alternative, and implication uses a construction transforming proofs.
BACKWARD-COMPUTATION.ALGORITHM	shortlex-unrank/1.0
BACKWARD-COMPUTATION.MATCHES-VALUE	true
ALPHABET	!"#\$%&'()*+,-./0123456789:;<=>?@ABCDEFGHIJKLMNOPQRSTUVWXYZ[\]^_`abcdefghijklmnopqrstuvwxyz{ }~
METRICS.ALPHABET-SYMBOLS	95
METRICS.VALUE-SYMBOLS	200
METRICS.SEQUENTIAL-ID-DIGITS	396
METRICS.NATIVE-INTEGER	false
INSTRUCTION.LANGUAGE	GPIL/1.0
INSTRUCTION.SOURCE-LINE	6
INSTRUCTION.OPERATION	literal
INSTRUCTION.SOURCE	event "BHK interpretation" : literal("The Brouwer-Heyting-Kolmogorov explanation of logical constants by their proofs: conjunction uses pairs, disjunction uses a tagged alternative, and implication uses a construction transforming proofs.")

Judgment

Complete event record 7 of 210

TIME-STAMP	2026-07-18T11:25:00.044Z
LABEL	Judgment
VALUE	A metalevel assertion licensed by formal rules, such as "A is a proposition", "A is true", or "a has type A". Judgments state what a proof system establishes.
ID	gpil_0007_fba64ff6395fb867
SEQUENTIAL-STRING ID	1082257954367440330058870542334668984280078660051641605489460100585138057487313470840783490252153273503762664280496891672793339058002219867217157978125981550305053873728083289816464529647770823010532421778790940719395785451664726147035802710478230749687630150230342836570610319649575515346421211059468022714724370
BACKWARD-COMPUTED VALUE	A metalevel assertion licensed by formal rules, such as "A is a proposition", "A is true", or "a has type A". Judgments state what a proof system establishes.
BACKWARD-COMPUTATION.ALGORITHM	shortlex-unrank/1.0
BACKWARD-COMPUTATION.MATCHES-VALUE	true
ALPHABET	!"#\$%&'()*+,-./0123456789:;<=>?@ABCDEFGHIJKLMNOPQRSTUVWXYZ[\]^_`abcdefghijklmnopqrstuvwxyz{ }~
METRICS.ALPHABET-SYMBOLS	95
METRICS.VALUE-SYMBOLS	158
METRICS.SEQUENTIAL-ID-DIGITS	313
METRICS.NATIVE-INTEGER	false
INSTRUCTION.LANGUAGE	GPIL/1.0
INSTRUCTION.SOURCE-LINE	7
INSTRUCTION.OPERATION	literal
INSTRUCTION.SOURCE	event "Judgment" : literal("A metalevel assertion licensed by formal rules, such as \"A is a proposition\", \"A is true\", or \"a has type A\". Judgments state what a proof system establishes.")

Proof object

Complete event record 8 of 210

TIME-STAMP

2026-07-18T11:25:00.044Z

LABEL

Proof object

VALUE

An explicit syntactic object, usually a term, that records the evidence and rule applications establishing a judgment and can often be checked or evaluated mechanically.

ID

gpil_0008_4e0deac514dec5e5

SEQUENTIAL-STRING ID

630291229071082659101218451116731590409530795261637186753954960436388317087049949357561519356167114
044613361273135686799660961223305804196042067180319498819783437684280066976625055118922165065110765
438126803040970021557829696910545202543312320302637673206191696828843956011903563566650141689728868
4746480853864852341188472030549031365

BACKWARD-COMPUTED VALUE

An explicit syntactic object, usually a term, that records the evidence and rule applications establishing a judgment and can often be checked or evaluated mechanically.

BACKWARD-COMPUTATION.ALGORITHM

shortlex-unrank/1.0

BACKWARD-COMPUTATION.MATCHES-VALUE

true

ALPHABET

!"#\$%&'()*+,-./0123456789:;<=>?@ABCDEFGHIJKLMNPOQRSTUVWXYZ[\]^_`abcdefghijklmnopqrstuvwxyz{|}~

METRICS.ALPHABET-SYMBOLS

95

METRICS.VALUE-SYMBOLS

169

METRICS.SEQUENTIAL-ID-DIGITS

334

METRICS.NATIVE-INTEGER

false

INSTRUCTION.LANGUAGE

GPIL/1.0

INSTRUCTION.SOURCE-LINE

8

INSTRUCTION.OPERATION

literal

INSTRUCTION.SOURCE

event "Proof object" : literal("An explicit syntactic object, usually a term, that records the evidence and rule applications establishing a judgment and can often be checked or evaluated mechanically.")

Witness

Complete event record 9 of 210

TIME-STAMP	2026-07-18T11:25:00.044Z
LABEL	Witness
VALUE	Concrete data that establishes an existential statement. A constructive proof of "there exists x such that A(x)" provides a term t together with evidence for A(t).
ID	gpil_0009_2987cc2491361c98
SEQUENTIAL-STRING ID	9071364465088435095604133626541632075371889356853149675234552759814649438409689804966875135052809758874867182598438370171097943113973052722030178991745596491358336683204041557497252349269719965286592836388749191354066739502883525192411721302592348275386257220445173564528846045277538358840977273062627206331446838671833840
BACKWARD-COMPUTED VALUE	Concrete data that establishes an existential statement. A constructive proof of "there exists x such that A(x)" provides a term t together with evidence for A(t).
BACKWARD-COMPUTATION.ALGORITHM	shortlex-unrank/1.0
BACKWARD-COMPUTATION.MATCHES-VALUE	true
ALPHABET	!"#\$%&'()*+,-./0123456789:;<=>?@ABCDEFGHIJKLMNOPQRSTUVWXYZ[\]^_`abcdefghijklmnopqrstuvwxyz{ }~
METRICS.ALPHABET-SYMBOLS	95
METRICS.VALUE-SYMBOLS	163
METRICS.SEQUENTIAL-ID-DIGITS	322
METRICS.NATIVE-INTEGER	false
INSTRUCTION.LANGUAGE	GPIL/1.0
INSTRUCTION.SOURCE-LINE	9
INSTRUCTION.OPERATION	literal
INSTRUCTION.SOURCE	event "Witness" : literal("Concrete data that establishes an existential statement. A constructive proof of \"there exists x such that A(x)\" provides a term t together with evidence for A(t).")

Decidability

Complete event record 10 of 210

TIME-STAMP	2026-07-18T11:25:00.044Z
LABEL	Decidability
VALUE	A proposition A is decidable when one can construct either a proof of A or a proof that A leads to contradiction, commonly written A or not A.
ID	gpil_0010_8fda7f31708deb9b
SEQUENTIAL-STRING ID	24589943916654354746704715838783746134711874102736616076094386333432815940513939006398962268386724332829516395438221468828951051438000655087973652628157270193918465007042988794536886622276373475899319955369671621678262676234997349536939303686589845915636974542784509955547798786020
BACKWARD-COMPUTED VALUE	A proposition A is decidable when one can construct either a proof of A or a proof that A leads to contradiction, commonly written A or not A.
BACKWARD-COMPUTATION.ALGORITHM	shortlex-unrank/1.0
BACKWARD-COMPUTATION.MATCHES-VALUE	true
ALPHABET	!"#\$%&'()*+,-./0123456789:;<=>?@ABCDEFGHIJKLMNOPQRSTUVWXYZ[\]^_`abcdefghijklmnopqrstuvwxyz{ }~
METRICS.ALPHABET-SYMBOLS	95
METRICS.VALUE-SYMBOLS	142
METRICS.SEQUENTIAL-ID-DIGITS	281
METRICS.NATIVE-INTEGER	false
INSTRUCTION.LANGUAGE	GPIL/1.0
INSTRUCTION.SOURCE-LINE	10
INSTRUCTION.OPERATION	literal
INSTRUCTION.SOURCE	event "Decidability" : literal("A proposition A is decidable when one can construct either a proof of A or a proof that A leads to contradiction, commonly written A or not A.")

Disjunction property

Complete event record 11 of 210

TIME-STAMP	2026-07-18T11:25:00.044Z
LABEL	Disjunction property
VALUE	The property that a closed proof of A or B yields a proof of A or a proof of B, so a proved disjunction contains a definite, inspectable choice of side.
ID	gpil_0011_1e722f196156c83e
SEQUENTIAL-STRING ID	2327310575733274741679258399007671596672562413909201175442752797705655767234797334027774248100877077186423325491715767421113179339710857445680750635041601488382826421426256282187694229071923173272149618405416921630047679614762311341037170171031827657665606342654756877244476815181229747176210939455765
BACKWARD-COMPUTED VALUE	The property that a closed proof of A or B yields a proof of A or a proof of B, so a proved disjunction contains a definite, inspectable choice of side.
BACKWARD-COMPUTATION.ALGORITHM	shortlex-unrank/1.0
BACKWARD-COMPUTATION.MATCHES-VALUE	true
ALPHABET	!"#\$%&'()*+,-./0123456789:;<=>?@ABCDEFGHIJKLMNOPQRSTUVWXYZ[\]^_`abcdefghijklmnopqrstuvwxyz{ }~
METRICS.ALPHABET-SYMBOLS	95
METRICS.VALUE-SYMBOLS	152
METRICS.SEQUENTIAL-ID-DIGITS	301
METRICS.NATIVE-INTEGER	false
INSTRUCTION.LANGUAGE	GPIL/1.0
INSTRUCTION.SOURCE-LINE	11
INSTRUCTION.OPERATION	literal
INSTRUCTION.SOURCE	event "Disjunction property" : literal("The property that a closed proof of A or B yields a proof of A or a proof of B, so a proved disjunction contains a definite, inspectable choice of side.")

Existence property

Complete event record 12 of 210

TIME-STAMP	2026-07-18T11:25:00.044Z
LABEL	Existence property
VALUE	The property that a proof of an existential statement permits extraction of a particular witness and a proof that the witness satisfies the stated condition.
ID	gpil_0012_f17fc7855d08e562
SEQUENTIAL-STRING ID	18008285591445580795616468882889051977655392528379907958150476901417398714844411318750659388199616184811224445423997411492798960282654230232513256121772171179337541217526110293879041642945446110691120614498976453113439374034316047047710287443239300845928157110832432698310035928431318561948367135480065043481520
BACKWARD-COMPUTED VALUE	The property that a proof of an existential statement permits extraction of a particular witness and a proof that the witness satisfies the stated condition.
BACKWARD-COMPUTATION.ALGORITHM	shortlex-unrank/1.0
BACKWARD-COMPUTATION.MATCHES-VALUE	true
ALPHABET	!"#\$%&'()*+,-./0123456789:;<=>?@ABCDEFGHIJKLMNOPQRSTUVWXYZ[\]^_`abcdefghijklmnopqrstuvwxyz{ }~
METRICS.ALPHABET-SYMBOLS	95
METRICS.VALUE-SYMBOLS	157
METRICS.SEQUENTIAL-ID-DIGITS	311
METRICS.NATIVE-INTEGER	false
INSTRUCTION.LANGUAGE	GPIL/1.0
INSTRUCTION.SOURCE-LINE	12
INSTRUCTION.OPERATION	literal
INSTRUCTION.SOURCE	event "Existence property" : literal("The property that a proof of an existential statement permits extraction of a particular witness and a proof that the witness satisfies the stated condition.")

Natural deduction

Complete event record 13 of 210

TIME-STAMP	2026-07-18T11:25:00.044Z
LABEL	Natural deduction
VALUE	A proof calculus organized around introduction and elimination rules for each logical connective, designed to represent the structure of ordinary deductive reasoning.
ID	gpil_0013_4831d16cec7c30bc
SEQUENTIAL-STRING ID	7179993732738933339879233466132829906671317034359848676873657029208589672817902254884443429215969157911527364005240662255410208295731435055305073941083486533038839332405189451774137958523472175197685758920561481510401585601484266726598278920523959840335622464845259863315319961630085772066639160587259167281645704574896927796205
BACKWARD-COMPUTED VALUE	A proof calculus organized around introduction and elimination rules for each logical connective, designed to represent the structure of ordinary deductive reasoning.
BACKWARD-COMPUTATION.ALGORITHM	shortlex-unrank/1.0
BACKWARD-COMPUTATION.MATCHES-VALUE	true
ALPHABET	!"#\$%&'()*+,-./0123456789:;<=>?@ABCDEFGHIJKLMNOPQRSTUVWXYZ[\]^_`abcdefghijklmnopqrstuvwxyz{ }~
METRICS.ALPHABET-SYMBOLS	95
METRICS.VALUE-SYMBOLS	166
METRICS.SEQUENTIAL-ID-DIGITS	328
METRICS.NATIVE-INTEGER	false
INSTRUCTION.LANGUAGE	GPIL/1.0
INSTRUCTION.SOURCE-LINE	13
INSTRUCTION.OPERATION	literal
INSTRUCTION.SOURCE	event "Natural deduction" : literal("A proof calculus organized around introduction and elimination rules for each logical connective, designed to represent the structure of ordinary deductive reasoning.")

Introduction rule

Complete event record 14 of 210

TIME-STAMP	2026-07-18T11:25:00.044Z
LABEL	Introduction rule
VALUE	A rule specifying how canonical evidence for a connective or type is constructed, such as building a pair to prove a conjunction.
ID	gpil_0014_5aac4b604708e1f9
SEQUENTIAL-STRING ID	479019885571311146911783271270839860603578819647385634475165107725189649612647893975320270418249652133256737537807983345474577230958451676277657605865410816142429121016111271912621420962221641265324375203167027025819325742608518777263557073262507787328395
BACKWARD-COMPUTED VALUE	A rule specifying how canonical evidence for a connective or type is constructed, such as building a pair to prove a conjunction.
BACKWARD-COMPUTATION.ALGORITHM	shortlex-unrank/1.0
BACKWARD-COMPUTATION.MATCHES-VALUE	true
ALPHABET	!"#\$%&'()*+,-./0123456789:;<=>?@ABCDEFGHIJKLMNOPQRSTUVWXYZ[\]^_`abcdefghijklmnopqrstuvwxyz{ }~
METRICS.ALPHABET-SYMBOLS	95
METRICS.VALUE-SYMBOLS	129
METRICS.SEQUENTIAL-ID-DIGITS	255
METRICS.NATIVE-INTEGER	false
INSTRUCTION.LANGUAGE	GPIL/1.0
INSTRUCTION.SOURCE-LINE	14
INSTRUCTION.OPERATION	literal
INSTRUCTION.SOURCE	event "Introduction rule" : literal("A rule specifying how canonical evidence for a connective or type is constructed, such as building a pair to prove a conjunction.")

Elimination rule

Complete event record 15 of 210

TIME-STAMP
2026-07-18T11:25:00.044Z

LABEL
Elimination rule

VALUE
A rule specifying how evidence for a connective or type may be used, such as projecting a component from evidence for a conjunction.

ID
gpil_0015_952700ad4a411aa9

SEQUENTIAL-STRING ID
410699674391702894583490182205836325484993391323731514003512565816253503475585107412703615194700513507406073015971986537960718762485576215968686050917046002917626928605452796528909632962576832906359045482540896575296458563901129830604548229910706190272679906520

BACKWARD-COMPUTED VALUE
A rule specifying how evidence for a connective or type may be used, such as projecting a component from evidence for a conjunction.

BACKWARD-COMPUTATION.ALGORITHM
shortlex-unrank/1.0

BACKWARD-COMPUTATION.MATCHES-VALUE
true

ALPHABET
!"#\$%&'()*+,-./0123456789:;<=>?@ABCDEFGHIJKLMNOPQRSTUVWXYZ[\]^_`abcdefghijklmnopqrstuvwxyz{|}~

METRICS.ALPHABET-SYMBOLS
95

METRICS.VALUE-SYMBOLS
132

METRICS.SEQUENTIAL-ID-DIGITS
261

METRICS.NATIVE-INTEGER
false

INSTRUCTION.LANGUAGE
GPIL/1.0

INSTRUCTION.SOURCE-LINE
15

INSTRUCTION.OPERATION
literal

INSTRUCTION.SOURCE
event "Elimination rule" : literal("A rule specifying how evidence for a connective or type may be used, such as projecting a component from evidence for a conjunction.")

Harmony

Complete event record 16 of 210

TIME-STAMP	2026-07-18T11:25:00.044Z
LABEL	Harmony
VALUE	The proof-theoretic balance in which elimination rules use exactly the information supplied by introduction rules, giving a connective neither excess nor deficient inferential strength.
ID	gpil_0016_78c79597d3362ca7
SEQUENTIAL-STRING ID	428285447100198428572971762789716876614317800589455153669161503349350264783352572071471385885905753801703266859019513535501589837199273679173344070022331357200815092242141070428981271623773228173403942901648550737804205599585259953343579953438386138517239414785686331581790951505392066018619527708786084208913865725546157827783204530434820125844137720774298349588825
BACKWARD-COMPUTED VALUE	The proof-theoretic balance in which elimination rules use exactly the information supplied by introduction rules, giving a connective neither excess nor deficient inferential strength.
BACKWARD-COMPUTATION.ALGORITHM	shortlex-unrank/1.0
BACKWARD-COMPUTATION.MATCHES-VALUE	true
ALPHABET	!"#\$%&'()*+,-./0123456789:;<=>?@ABCDEFGHIJKLMNOPQRSTUVWXYZ[\]^_`abcdefghijklmnopqrstuvwxyz{ }~
METRICS.ALPHABET-SYMBOLS	95
METRICS.VALUE-SYMBOLS	185
METRICS.SEQUENTIAL-ID-DIGITS	366
METRICS.NATIVE-INTEGER	false
INSTRUCTION.LANGUAGE	GPIL/1.0
INSTRUCTION.SOURCE-LINE	16
INSTRUCTION.OPERATION	literal
INSTRUCTION.SOURCE	event "Harmony" : literal("The proof-theoretic balance in which elimination rules use exactly the information supplied by introduction rules, giving a connective neither excess nor deficient inferential strength.")

Normalization

Complete event record 17 of 210

TIME-STAMP	2026-07-18T11:25:00.044Z
LABEL	Normalization
VALUE	The process or theorem that removes proof detours, especially an introduction followed immediately by a matching elimination, until the derivation reaches normal form.
ID	gpil_0017_cfd7520aace50ae6
SEQUENTIAL-STRING ID	1078222579595268972895970593262364207011412318547043847104531806713450865322388166062756003667642043108356914404758276424177932635355308724733099401650921306246871609988423113169370940743499096232863216617002580841454618795882616685608019098197240213797416994069266025114211110329891471357475836293494207038985476368096673844556500
BACKWARD-COMPUTED VALUE	The process or theorem that removes proof detours, especially an introduction followed immediately by a matching elimination, until the derivation reaches normal form.
BACKWARD-COMPUTATION.ALGORITHM	shortlex-unrank/1.0
BACKWARD-COMPUTATION.MATCHES-VALUE	true
ALPHABET	!"#\$%&'()*+,-./0123456789:;<=>?@ABCDEFGHIJKLMNOPQRSTUVWXYZ[\]^_`abcdefghijklmnopqrstuvwxyz{ }~
METRICS.ALPHABET-SYMBOLS	95
METRICS.VALUE-SYMBOLS	167
METRICS.SEQUENTIAL-ID-DIGITS	331
METRICS.NATIVE-INTEGER	false
INSTRUCTION.LANGUAGE	GPIL/1.0
INSTRUCTION.SOURCE-LINE	17
INSTRUCTION.OPERATION	literal
INSTRUCTION.SOURCE	event "Normalization" : literal("The process or theorem that removes proof detours, especially an introduction followed immediately by a matching elimination, until the derivation reaches normal form.")

Normal form

Complete event record 18 of 210

TIME-STAMP	2026-07-18T11:25:00.044Z
LABEL	Normal form
VALUE	A proof or proof term with no applicable computational reductions or introduction-elimination detours; its shape makes the evidence for the conclusion more explicit.
ID	gpil_0018_fa7e528d79edf0fd
SEQUENTIAL-STRING ID	75578881397251933969072850855154293454513548832609745931644065254641211401420890267290376155722712223407554182789605395363240212904029632907163383926163716186659245758808009497619206614373133422016482244406603685966984244461225359039143313886094782038277999962092130622984872140100372642951856759703859965215325588569182270565
BACKWARD-COMPUTED VALUE	A proof or proof term with no applicable computational reductions or introduction-elimination detours; its shape makes the evidence for the conclusion more explicit.
BACKWARD-COMPUTATION.ALGORITHM	shortlex-unrank/1.0
BACKWARD-COMPUTATION.MATCHES-VALUE	true
ALPHABET	!"#\$%&'()*+,-./0123456789:;<=>?@ABCDEFGHIJKLMNOPQRSTUVWXYZ[\]^_`abcdefghijklmnopqrstuvwxyz{ }~
METRICS.ALPHABET-SYMBOLS	95
METRICS.VALUE-SYMBOLS	165
METRICS.SEQUENTIAL-ID-DIGITS	326
METRICS.NATIVE-INTEGER	false
INSTRUCTION.LANGUAGE	GPIL/1.0
INSTRUCTION.SOURCE-LINE	18
INSTRUCTION.OPERATION	literal
INSTRUCTION.SOURCE	event "Normal form" : literal("A proof or proof term with no applicable computational reductions or introduction-elimination detours; its shape makes the evidence for the conclusion more explicit.")

Sequent calculus

Complete event record 19 of 210

TIME-STAMP	2026-07-18T11:25:00.044Z
LABEL	Sequent calculus
VALUE	A formal calculus whose basic judgments are sequents, typically a context of assumptions followed by a conclusion, and whose rules expose logical and structural proof steps.
ID	gpil_0019_c4b50d8590a0220f
SEQUENTIAL-STRING ID	501389358891945970270323234372788909361053852049959208113718058628227889972316931303371287255858285305311028917377423820522045339534853939279364979318199507885294660174290407145050233406901963224786771598197405186773720451339934275269642166374044529229383149436172773324975373527192729953958628432861355820609162497501433179550984298994827145
BACKWARD-COMPUTED VALUE	A formal calculus whose basic judgments are sequents, typically a context of assumptions followed by a conclusion, and whose rules expose logical and structural proof steps.
BACKWARD-COMPUTATION.ALGORITHM	shortlex-unrank/1.0
BACKWARD-COMPUTATION.MATCHES-VALUE	true
ALPHABET	!"#\$%&'()*+,-./0123456789:;<=>?@ABCDEFGHIJKLMNOPQRSTUVWXYZ[\]^_`abcdefghijklmnopqrstuvwxyz{ }~
METRICS.ALPHABET-SYMBOLS	95
METRICS.VALUE-SYMBOLS	173
METRICS.SEQUENTIAL-ID-DIGITS	342
METRICS.NATIVE-INTEGER	false
INSTRUCTION.LANGUAGE	GPIL/1.0
INSTRUCTION.SOURCE-LINE	19
INSTRUCTION.OPERATION	literal
INSTRUCTION.SOURCE	event "Sequent calculus" : literal("A formal calculus whose basic judgments are sequents, typically a context of assumptions followed by a conclusion, and whose rules expose logical and structural proof steps.")

Cut rule

Complete event record 20 of 210

TIME-STAMP	2026-07-18T11:25:00.044Z
LABEL	Cut rule
VALUE	A rule that composes a proof of an intermediate formula with a proof using that formula as an assumption, thereby formalizing the use of a lemma.
ID	gpil_0020_f7554fa1d1153eb1
SEQUENTIAL-STRING ID	21082942645445126267187376748509744416252698989276867713968444727483566971279631027254427118114377503902194517462685675739488600591389948920940505435344029555943298514139304215215125727498112851916708634483869521162057598836436160629837985810795736272718629591622746150020115543817185485
BACKWARD-COMPUTED VALUE	A rule that composes a proof of an intermediate formula with a proof using that formula as an assumption, thereby formalizing the use of a lemma.
BACKWARD-COMPUTATION.ALGORITHM	shortlex-unrank/1.0
BACKWARD-COMPUTATION.MATCHES-VALUE	true
ALPHABET	!"#\$%&'()*+,-./0123456789:;<=>?@ABCDEFGHIJKLMNOPQRSTUVWXYZ[\]^_`abcdefghijklmnopqrstuvwxyz{ }~
METRICS.ALPHABET-SYMBOLS	95
METRICS.VALUE-SYMBOLS	145
METRICS.SEQUENTIAL-ID-DIGITS	287
METRICS.NATIVE-INTEGER	false
INSTRUCTION.LANGUAGE	GPIL/1.0
INSTRUCTION.SOURCE-LINE	20
INSTRUCTION.OPERATION	literal
INSTRUCTION.SOURCE	event "Cut rule" : literal("A rule that composes a proof of an intermediate formula with a proof using that formula as an assumption, thereby formalizing the use of a lemma.")

Cut elimination

Complete event record 21 of 210

TIME-STAMP	2026-07-18T11:25:00.044Z
LABEL	Cut elimination
VALUE	The theorem or transformation that removes uses of cut from a derivation. It exposes direct proof structure and underlies consistency, normalization, and proof analysis.
ID	gpil_0021_8cc55a3c0396b095
SEQUENTIAL-STRING ID	9730958789497527255081204926629524039966016238505375698120458156988239022543901449597107541288250445373400100009192734015824547557166393217861707843410704421472804763321262286508063843643403181481093045360985391139475015583607126065333825465488791570558898837022287428046788426509422189719228422255501397570463206741708729408772401595
BACKWARD-COMPUTED VALUE	The theorem or transformation that removes uses of cut from a derivation. It exposes direct proof structure and underlies consistency, normalization, and proof analysis.
BACKWARD-COMPUTATION.ALGORITHM	shortlex-unrank/1.0
BACKWARD-COMPUTATION.MATCHES-VALUE	true
ALPHABET	!"#\$%&'()*+,-./0123456789:;<=>?@ABCDEFGHIJKLMNOPQRSTUVWXYZ[\]^_`abcdefghijklmnopqrstuvwxyz{ }~
METRICS.ALPHABET-SYMBOLS	95
METRICS.VALUE-SYMBOLS	169
METRICS.SEQUENTIAL-ID-DIGITS	334
METRICS.NATIVE-INTEGER	false
INSTRUCTION.LANGUAGE	GPIL/1.0
INSTRUCTION.SOURCE-LINE	21
INSTRUCTION.OPERATION	literal
INSTRUCTION.SOURCE	event "Cut elimination" : literal("The theorem or transformation that removes uses of cut from a derivation. It exposes direct proof structure and underlies consistency, normalization, and proof analysis.")

Subformula property

Complete event record 22 of 210

TIME-STAMP	2026-07-18T11:25:00.044Z
LABEL	Subformula property
VALUE	The analytic property, typically obtained for cut-free systems, that formulas in a derivation are built from formulas already present in its premises or conclusion.
ID	gpil_0022_1f389ebc3a176377
SEQUENTIAL-STRING ID	1257585742402180001858336047396377066753443118609451056537037835580622480853140007979017448330154710783787118691517283625075287224950870740633861611605868548581390489148896738091782019857187687652321220836708991061630464752215817778266561542197572366492244453372567211697277310672144851040448671239850830688333125773335321520
BACKWARD-COMPUTED VALUE	The analytic property, typically obtained for cut-free systems, that formulas in a derivation are built from formulas already present in its premises or conclusion.
BACKWARD-COMPUTATION.ALGORITHM	shortlex-unrank/1.0
BACKWARD-COMPUTATION.MATCHES-VALUE	true
ALPHABET	!"#\$%&'()*+,-./0123456789:;<=>?@ABCDEFGHIJKLMNOPQRSTUVWXYZ[\]^_`abcdefghijklmnopqrstuvwxyz{ }~
METRICS.ALPHABET-SYMBOLS	95
METRICS.VALUE-SYMBOLS	164
METRICS.SEQUENTIAL-ID-DIGITS	325
METRICS.NATIVE-INTEGER	false
INSTRUCTION.LANGUAGE	GPIL/1.0
INSTRUCTION.SOURCE-LINE	22
INSTRUCTION.OPERATION	literal
INSTRUCTION.SOURCE	event "Subformula property" : literal("The analytic property, typically obtained for cut-free systems, that formulas in a derivation are built from formulas already present in its premises or conclusion.")

Curry-Howard correspondence

Complete event record 23 of 210

TIME-STAMP

2026-07-18T11:25:00.044Z

LABEL

Curry-Howard correspondence

VALUE

The correspondence between propositions and types, proofs and typed terms, and proof normalization and program evaluation, revealing computational content in constructive derivations.

ID

gpil_0023_a5b02eb2fbecb62d

SEQUENTIAL-STRING ID

474554510610533690154988992992240775122878854997582178216488204481368592396325513870885045790269524
883491548263108471522118284080880403996262297332574458742727062035877569436515723466022552575520711
087587963466509034354538612088420449030249980369005893982352153706002696739292437898658455388010896
86968652071599324941375481302518988899608282162325026402659454095

BACKWARD-COMPUTED VALUE

The correspondence between propositions and types, proofs and typed terms, and proof normalization and program evaluation, revealing computational content in constructive derivations.

BACKWARD-COMPUTATION.ALGORITHM

shortlex-unrank/1.0

BACKWARD-COMPUTATION.MATCHES-VALUE

true

ALPHABET

!"#\$%&'()*+,-./0123456789:;<=>?@ABCDEFGHIJKLMNOPQRSTUVWXYZ[\]^_`abcdefghijklmnopqrstuvwxyz{|}~

METRICS.ALPHABET-SYMBOLS

95

METRICS.VALUE-SYMBOLS

183

METRICS.SEQUENTIAL-ID-DIGITS

362

METRICS.NATIVE-INTEGER

false

INSTRUCTION.LANGUAGE

GPIL/1.0

INSTRUCTION.SOURCE-LINE

23

INSTRUCTION.OPERATION

literal

INSTRUCTION.SOURCE

event "Curry-Howard correspondence" : literal("The correspondence between propositions and types, proofs and typed terms, and proof normalization and program evaluation, revealing computational content in constructive derivations.")

Propositions as types

Complete event record 24 of 210

TIME-STAMP	2026-07-18T11:25:00.044Z
LABEL	Propositions as types
VALUE	The reading in which a proposition is represented by a type and a proof by an inhabitant of that type; proving the proposition means constructing such a term.
ID	gpil_0024_27bcd996ec2d8020
SEQUENTIAL-STRING ID	1710787131914433744894321937167097215400259931785433039854244948505554932786049684585584884868754962682363203771362269107907867811057201511074978745283665965341008184154698260145692225663751948321279214408835206452296647522882087627974388574539027891923808291693467454644448875432131238721343742683741392007682125
BACKWARD-COMPUTED VALUE	The reading in which a proposition is represented by a type and a proof by an inhabitant of that type; proving the proposition means constructing such a term.
BACKWARD-COMPUTATION.ALGORITHM	shortlex-unrank/1.0
BACKWARD-COMPUTATION.MATCHES-VALUE	true
ALPHABET	!"#\$%&'()*+,-./0123456789:;<=>?@ABCDEFGHIJKLMNOPQRSTUVWXYZ[\]^_`abcdefghijklmnopqrstuvwxyz{ }~
METRICS.ALPHABET-SYMBOLS	95
METRICS.VALUE-SYMBOLS	158
METRICS.SEQUENTIAL-ID-DIGITS	313
METRICS.NATIVE-INTEGER	false
INSTRUCTION.LANGUAGE	GPIL/1.0
INSTRUCTION.SOURCE-LINE	24
INSTRUCTION.OPERATION	literal
INSTRUCTION.SOURCE	event "Propositions as types" : literal("The reading in which a proposition is represented by a type and a proof by an inhabitant of that type; proving the proposition means constructing such a term.")

Typed lambda calculus

Complete event record 25 of 210

TIME-STAMP	2026-07-18T11:25:00.044Z
LABEL	Typed lambda calculus
VALUE	A formal language of typed functions and computation whose term constructors and reductions provide proof objects and normalization behavior for corresponding logical rules.
ID	gpil_0025_2069e01947081282
SEQUENTIAL-STRING ID	501389358891945970480798511521166679718497975505303209654019822679331530758143961164378061220673443941876942613409590477438567927873271961399681631535315757823633880666916188154155441827160678339829886080215615260074168076721439307201568578296949898819414639263690258320691241468071027053155571998344031226276082406674336206357001088531011370
BACKWARD-COMPUTED VALUE	A formal language of typed functions and computation whose term constructors and reductions provide proof objects and normalization behavior for corresponding logical rules.
BACKWARD-COMPUTATION.ALGORITHM	shortlex-unrank/1.0
BACKWARD-COMPUTATION.MATCHES-VALUE	true
ALPHABET	!"#\$%&'()*+,-./0123456789:;<=>?@ABCDEFGHIJKLMNOPQRSTUVWXYZ[\]^_`abcdefghijklmnopqrstuvwxyz{ }~
METRICS.ALPHABET-SYMBOLS	95
METRICS.VALUE-SYMBOLS	173
METRICS.SEQUENTIAL-ID-DIGITS	342
METRICS.NATIVE-INTEGER	false
INSTRUCTION.LANGUAGE	GPIL/1.0
INSTRUCTION.SOURCE-LINE	25
INSTRUCTION.OPERATION	literal
INSTRUCTION.SOURCE	event "Typed lambda calculus" : literal("A formal language of typed functions and computation whose term constructors and reductions provide proof objects and normalization behavior for corresponding logical rules.")

Intuitionistic type theory

Complete event record 26 of 210

TIME-STAMP	2026-07-18T11:25:00.044Z
LABEL	Intuitionistic type theory
VALUE	A constructive foundation, associated with Martin-Lof, that combines dependent types, formal judgments, and computational equality so that proofs are checkable terms.
ID	gpil_0026_9ed529d8a3edf3c2
SEQUENTIAL-STRING ID	7179688979568384683357426634261950609498233702500720618615499166827473686253335789839437730959054961947667361875858219606083908713920887666140169332340874250304009085502088449792200690845972468997189823132022501642283155712853952905958518032512512837076223068950834306048482967023214074588946859780748631194531038604706584183445
BACKWARD-COMPUTED VALUE	A constructive foundation, associated with Martin-Lof, that combines dependent types, formal judgments, and computational equality so that proofs are checkable terms.
BACKWARD-COMPUTATION.ALGORITHM	shortlex-unrank/1.0
BACKWARD-COMPUTATION.MATCHES-VALUE	true
ALPHABET	!"#\$%&'()*+,-./0123456789:;<=>?@ABCDEFGHIJKLMNOPQRSTUVWXYZ[\]^_`abcdefghijklmnopqrstuvwxyz{ }~
METRICS.ALPHABET-SYMBOLS	95
METRICS.VALUE-SYMBOLS	166
METRICS.SEQUENTIAL-ID-DIGITS	328
METRICS.NATIVE-INTEGER	false
INSTRUCTION.LANGUAGE	GPIL/1.0
INSTRUCTION.SOURCE-LINE	26
INSTRUCTION.OPERATION	literal
INSTRUCTION.SOURCE	event "Intuitionistic type theory" : literal("A constructive foundation, associated with Martin-Lof, that combines dependent types, formal judgments, and computational equality so that proofs are checkable terms.")

Dependent type

Complete event record 27 of 210

TIME-STAMP	2026-07-18T11:25:00.044Z
LABEL	Dependent type
VALUE	A type that may vary with a term. Dependent function types model universal quantification, while dependent pair types model existential evidence with its witness.
ID	gpil_0027_aa39b9096908f6be
SEQUENTIAL-STRING ID	88152656905941120863449734428012214000864183182815563415886315079280643810516235668395632832344945125051156774711160283511562252834979848037115101047172233180839723210799272697731398164247055459093059598943341714390553614437236887396921192273504090495954728808131348906120216271055800152019800021643653416620770756116095
BACKWARD-COMPUTED VALUE	A type that may vary with a term. Dependent function types model universal quantification, while dependent pair types model existential evidence with its witness.
BACKWARD-COMPUTATION.ALGORITHM	shortlex-unrank/1.0
BACKWARD-COMPUTATION.MATCHES-VALUE	true
ALPHABET	!"#\$%&'()*+,-./0123456789:;<=>?@ABCDEFGHIJKLMNOPQRSTUVWXYZ[\]^_`abcdefghijklmnopqrstuvwxyz{ }~
METRICS.ALPHABET-SYMBOLS	95
METRICS.VALUE-SYMBOLS	162
METRICS.SEQUENTIAL-ID-DIGITS	320
METRICS.NATIVE-INTEGER	false
INSTRUCTION.LANGUAGE	GPIL/1.0
INSTRUCTION.SOURCE-LINE	27
INSTRUCTION.OPERATION	literal
INSTRUCTION.SOURCE	event "Dependent type" : literal("A type that may vary with a term. Dependent function types model universal quantification, while dependent pair types model existential evidence with its witness.")

Canonical form

Complete event record 28 of 210

TIME-STAMP	2026-07-18T11:25:00.044Z
LABEL	Canonical form
VALUE	A fully introduced, evaluated form that visibly displays the kind of evidence present, such as a pair, an injection, a lambda abstraction, or a numeral.
ID	gpil_0028_9ebe406732926a6a
SEQUENTIAL-STRING ID	1472242972330064506163774394120754499075948170847532213341859096731873213619359594182815089811886276818606294367108863710892868399110734559148618027232360191272672802826712981294645769078750252546223786102225428864919179838430755127149008185599090721051263073467654950283487328365461530971225462221355
BACKWARD-COMPUTED VALUE	A fully introduced, evaluated form that visibly displays the kind of evidence present, such as a pair, an injection, a lambda abstraction, or a numeral.
BACKWARD-COMPUTATION.ALGORITHM	shortlex-unrank/1.0
BACKWARD-COMPUTATION.MATCHES-VALUE	true
ALPHABET	!"#\$%&'()*+,-./0123456789:;<=>?@ABCDEFGHIJKLMNOPQRSTUVWXYZ[\]^_`abcdefghijklmnopqrstuvwxyz{ }~
METRICS.ALPHABET-SYMBOLS	95
METRICS.VALUE-SYMBOLS	152
METRICS.SEQUENTIAL-ID-DIGITS	301
METRICS.NATIVE-INTEGER	false
INSTRUCTION.LANGUAGE	GPIL/1.0
INSTRUCTION.SOURCE-LINE	28
INSTRUCTION.OPERATION	literal
INSTRUCTION.SOURCE	event "Canonical form" : literal("A fully introduced, evaluated form that visibly displays the kind of evidence present, such as a pair, an injection, a lambda abstraction, or a numeral.")

Canonicity

Complete event record 29 of 210

TIME-STAMP	2026-07-18T11:25:00.044Z
LABEL	Canonicity
VALUE	The property that every closed term of a suitable data type evaluates to a canonical constructor form, ensuring that a closed proof reveals concrete computational evidence.
ID	gpil_0029_5ff306e67ec276d5
SEQUENTIAL-STRING ID	8343080784728984845039041244709374276247336617671935936358499418678847198699826903864609529839865951416326833274660951896804698753479233392960447860851826648387382466183933843316960313013059812229952366859249138403714981036085049518401946513626134487022634311242986775339312247469745114007014995248511715984835023678172547583097867122634240
BACKWARD-COMPUTED VALUE	The property that every closed term of a suitable data type evaluates to a canonical constructor form, ensuring that a closed proof reveals concrete computational evidence.
BACKWARD-COMPUTATION.ALGORITHM	shortlex-unrank/1.0
BACKWARD-COMPUTATION.MATCHES-VALUE	true
ALPHABET	!"#\$%&'()*+,-./0123456789:;<=>?@ABCDEFGHIJKLMNOPQRSTUVWXYZ[\]^_`abcdefghijklmnopqrstuvwxyz{ }~
METRICS.ALPHABET-SYMBOLS	95
METRICS.VALUE-SYMBOLS	172
METRICS.SEQUENTIAL-ID-DIGITS	340
METRICS.NATIVE-INTEGER	false
INSTRUCTION.LANGUAGE	GPIL/1.0
INSTRUCTION.SOURCE-LINE	29
INSTRUCTION.OPERATION	literal
INSTRUCTION.SOURCE	event "Canonicity" : literal("The property that every closed term of a suitable data type evaluates to a canonical constructor form, ensuring that a closed proof reveals concrete computational evidence.")

Proof relevance

Complete event record 30 of 210

TIME-STAMP	2026-07-18T11:25:00.044Z
LABEL	Proof relevance
VALUE	The principle that proof objects may carry distinct data or computational behavior and are not automatically identified merely because they establish the same proposition.
ID	gpil_0030_4e4eed285eb7b434
SEQUENTIAL-STRING ID	87821902997133831842518979355201879268315988487547585075870280506518322301111538180745501334200599722982897348595998865112091537280687002021903135178568261857965061995268373135116007286042804700323093411035452790287924980057843079206589458111512793082588799668424063269738564064947112542179717733771880006384476118829171019615342028794020
BACKWARD-COMPUTED VALUE	The principle that proof objects may carry distinct data or computational behavior and are not automatically identified merely because they establish the same proposition.
BACKWARD-COMPUTATION.ALGORITHM	shortlex-unrank/1.0
BACKWARD-COMPUTATION.MATCHES-VALUE	true
ALPHABET	!"#\$%&'()*+,-./0123456789:;<=>?@ABCDEFGHIJKLMNOPQRSTUVWXYZ[\]^_`abcdefghijklmnopqrstuvwxyz{ }~
METRICS.ALPHABET-SYMBOLS	95
METRICS.VALUE-SYMBOLS	171
METRICS.SEQUENTIAL-ID-DIGITS	338
METRICS.NATIVE-INTEGER	false
INSTRUCTION.LANGUAGE	GPIL/1.0
INSTRUCTION.SOURCE-LINE	30
INSTRUCTION.OPERATION	literal
INSTRUCTION.SOURCE	event "Proof relevance" : literal("The principle that proof objects may carry distinct data or computational behavior and are not automatically identified merely because they establish the same proposition.")

Realizability

Complete event record 31 of 210

TIME-STAMP	2026-07-18T11:25:00.044Z
LABEL	Realizability
VALUE	An interpretation that assigns computational objects, called realizers, to formulas according to what counts as effective evidence, supporting constructive soundness and program extraction.
ID	gpil_0031_5f5e0aa9e89a3456
SEQUENTIAL-STRING ID	22595056211047576367613750748124887891881568461059691565601357990225103205123669395697162994423847301816169431103177641576069488750723289744998435528653473991584253993874498086866629647351367070229202232206721285263051373076793702110665718649331794486791736007343596481150819362147923089742406706598958955763189575517429507036817132647291375290038155338076973205534870719020
BACKWARD-COMPUTED VALUE	An interpretation that assigns computational objects, called realizers, to formulas according to what counts as effective evidence, supporting constructive soundness and program extraction.
BACKWARD-COMPUTATION.ALGORITHM	shortlex-unrank/1.0
BACKWARD-COMPUTATION.MATCHES-VALUE	true
ALPHABET	!"#\$%&'()*+,-./0123456789:;<=>?@ABCDEFGHIJKLMNOPQRSTUVWXYZ[\]^_`abcdefghijklmnopqrstuvwxyz{ }~
METRICS.ALPHABET-SYMBOLS	95
METRICS.VALUE-SYMBOLS	189
METRICS.SEQUENTIAL-ID-DIGITS	374
METRICS.NATIVE-INTEGER	false
INSTRUCTION.LANGUAGE	GPIL/1.0
INSTRUCTION.SOURCE-LINE	31
INSTRUCTION.OPERATION	literal
INSTRUCTION.SOURCE	event "Realizability" : literal("An interpretation that assigns computational objects, called realizers, to formulas according to what counts as effective evidence, supporting constructive soundness and program extraction.")

Law of excluded middle

Complete event record 32 of 210

TIME-STAMP	2026-07-18T11:25:00.044Z
LABEL	Law of excluded middle
VALUE	The schema A or not A. Constructive systems accept it for propositions equipped with a decision method, but do not assume it uniformly for every proposition.
ID	gpil_0032_541a7102313e8cb7
SEQUENTIAL-STRING ID	18008285603127324652140449623185087614171509266732906640395993606686614044242937523735773370193406612901280086314404528849619291560131081972315420655843036823846208017330222445991750817037765092675602220075239391846393302365689107532365664947285915198257865998361293749274171323856124293875296694586478503403395
BACKWARD-COMPUTED VALUE	The schema A or not A. Constructive systems accept it for propositions equipped with a decision method, but do not assume it uniformly for every proposition.
BACKWARD-COMPUTATION.ALGORITHM	shortlex-unrank/1.0
BACKWARD-COMPUTATION.MATCHES-VALUE	true
ALPHABET	!"#\$%&'()*+,-./0123456789:;<=>?@ABCDEFGHIJKLMNOPQRSTUVWXYZ[\]^_`abcdefghijklmnopqrstuvwxyz{ }~
METRICS.ALPHABET-SYMBOLS	95
METRICS.VALUE-SYMBOLS	157
METRICS.SEQUENTIAL-ID-DIGITS	311
METRICS.NATIVE-INTEGER	false
INSTRUCTION.LANGUAGE	GPIL/1.0
INSTRUCTION.SOURCE-LINE	32
INSTRUCTION.OPERATION	literal
INSTRUCTION.SOURCE	event "Law of excluded middle" : literal("The schema A or not A. Constructive systems accept it for propositions equipped with a decision method, but do not assume it uniformly for every proposition.")

Double-negation elimination

Complete event record 33 of 210

TIME-STAMP	2026-07-18T11:25:00.044Z
LABEL	Double-negation elimination
VALUE	The inference from not-not-A to A. It is not generally valid intuitionistically because refuting every refutation of A need not construct evidence for A.
ID	gpil_0033_e0e30fe0922faffd
SEQUENTIAL-STRING ID	221094504339145737657939737276133430144367814252944126393223342254241982943774317410854342542900447206989211968592434494112890369479282174971822290903795943184215730882650851078460318664891409526289095403747222484457177417435505370354692376800531152506218190201808923735479757519062345416773410483111895
BACKWARD-COMPUTED VALUE	The inference from not-not-A to A. It is not generally valid intuitionistically because refuting every refutation of A need not construct evidence for A.
BACKWARD-COMPUTATION.ALGORITHM	shortlex-unrank/1.0
BACKWARD-COMPUTATION.MATCHES-VALUE	true
ALPHABET	!"#\$%&'()*+,-./0123456789:;<=>?@ABCDEFGHIJKLMNOPQRSTUVWXYZ[\]^_`abcdefghijklmnopqrstuvwxyz{ }~
METRICS.ALPHABET-SYMBOLS	95
METRICS.VALUE-SYMBOLS	153
METRICS.SEQUENTIAL-ID-DIGITS	303
METRICS.NATIVE-INTEGER	false
INSTRUCTION.LANGUAGE	GPIL/1.0
INSTRUCTION.SOURCE-LINE	33
INSTRUCTION.OPERATION	literal
INSTRUCTION.SOURCE	event "Double-negation elimination" : literal("The inference from not-not-A to A. It is not generally valid intuitionistically because refuting every refutation of A need not construct evidence for A.")

Minimal logic

Complete event record 34 of 210

TIME-STAMP	2026-07-18T11:25:00.044Z
LABEL	Minimal logic
VALUE	The intuitionistic core obtained without the rule that arbitrary conclusions follow from falsity, allowing constructive implication and negation to be studied separately from explosion.
ID	gpil_0034_0a2f76a3516bf171
SEQUENTIAL-STRING ID	428285446411677950150371540162339011016582407554695943405547758471181760627444037200428206240925866665876528526856272664064328171148800690553246748997029828535862515782114177184341958367798521932700929257706617134215020041759572895592239991950600486132783561832473876504634029769238795992078178293182730766755567122901950639129398003594046493456383524192379909012145
BACKWARD-COMPUTED VALUE	The intuitionistic core obtained without the rule that arbitrary conclusions follow from falsity, allowing constructive implication and negation to be studied separately from explosion.
BACKWARD-COMPUTATION.ALGORITHM	shortlex-unrank/1.0
BACKWARD-COMPUTATION.MATCHES-VALUE	true
ALPHABET	!"#\$%&'()*+,-./0123456789:;<=>?@ABCDEFGHIJKLMNOPQRSTUVWXYZ[\]^_`abcdefghijklmnopqrstuvwxyz{ }~
METRICS.ALPHABET-SYMBOLS	95
METRICS.VALUE-SYMBOLS	185
METRICS.SEQUENTIAL-ID-DIGITS	366
METRICS.NATIVE-INTEGER	false
INSTRUCTION.LANGUAGE	GPIL/1.0
INSTRUCTION.SOURCE-LINE	34
INSTRUCTION.OPERATION	literal
INSTRUCTION.SOURCE	event "Minimal logic" : literal("The intuitionistic core obtained without the rule that arbitrary conclusions follow from falsity, allowing constructive implication and negation to be studied separately from explosion.")

Heyting arithmetic

Complete event record 35 of 210

TIME-STAMP	2026-07-18T11:25:00.044Z
LABEL	Heyting arithmetic
VALUE	Peano arithmetic formulated with intuitionistic logic. It is a standard formal setting for studying constructive number theory, realizability, and witness extraction.
ID	gpil_0035_326087f9c682cdfb
SEQUENTIAL-STRING ID	10498753636867967270296557861353321652236818344706376997133008499346984470999866422302643452230146805982097910843069318269015675159260334410907491734580172956809162127579084303082543569996493830406662023818283132404479783773723002839731363564110563952095326203318792660944382973820947288422880555511700093656230170856798542594020
BACKWARD-COMPUTED VALUE	Peano arithmetic formulated with intuitionistic logic. It is a standard formal setting for studying constructive number theory, realizability, and witness extraction.
BACKWARD-COMPUTATION.ALGORITHM	shortlex-unrank/1.0
BACKWARD-COMPUTATION.MATCHES-VALUE	true
ALPHABET	!"#\$%&'()*+,-./0123456789:;<=>?@ABCDEFGHIJKLMNOPQRSTUVWXYZ[\]^_`abcdefghijklmnopqrstuvwxyz{ }~
METRICS.ALPHABET-SYMBOLS	95
METRICS.VALUE-SYMBOLS	166
METRICS.SEQUENTIAL-ID-DIGITS	329
METRICS.NATIVE-INTEGER	false
INSTRUCTION.LANGUAGE	GPIL/1.0
INSTRUCTION.SOURCE-LINE	35
INSTRUCTION.OPERATION	literal
INSTRUCTION.SOURCE	event "Heyting arithmetic" : literal("Peano arithmetic formulated with intuitionistic logic. It is a standard formal setting for studying constructive number theory, realizability, and witness extraction.")

Constructive set theory

Complete event record 36 of 210

TIME-STAMP	2026-07-18T11:25:00.044Z
LABEL	Constructive set theory
VALUE	A family of set-theoretic foundations using intuitionistic logic and constructively justified axioms, often emphasizing predicative generation and explicit existence.
ID	gpil_0036_f7b87461332e4fa1
SEQUENTIAL-STRING ID	7179755687071883201820472392934864223779431498930066068228400068980631477397745195595187063680477041744477540485063712100750014811178178020477678634733545083778565900943605835834485899155946425184361257302304596533814077633963104135655642393022122782304855490851156347903116926657196336628546161543893586747898833271426118990490
BACKWARD-COMPUTED VALUE	A family of set-theoretic foundations using intuitionistic logic and constructively justified axioms, often emphasizing predicative generation and explicit existence.
BACKWARD-COMPUTATION.ALGORITHM	shortlex-unrank/1.0
BACKWARD-COMPUTATION.MATCHES-VALUE	true
ALPHABET	!"#\$%&'()*+,-./0123456789:;<=>?@ABCDEFGHIJKLMNOPQRSTUVWXYZ[\]^_`abcdefghijklmnopqrstuvwxyz{ }~
METRICS.ALPHABET-SYMBOLS	95
METRICS.VALUE-SYMBOLS	166
METRICS.SEQUENTIAL-ID-DIGITS	328
METRICS.NATIVE-INTEGER	false
INSTRUCTION.LANGUAGE	GPIL/1.0
INSTRUCTION.SOURCE-LINE	36
INSTRUCTION.OPERATION	literal
INSTRUCTION.SOURCE	event "Constructive set theory" : literal("A family of set-theoretic foundations using intuitionistic logic and constructively justified axioms, often emphasizing predicative generation and explicit existence.")

Predicativity

Complete event record 37 of 210

TIME-STAMP	2026-07-18T11:25:00.044Z
LABEL	Predicativity
VALUE	The discipline of avoiding definitions that quantify over a totality containing the object being defined, motivating stratified universes and controlled inductive constructions.
ID	gpil_0037_e63120160c3ddb1a
SEQUENTIAL-STRING ID	64557168534203525536028308610296743701573665626145218102217563504737917187926377061056028913131074574770948540682022792202852792492104209041146897521698768440409262226689230242967270898971399404756197626158214942571984371840761949972352563094563503406957438521344662615183024368292813604535295293295045903031167258422315191152764259036561434576125970
BACKWARD-COMPUTED VALUE	The discipline of avoiding definitions that quantify over a totality containing the object being defined, motivating stratified universes and controlled inductive constructions.
BACKWARD-COMPUTATION.ALGORITHM	shortlex-unrank/1.0
BACKWARD-COMPUTATION.MATCHES-VALUE	true
ALPHABET	!"#\$%&'()*+,-./0123456789:;<=>?@ABCDEFGHIJKLMNOPQRSTUVWXYZ[\]^_`abcdefghijklmnopqrstuvwxyz{ }~
METRICS.ALPHABET-SYMBOLS	95
METRICS.VALUE-SYMBOLS	177
METRICS.SEQUENTIAL-ID-DIGITS	350
METRICS.NATIVE-INTEGER	false
INSTRUCTION.LANGUAGE	GPIL/1.0
INSTRUCTION.SOURCE-LINE	37
INSTRUCTION.OPERATION	literal
INSTRUCTION.SOURCE	event "Predicativity" : literal("The discipline of avoiding definitions that quantify over a totality containing the object being defined, motivating stratified universes and controlled inductive constructions.")

Choice sequence

Complete event record 38 of 210

TIME-STAMP	2026-07-18T11:25:00.044Z
LABEL	Choice sequence
VALUE	A potentially infinite sequence generated stage by stage, possibly without a fixed completed law; it is a central object in Brouwerian accounts of constructive analysis.
ID	gpil_0038_83e991b7dbbc6e21
SEQUENTIAL-STRING ID	6155946504319639098223720506691517305365666409217578446974995428654090035593726458604276199214223344052026480314220640459522322774715035487824246942059023349724965599629567380914696443944516932830062461787242136121110453156832311866116717673119100363620506625232093038038894352932469372593831102621366834381165605415225844183811464095
BACKWARD-COMPUTED VALUE	A potentially infinite sequence generated stage by stage, possibly without a fixed completed law; it is a central object in Brouwerian accounts of constructive analysis.
BACKWARD-COMPUTATION.ALGORITHM	shortlex-unrank/1.0
BACKWARD-COMPUTATION.MATCHES-VALUE	true
ALPHABET	!"#\$%&'()*+,-./0123456789:;<=>?@ABCDEFGHIJKLMNOPQRSTUVWXYZ[\]^_`abcdefghijklmnopqrstuvwxyz{ }~
METRICS.ALPHABET-SYMBOLS	95
METRICS.VALUE-SYMBOLS	169
METRICS.SEQUENTIAL-ID-DIGITS	334
METRICS.NATIVE-INTEGER	false
INSTRUCTION.LANGUAGE	GPIL/1.0
INSTRUCTION.SOURCE-LINE	38
INSTRUCTION.OPERATION	literal
INSTRUCTION.SOURCE	event "Choice sequence" : literal("A potentially infinite sequence generated stage by stage, possibly without a fixed completed law; it is a central object in Brouwerian accounts of constructive analysis.")

Bar induction

Complete event record 39 of 210

TIME-STAMP	2026-07-18T11:25:00.044Z
LABEL	Bar induction
VALUE	An induction principle on finite sequences that meet a bar, used to justify conclusions about all continuing choice sequences from well-founded local information.
ID	gpil_0039_b2fc433fb485c95f
SEQUENTIAL-STRING ID	90256000183448202954487552012134107785168678013854334808593384946562912878409576759086873600752071244084274486993215851080394336489914245963898234102366535978755876570462542553661165350106090808783742651910907279519175203900800339184675169162075902294478164917990351665099728742364403895217298774113158340485823470444020
BACKWARD-COMPUTED VALUE	An induction principle on finite sequences that meet a bar, used to justify conclusions about all continuing choice sequences from well-founded local information.
BACKWARD-COMPUTATION.ALGORITHM	shortlex-unrank/1.0
BACKWARD-COMPUTATION.MATCHES-VALUE	true
ALPHABET	!"#\$%&'()*+,-./0123456789:;<=>?@ABCDEFGHIJKLMNOPQRSTUVWXYZ[\]^_`abcdefghijklmnopqrstuvwxyz{ }~
METRICS.ALPHABET-SYMBOLS	95
METRICS.VALUE-SYMBOLS	162
METRICS.SEQUENTIAL-ID-DIGITS	320
METRICS.NATIVE-INTEGER	false
INSTRUCTION.LANGUAGE	GPIL/1.0
INSTRUCTION.SOURCE-LINE	39
INSTRUCTION.OPERATION	literal
INSTRUCTION.SOURCE	event "Bar induction" : literal("An induction principle on finite sequences that meet a bar, used to justify conclusions about all continuing choice sequences from well-founded local information.")

Fan theorem

Complete event record 40 of 210

TIME-STAMP	2026-07-18T11:25:00.044Z
LABEL	Fan theorem
VALUE	A Brouwerian compactness principle asserting a uniform bound for suitable bars on finitely branching trees; it plays an important role in intuitionistic analysis.
ID	gpil_0040_2e3c7cc8b7e24811
SEQUENTIAL-STRING ID	88138280061390429207542653031109569388371317941781546002295821932354523652795314714039137436696929775670631663733705152347476748850314299994360977353056937054711319240733670544689981189528517759492427081203163887769500059859727013221141505573963170257221056766991785431764742993894219520335158151826120650712624368104720
BACKWARD-COMPUTED VALUE	A Brouwerian compactness principle asserting a uniform bound for suitable bars on finitely branching trees; it plays an important role in intuitionistic analysis.
BACKWARD-COMPUTATION.ALGORITHM	shortlex-unrank/1.0
BACKWARD-COMPUTATION.MATCHES-VALUE	true
ALPHABET	!"#\$%&'()*+,-./0123456789:;<=>?@ABCDEFGHIJKLMNOPQRSTUVWXYZ[\]^_`abcdefghijklmnopqrstuvwxyz{ }~
METRICS.ALPHABET-SYMBOLS	95
METRICS.VALUE-SYMBOLS	162
METRICS.SEQUENTIAL-ID-DIGITS	320
METRICS.NATIVE-INTEGER	false
INSTRUCTION.LANGUAGE	GPIL/1.0
INSTRUCTION.SOURCE-LINE	40
INSTRUCTION.OPERATION	literal
INSTRUCTION.SOURCE	event "Fan theorem" : literal("A Brouwerian compactness principle asserting a uniform bound for suitable bars on finitely branching trees; it plays an important role in intuitionistic analysis.")

Markov's principle

Complete event record 41 of 210

TIME-STAMP	2026-07-18T11:25:00.044Z
LABEL	Markov's principle
VALUE	For a decidable predicate on natural numbers, this principle turns the impossibility of universal failure into an existential witness. It is accepted in some constructive schools but not all.
ID	gpil_0041_a82f9fcf6c070cbf
SEQUENTIAL-STRING ID	233306877227753752916020005444951033925559373998588410017947324751537802882615377624264247078563710451312070599771535521170699794548815759391037802033697144468534363603373241749395086893713189960270226867344434322824295465190559457051836456489026840009115638755057176788538403195807587901548517452194558806003064671093039992938384178677376983588564768072966006141107952554255255
BACKWARD-COMPUTED VALUE	For a decidable predicate on natural numbers, this principle turns the impossibility of universal failure into an existential witness. It is accepted in some constructive schools but not all.
BACKWARD-COMPUTATION.ALGORITHM	shortlex-unrank/1.0
BACKWARD-COMPUTATION.MATCHES-VALUE	true
ALPHABET	!"#\$%&'()*+,-./0123456789:;<=>?@ABCDEFGHIJKLMNOPQRSTUVWXYZ[\]^_`abcdefghijklmnopqrstuvwxyz{ }~
METRICS.ALPHABET-SYMBOLS	95
METRICS.VALUE-SYMBOLS	191
METRICS.SEQUENTIAL-ID-DIGITS	378
METRICS.NATIVE-INTEGER	false
INSTRUCTION.LANGUAGE	GPIL/1.0
INSTRUCTION.SOURCE-LINE	41
INSTRUCTION.OPERATION	literal
INSTRUCTION.SOURCE	event "Markov's principle" : literal("For a decidable predicate on natural numbers, this principle turns the impossibility of universal failure into an existential witness. It is accepted in some constructive schools but not all.")

Church's thesis

Complete event record 42 of 210

TIME-STAMP	2026-07-18T11:25:00.044Z
LABEL	Church's thesis
VALUE	The thesis that every total numerical function is computable. Formal variants are compatible with some constructive systems but are not principles of constructivism in general.
ID	gpil_0042_aa045656d1297f62
SEQUENTIAL-STRING ID	679549144945742083785071263842978751290523550671323176877336640460428089068192629083773170568873312596754965603558211171925176977081124020155413062993675375182765504796064718746219159408234611886253382376263330770344002966236344201960915887327567175462732332001893942228577624849720157202125119432212281054370816113353337053509650019071281212358855
BACKWARD-COMPUTED VALUE	The thesis that every total numerical function is computable. Formal variants are compatible with some constructive systems but are not principles of constructivism in general.
BACKWARD-COMPUTATION.ALGORITHM	shortlex-unrank/1.0
BACKWARD-COMPUTATION.MATCHES-VALUE	true
ALPHABET	!"#\$%&'()*+,-./0123456789:;<=>?@ABCDEFGHIJKLMNOPQRSTUVWXYZ[\]^_`abcdefghijklmnopqrstuvwxyz{ }~
METRICS.ALPHABET-SYMBOLS	95
METRICS.VALUE-SYMBOLS	176
METRICS.SEQUENTIAL-ID-DIGITS	348
METRICS.NATIVE-INTEGER	false
INSTRUCTION.LANGUAGE	GPIL/1.0
INSTRUCTION.SOURCE-LINE	42
INSTRUCTION.OPERATION	literal
INSTRUCTION.SOURCE	event "Church's thesis" : literal("The thesis that every total numerical function is computable. Formal variants are compatible with some constructive systems but are not principles of constructivism in general.")

Axiom of choice

Complete event record 43 of 210

TIME-STAMP	2026-07-18T11:25:00.044Z
LABEL	Axiom of choice
VALUE	A principle selecting an element from every inhabited member of a family. Under a type-theoretic reading it can be constructive because evidence of inhabitation already contains witnesses.
ID	gpil_0043_db78aaaf1e7865c4
SEQUENTIAL-STRING ID	232296882020297616762920701735402657655043900713606608268701547711068684576407433157248564840694601353401494961179150198635364896234858180226178662937437226325582265112265773719460962869153558011456384683569527233913192566250205808341946221075837223118620235508221844786551323911188477810312578017663971880166324287653518239248170767560261982882732751323390907620334980495
BACKWARD-COMPUTED VALUE	A principle selecting an element from every inhabited member of a family. Under a type-theoretic reading it can be constructive because evidence of inhabitation already contains witnesses.
BACKWARD-COMPUTATION.ALGORITHM	shortlex-unrank/1.0
BACKWARD-COMPUTATION.MATCHES-VALUE	true
ALPHABET	!"#\$%&'()*+,-./0123456789:;<=>?@ABCDEFGHIJKLMNOPQRSTUVWXYZ[\]^_`abcdefghijklmnopqrstuvwxyz{ }~
METRICS.ALPHABET-SYMBOLS	95
METRICS.VALUE-SYMBOLS	188
METRICS.SEQUENTIAL-ID-DIGITS	372
METRICS.NATIVE-INTEGER	false
INSTRUCTION.LANGUAGE	GPIL/1.0
INSTRUCTION.SOURCE-LINE	43
INSTRUCTION.OPERATION	literal
INSTRUCTION.SOURCE	event "Axiom of choice" : literal("A principle selecting an element from every inhabited member of a family. Under a type-theoretic reading it can be constructive because evidence of inhabitation already contains witnesses.")

Countable choice

Complete event record 44 of 210

TIME-STAMP	2026-07-18T11:25:00.044Z
LABEL	Countable choice
VALUE	The restricted choice principle for families indexed by the natural numbers. Its constructive status depends on the underlying theory and the meaning assigned to existence.
ID	gpil_0044_873cb26b232528fb
SEQUENTIAL-STRING ID	8343080788278721763338279296292981968095633957158613023103930921420878318063439875840427375561452313033576569572609648820317697116085663939129682415118609953619953053404038347013680910631727853928987488871968376334753975133755525355248912268986661970147937260114978933302862720173854190657531869436185271585190632660788939582061993013521740
BACKWARD-COMPUTED VALUE	The restricted choice principle for families indexed by the natural numbers. Its constructive status depends on the underlying theory and the meaning assigned to existence.
BACKWARD-COMPUTATION.ALGORITHM	shortlex-unrank/1.0
BACKWARD-COMPUTATION.MATCHES-VALUE	true
ALPHABET	!"#\$%&'()*+,-./0123456789:;<=>?@ABCDEFGHIJKLMNOPQRSTUVWXYZ[\]^_`abcdefghijklmnopqrstuvwxyz{ }~
METRICS.ALPHABET-SYMBOLS	95
METRICS.VALUE-SYMBOLS	172
METRICS.SEQUENTIAL-ID-DIGITS	340
METRICS.NATIVE-INTEGER	false
INSTRUCTION.LANGUAGE	GPIL/1.0
INSTRUCTION.SOURCE-LINE	44
INSTRUCTION.OPERATION	literal
INSTRUCTION.SOURCE	event "Countable choice" : literal("The restricted choice principle for families indexed by the natural numbers. Its constructive status depends on the underlying theory and the meaning assigned to existence.")

Church's rule

Complete event record 45 of 210

TIME-STAMP	2026-07-18T11:25:00.044Z
LABEL	Church's rule
VALUE	A metatheoretic extraction rule connecting derivable existential numerical claims with recursive witnesses, expressing the effective content of intuitionistic arithmetic.
ID	gpil_0045_07478dd006589428
SEQUENTIAL-STRING ID	584809003095763989811417400238144067633853602915207060335941203490447305943931480487268726776952273681850696172340564386061139313482895717922323753465744971515953863867458987152185021900350789514244740397003195526596640808196064828904235319904292522001468087756070873824918673067851025109244094752608253832420122367893145551986376773075
BACKWARD-COMPUTED VALUE	A metatheoretic extraction rule connecting derivable existential numerical claims with recursive witnesses, expressing the effective content of intuitionistic arithmetic.
BACKWARD-COMPUTATION.ALGORITHM	shortlex-unrank/1.0
BACKWARD-COMPUTATION.MATCHES-VALUE	true
ALPHABET	!"#\$%&'()*+,-./0123456789:;<=>?@ABCDEFGHIJKLMNOPQRSTUVWXYZ[\]^_`abcdefghijklmnopqrstuvwxyz{ }~
METRICS.ALPHABET-SYMBOLS	95
METRICS.VALUE-SYMBOLS	170
METRICS.SEQUENTIAL-ID-DIGITS	336
METRICS.NATIVE-INTEGER	false
INSTRUCTION.LANGUAGE	GPIL/1.0
INSTRUCTION.SOURCE-LINE	45
INSTRUCTION.OPERATION	literal
INSTRUCTION.SOURCE	event "Church's rule" : literal("A metatheoretic extraction rule connecting derivable existential numerical claims with recursive witnesses, expressing the effective content of intuitionistic arithmetic.")

Continuity principle

Complete event record 46 of 210

TIME-STAMP	2026-07-18T11:25:00.044Z
LABEL	Continuity principle
VALUE	The constructive principle that a total operation on an infinite choice sequence depends on only a finite initial segment, a characteristic idea in intuitionistic analysis.
ID	gpil_0046_a9c9b700e297c34c
SEQUENTIAL-STRING ID	8343080759906646414647193198050062031369092311988002388831321193950478351268389854194486136358561156061324067500260965004423152183515439161670019416958794797068290833384848885107406834651242942414230487213160501067048106377576986627046089924292484769434681893949740791061172589914044776558837446930940969999558715614426031446401253518495345
BACKWARD-COMPUTED VALUE	The constructive principle that a total operation on an infinite choice sequence depends on only a finite initial segment, a characteristic idea in intuitionistic analysis.
BACKWARD-COMPUTATION.ALGORITHM	shortlex-unrank/1.0
BACKWARD-COMPUTATION.MATCHES-VALUE	true
ALPHABET	!"#\$%&'()*+,-./0123456789:;<=>?@ABCDEFGHIJKLMNOPQRSTUVWXYZ[\]^_`abcdefghijklmnopqrstuvwxyz{ }~
METRICS.ALPHABET-SYMBOLS	95
METRICS.VALUE-SYMBOLS	172
METRICS.SEQUENTIAL-ID-DIGITS	340
METRICS.NATIVE-INTEGER	false
INSTRUCTION.LANGUAGE	GPIL/1.0
INSTRUCTION.SOURCE-LINE	46
INSTRUCTION.OPERATION	literal
INSTRUCTION.SOURCE	event "Continuity principle" : literal("The constructive principle that a total operation on an infinite choice sequence depends on only a finite initial segment, a characteristic idea in intuitionistic analysis.")

Kripke semantics

Complete event record 47 of 210

TIME-STAMP	2026-07-18T11:25:00.044Z
LABEL	Kripke semantics
VALUE	A semantics using ordered stages of information. A formula may become verified at a later stage, while established information persists along the ordering.
ID	gpil_0047_cff56c88fa38ffeb
SEQUENTIAL-STRING ID	1262317074347605024613210892258289737608573328767133985631524445598131145958280686213610001074805067323225798806279649079990747000412110737093144100736762759232564438631346156611477242242477771258020610387955850835881543459664260624456041983606366238027861897610645203700538925449734750839274215333548942455
BACKWARD-COMPUTED VALUE	A semantics using ordered stages of information. A formula may become verified at a later stage, while established information persists along the ordering.
BACKWARD-COMPUTATION.ALGORITHM	shortlex-unrank/1.0
BACKWARD-COMPUTATION.MATCHES-VALUE	true
ALPHABET	!"#\$%&'()*+,-./0123456789:;<=>?@ABCDEFGHIJKLMNOPQRSTUVWXYZ[\]^_`abcdefghijklmnopqrstuvwxyz{ }~
METRICS.ALPHABET-SYMBOLS	95
METRICS.VALUE-SYMBOLS	155
METRICS.SEQUENTIAL-ID-DIGITS	307
METRICS.NATIVE-INTEGER	false
INSTRUCTION.LANGUAGE	GPIL/1.0
INSTRUCTION.SOURCE-LINE	47
INSTRUCTION.OPERATION	literal
INSTRUCTION.SOURCE	event "Kripke semantics" : literal("A semantics using ordered stages of information. A formula may become verified at a later stage, while established information persists along the ordering.")

Kripke frame

Complete event record 48 of 210

TIME-STAMP	2026-07-18T11:25:00.044Z
LABEL	Kripke frame
VALUE	The partially ordered collection of information states underlying a Kripke model, where the accessibility relation represents possible growth of constructive knowledge.
ID	gpil_0048_4b33e2b33d0708ea
SEQUENTIAL-STRING ID	102431145057374007457121738169548244362802752229850424734394632324728178649990731441272321395497112055520626372769676146991318297652806210758244019648585267789895420771507067133063091620062091369283535830692567925897523517342768468069803579908672303834243214978203690576587347976457767648298566250741558866539185319665320129642649715
BACKWARD-COMPUTED VALUE	The partially ordered collection of information states underlying a Kripke model, where the accessibility relation represents possible growth of constructive knowledge.
BACKWARD-COMPUTATION.ALGORITHM	shortlex-unrank/1.0
BACKWARD-COMPUTATION.MATCHES-VALUE	true
ALPHABET	!"#\$%&'()*+,-./0123456789:;<=>?@ABCDEFGHIJKLMNOPQRSTUVWXYZ[\]^_`abcdefghijklmnopqrstuvwxyz{ }~
METRICS.ALPHABET-SYMBOLS	95
METRICS.VALUE-SYMBOLS	168
METRICS.SEQUENTIAL-ID-DIGITS	333
METRICS.NATIVE-INTEGER	false
INSTRUCTION.LANGUAGE	GPIL/1.0
INSTRUCTION.SOURCE-LINE	48
INSTRUCTION.OPERATION	literal
INSTRUCTION.SOURCE	event "Kripke frame" : literal("The partially ordered collection of information states underlying a Kripke model, where the accessibility relation represents possible growth of constructive knowledge.")

Monotonic forcing

Complete event record 49 of 210

TIME-STAMP	2026-07-18T11:25:00.044Z
LABEL	Monotonic forcing
VALUE	The persistence condition that if a world forces a formula, every accessible world with more information also forces it. This models the durability of constructed evidence.
ID	gpil_0049_e5ce22effc81d9c0
SEQUENTIAL-STRING ID	8343080784468972306283975796841491841237936286418936243369273582298554480011064573484688336418939891282218672660360298977612688103173596484178139806302738261599041982093136629118892996894506579942367704445898272381738869889900675080344955955678638239467572493197170544330916816621216570315715964294479414442725770246105120274321116605056115
BACKWARD-COMPUTED VALUE	The persistence condition that if a world forces a formula, every accessible world with more information also forces it. This models the durability of constructed evidence.
BACKWARD-COMPUTATION.ALGORITHM	shortlex-unrank/1.0
BACKWARD-COMPUTATION.MATCHES-VALUE	true
ALPHABET	!"#\$%&'()*+,-./0123456789:;<=>?@ABCDEFGHIJKLMNOPQRSTUVWXYZ[\]^_`abcdefghijklmnopqrstuvwxyz{ }~
METRICS.ALPHABET-SYMBOLS	95
METRICS.VALUE-SYMBOLS	172
METRICS.SEQUENTIAL-ID-DIGITS	340
METRICS.NATIVE-INTEGER	false
INSTRUCTION.LANGUAGE	GPIL/1.0
INSTRUCTION.SOURCE-LINE	49
INSTRUCTION.OPERATION	literal
INSTRUCTION.SOURCE	event "Monotonic forcing" : literal("The persistence condition that if a world forces a formula, every accessible world with more information also forces it. This models the durability of constructed evidence.")

Heyting algebra

Complete event record 50 of 210

TIME-STAMP	2026-07-18T11:25:00.044Z
LABEL	Heyting algebra
VALUE	A bounded distributive lattice whose implication is right adjoint to conjunction. Such algebras provide the standard algebraic semantics for intuitionistic propositional logic.
ID	gpil_0050_2f258ecf9c102099
SEQUENTIAL-STRING ID	429873101329097984695985663785957359425016368345008475309166290679346550737085117680830276206506920286180450817224165987120765953826494750351050843480556329872716925286170871534879021120953706781910254269890354108135031274467185193628930700825226536948876130439577721639385177808504463956023948369953055878603645022670624615852205424306445515495950
BACKWARD-COMPUTED VALUE	A bounded distributive lattice whose implication is right adjoint to conjunction. Such algebras provide the standard algebraic semantics for intuitionistic propositional logic.
BACKWARD-COMPUTATION.ALGORITHM	shortlex-unrank/1.0
BACKWARD-COMPUTATION.MATCHES-VALUE	true
ALPHABET	!"#\$%&'()*+,-./0123456789:;<=>?@ABCDEFGHIJKLMNOPQRSTUVWXYZ[\]^_`abcdefghijklmnopqrstuvwxyz{ }~
METRICS.ALPHABET-SYMBOLS	95
METRICS.VALUE-SYMBOLS	176
METRICS.SEQUENTIAL-ID-DIGITS	348
METRICS.NATIVE-INTEGER	false
INSTRUCTION.LANGUAGE	GPIL/1.0
INSTRUCTION.SOURCE-LINE	50
INSTRUCTION.OPERATION	literal
INSTRUCTION.SOURCE	event "Heyting algebra" : literal("A bounded distributive lattice whose implication is right adjoint to conjunction. Such algebras provide the standard algebraic semantics for intuitionistic propositional logic.")

Topological semantics

Complete event record 51 of 210

TIME-STAMP	2026-07-18T11:25:00.044Z
LABEL	Topological semantics
VALUE	An interpretation of propositions as open sets, with conjunction as intersection, disjunction as union, and implication determined by the interior operation.
ID	gpil_0051_e56cbbde1c249b1f
SEQUENTIAL-STRING ID	11664283237625035745480131005158558259032513743013791232409818315393197008408962617321147187110891215633507022397883959267696602203164519245085445344071861203665388847026916194264372240691292297888636270720059923206398703181716125608098722135136372445261939302194243308945419240368283943260425661248965140678395
BACKWARD-COMPUTED VALUE	An interpretation of propositions as open sets, with conjunction as intersection, disjunction as union, and implication determined by the interior operation.
BACKWARD-COMPUTATION.ALGORITHM	shortlex-unrank/1.0
BACKWARD-COMPUTATION.MATCHES-VALUE	true
ALPHABET	!"#\$%&'()*+,-./0123456789:;<=>?@ABCDEFGHIJKLMNOPQRSTUVWXYZ[\]^_`abcdefghijklmnopqrstuvwxyz{ }~
METRICS.ALPHABET-SYMBOLS	95
METRICS.VALUE-SYMBOLS	157
METRICS.SEQUENTIAL-ID-DIGITS	311
METRICS.NATIVE-INTEGER	false
INSTRUCTION.LANGUAGE	GPIL/1.0
INSTRUCTION.SOURCE-LINE	51
INSTRUCTION.OPERATION	literal
INSTRUCTION.SOURCE	event "Topological semantics" : literal("An interpretation of propositions as open sets, with conjunction as intersection, disjunction as union, and implication determined by the interior operation.")

Beth semantics

Complete event record 52 of 210

TIME-STAMP	2026-07-18T11:25:00.044Z
LABEL	Beth semantics
VALUE	A branching-tree semantics in which bars capture when information along every future branch suffices, giving a constructive interpretation of logical connectives and quantifiers.
ID	gpil_0052_c53fdb6a390d3130
SEQUENTIAL-STRING ID	3879605110542526756416452973595094282520696583466857460092954918639273070038968111411001786224563443166712187157811486404123877228799810829737164391074100078717816718067558410121026842447760284496235412300409143924957782090347577135922611935144837075664389001925139272144010203009794269244533763683395000715532594766629465527553773711080412154429522695
BACKWARD-COMPUTED VALUE	A branching-tree semantics in which bars capture when information along every future branch suffices, giving a constructive interpretation of logical connectives and quantifiers.
BACKWARD-COMPUTATION.ALGORITHM	shortlex-unrank/1.0
BACKWARD-COMPUTATION.MATCHES-VALUE	true
ALPHABET	!"#\$%&'()*+,-./0123456789:;<=>?@ABCDEFGHIJKLMNOPQRSTUVWXYZ[\]^_`abcdefghijklmnopqrstuvwxyz{ }~
METRICS.ALPHABET-SYMBOLS	95
METRICS.VALUE-SYMBOLS	178
METRICS.SEQUENTIAL-ID-DIGITS	352
METRICS.NATIVE-INTEGER	false
INSTRUCTION.LANGUAGE	GPIL/1.0
INSTRUCTION.SOURCE-LINE	52
INSTRUCTION.OPERATION	literal
INSTRUCTION.SOURCE	event "Beth semantics" : literal("A branching-tree semantics in which bars capture when information along every future branch suffices, giving a constructive interpretation of logical connectives and quantifiers.")

Categorical semantics

Complete event record 53 of 210

TIME-STAMP	2026-07-18T11:25:00.044Z
LABEL	Categorical semantics
VALUE	An interpretation of types, terms, contexts, and substitution using objects and morphisms, making structural relationships in constructive calculi mathematically explicit.
ID	gpil_0053_c14f2995aaa708ab
SEQUENTIAL-STRING ID	56883790843064143149326742217964170037032151256704128762643914924296040729702096537169421972129405086314387630178262310882800418348330621565198887830777888919424549642774346519674217377692069195221553019217388422642988252006763461847248476716434621836819800568043376149372616120342536659962944037463217659623806987037711931441909026020565
BACKWARD-COMPUTED VALUE	An interpretation of types, terms, contexts, and substitution using objects and morphisms, making structural relationships in constructive calculi mathematically explicit.
BACKWARD-COMPUTATION.ALGORITHM	shortlex-unrank/1.0
BACKWARD-COMPUTATION.MATCHES-VALUE	true
ALPHABET	!"#\$%&'()*+,-./0123456789:;<=>?@ABCDEFGHIJKLMNOPQRSTUVWXYZ[\]^_`abcdefghijklmnopqrstuvwxyz{ }~
METRICS.ALPHABET-SYMBOLS	95
METRICS.VALUE-SYMBOLS	171
METRICS.SEQUENTIAL-ID-DIGITS	338
METRICS.NATIVE-INTEGER	false
INSTRUCTION.LANGUAGE	GPIL/1.0
INSTRUCTION.SOURCE-LINE	53
INSTRUCTION.OPERATION	literal
INSTRUCTION.SOURCE	event "Categorical semantics" : literal("An interpretation of types, terms, contexts, and substitution using objects and morphisms, making structural relationships in constructive calculi mathematically explicit.")

Cartesian closed category

Complete event record 54 of 210

TIME-STAMP	2026-07-18T11:25:00.044Z
LABEL	Cartesian closed category
VALUE	A category with finite products and function-space objects. It models typed functions and the conjunction-implication fragment of intuitionistic logic.
ID	gpil_0054_ab4e28af29f88f6f
SEQUENTIAL-STRING ID	15497132432146584750987104805939171450261709472650496608352067424384016774725652951169493530264380396426117425904806437597569380508730523166398936304500463298438118791960341352332947930340317004614050399492854949719075096564162970912686162023662726212759369614527963378576771275489912334229174792825
BACKWARD-COMPUTED VALUE	A category with finite products and function-space objects. It models typed functions and the conjunction-implication fragment of intuitionistic logic.
BACKWARD-COMPUTATION.ALGORITHM	shortlex-unrank/1.0
BACKWARD-COMPUTATION.MATCHES-VALUE	true
ALPHABET	!"#\$%&'()*+,-./0123456789:;<=>?@ABCDEFGHIJKLMNOPQRSTUVWXYZ[\]^_`abcdefghijklmnopqrstuvwxyz{ }~
METRICS.ALPHABET-SYMBOLS	95
METRICS.VALUE-SYMBOLS	151
METRICS.SEQUENTIAL-ID-DIGITS	299
METRICS.NATIVE-INTEGER	false
INSTRUCTION.LANGUAGE	GPIL/1.0
INSTRUCTION.SOURCE-LINE	54
INSTRUCTION.OPERATION	literal
INSTRUCTION.SOURCE	event "Cartesian closed category" : literal("A category with finite products and function-space objects. It models typed functions and the conjunction-implication fragment of intuitionistic logic.")

Locally cartesian closed category

Complete event record 55 of 210

TIME-STAMP	2026-07-18T11:25:00.044Z
LABEL	Locally cartesian closed category
VALUE	A category whose slice over every object is cartesian closed, providing categorical structure for interpreting dependent types and substitution.
ID	gpil_0055_b4460576c23481c3
SEQUENTIAL-STRING ID	221914718271987212711996178425922890842911110365638317406565731596155914669273614823018722239459356387770148487329529529545541797451283319868693220024367387374605327136697583921791671156520944679903947930334082567032819356571663014954819244601415425361402663071327796437804715631506520
BACKWARD-COMPUTED VALUE	A category whose slice over every object is cartesian closed, providing categorical structure for interpreting dependent types and substitution.
BACKWARD-COMPUTATION.ALGORITHM	shortlex-unrank/1.0
BACKWARD-COMPUTATION.MATCHES-VALUE	true
ALPHABET	!"#\$%&'()*+,-./0123456789:;<=>?@ABCDEFGHIJKLMNOPQRSTUVWXYZ[\]^_`abcdefghijklmnopqrstuvwxyz{ }~
METRICS.ALPHABET-SYMBOLS	95
METRICS.VALUE-SYMBOLS	144
METRICS.SEQUENTIAL-ID-DIGITS	285
METRICS.NATIVE-INTEGER	false
INSTRUCTION.LANGUAGE	GPIL/1.0
INSTRUCTION.SOURCE-LINE	55
INSTRUCTION.OPERATION	literal
INSTRUCTION.SOURCE	event "Locally cartesian closed category" : literal("A category whose slice over every object is cartesian closed, providing categorical structure for interpreting dependent types and substitution.")

Topos

Complete event record 56 of 210

TIME-STAMP	2026-07-18T11:25:00.044Z
LABEL	Topos
VALUE	A category with finite limits, exponentials, and a subobject classifier. Its internal language generally obeys intuitionistic higher-order logic.
ID	gpil_0056_940422ae79a6bde2
SEQUENTIAL-STRING ID	21081898235838785207639638156209827866051640232027699387558229679341607649118637391988894785289971759275790269343446623459209900385859472639145530970280822060215831562939172308834185260703750174163391794224738175753836958483472129990836024419272678034232530910546566828573435232820495950
BACKWARD-COMPUTED VALUE	A category with finite limits, exponentials, and a subobject classifier. Its internal language generally obeys intuitionistic higher-order logic.
BACKWARD-COMPUTATION.ALGORITHM	shortlex-unrank/1.0
BACKWARD-COMPUTATION.MATCHES-VALUE	true
ALPHABET	!"#\$%&'()*+,-./0123456789:;<=>?@ABCDEFGHIJKLMNOPQRSTUVWXYZ[\]^_`abcdefghijklmnopqrstuvwxyz{ }~
METRICS.ALPHABET-SYMBOLS	95
METRICS.VALUE-SYMBOLS	145
METRICS.SEQUENTIAL-ID-DIGITS	287
METRICS.NATIVE-INTEGER	false
INSTRUCTION.LANGUAGE	GPIL/1.0
INSTRUCTION.SOURCE-LINE	56
INSTRUCTION.OPERATION	literal
INSTRUCTION.SOURCE	event "Topos" : literal("A category with finite limits, exponentials, and a subobject classifier. Its internal language generally obeys intuitionistic higher-order logic.")

Partial combinatory algebra

Complete event record 57 of 210

TIME-STAMP	2026-07-18T11:25:00.044Z
LABEL	Partial combinatory algebra
VALUE	A set equipped with partial application and enough combinators to represent computation, serving as a foundation for many realizability models.
ID	gpil_0057_40487c13092cf48f
SEQUENTIAL-STRING ID	2336066460209105957239355502549456751384359376295748222693310426864398321260553806567046214264080498072645022690336564590435350663694766624492868447832838459702690268429317509320920315214840956455230894585011213508565059860664182743437678662549538512977662816847737115480349358499795
BACKWARD-COMPUTED VALUE	A set equipped with partial application and enough combinators to represent computation, serving as a foundation for many realizability models.
BACKWARD-COMPUTATION.ALGORITHM	shortlex-unrank/1.0
BACKWARD-COMPUTATION.MATCHES-VALUE	true
ALPHABET	!"#\$%&'()*+,-./0123456789:;<=>?@ABCDEFGHIJKLMNPOQRSTUVWXYZ[\]^_`abcdefghijklmnopqrstuvwxyz{ }~
METRICS.ALPHABET-SYMBOLS	95
METRICS.VALUE-SYMBOLS	143
METRICS.SEQUENTIAL-ID-DIGITS	283
METRICS.NATIVE-INTEGER	false
INSTRUCTION.LANGUAGE	GPIL/1.0
INSTRUCTION.SOURCE-LINE	57
INSTRUCTION.OPERATION	literal
INSTRUCTION.SOURCE	event "Partial combinatory algebra" : literal("A set equipped with partial application and enough combinators to represent computation, serving as a foundation for many realizability models.")

Realizer

Complete event record 58 of 210

TIME-STAMP	2026-07-18T11:25:00.044Z
LABEL	Realizer
VALUE	A computational object that satisfies the evidence clause assigned to a formula by a realizability interpretation, such as a code, function, pair, or tagged value.
ID	gpil_0058_5429492c6b4be3ae
SEQUENTIAL-STRING ID	8374035838338711553213924846717605762881027196424113156305901909298428425064105197701173582760080145580445309665501663764053825444042827395236589262786502500822616436323117007475412938158604335721062391380782708076041377704173196291765543737666509444790297319324928635348761871372457063085650907211579872160653252584473190
BACKWARD-COMPUTED VALUE	A computational object that satisfies the evidence clause assigned to a formula by a realizability interpretation, such as a code, function, pair, or tagged value.
BACKWARD-COMPUTATION.ALGORITHM	shortlex-unrank/1.0
BACKWARD-COMPUTATION.MATCHES-VALUE	true
ALPHABET	!"#\$%&'()*+,-./0123456789:;<=>?@ABCDEFGHIJKLMNOPQRSTUVWXYZ[\]^_`abcdefghijklmnopqrstuvwxyz{ }~
METRICS.ALPHABET-SYMBOLS	95
METRICS.VALUE-SYMBOLS	163
METRICS.SEQUENTIAL-ID-DIGITS	322
METRICS.NATIVE-INTEGER	false
INSTRUCTION.LANGUAGE	GPIL/1.0
INSTRUCTION.SOURCE-LINE	58
INSTRUCTION.OPERATION	literal
INSTRUCTION.SOURCE	event "Realizer" : literal("A computational object that satisfies the evidence clause assigned to a formula by a realizability interpretation, such as a code, function, pair, or tagged value.")

Modified realizability

Complete event record 59 of 210

TIME-STAMP	2026-07-18T11:25:00.044Z
LABEL	Modified realizability
VALUE	A realizability interpretation developed to extract finite-type computational data while controlling which logical components contribute operational content.
ID	gpil_0059_d0bd19e0b60d4de3
SEQUENTIAL-STRING ID	11392374441886943598316449706777253303955404783847581796958268250276856096799249844933945142531663602156646983237202789640643179745962741849483387195683350635018778033555293788459603005583321591389909738831736450214893766143303653816688093004784282548368088740347939053862535822032333081709106809495087636449815
BACKWARD-COMPUTED VALUE	A realizability interpretation developed to extract finite-type computational data while controlling which logical components contribute operational content.
BACKWARD-COMPUTATION.ALGORITHM	shortlex-unrank/1.0
BACKWARD-COMPUTATION.MATCHES-VALUE	true
ALPHABET	!"#\$%&'()*+,-./0123456789:;<=>?@ABCDEFGHIJKLMNOPQRSTUVWXYZ[\]^_`abcdefghijklmnopqrstuvwxyz{ }~
METRICS.ALPHABET-SYMBOLS	95
METRICS.VALUE-SYMBOLS	157
METRICS.SEQUENTIAL-ID-DIGITS	311
METRICS.NATIVE-INTEGER	false
INSTRUCTION.LANGUAGE	GPIL/1.0
INSTRUCTION.SOURCE-LINE	59
INSTRUCTION.OPERATION	literal
INSTRUCTION.SOURCE	event "Modified realizability" : literal("A realizability interpretation developed to extract finite-type computational data while controlling which logical components contribute operational content.")

Dialectica interpretation

Complete event record 60 of 210

TIME-STAMP	2026-07-18T11:25:00.044Z
LABEL	Dialectica interpretation
VALUE	A functional interpretation that translates formulas into existential-universal statements involving witnessing functionals and counterexample functionals.
ID	gpil_0060_ba64c8b05afdc770
SEQUENTIAL-STRING ID	1262264319269227601364133993168670157672069760224301241079925779643717318071130251044543038722474062036879924052299691354209982778650838968986427557121992849057169925595817556682627824797468118361505185817009585548306946743147814743729580123482715366090062904298482136316365002476015496589185694792270060920
BACKWARD-COMPUTED VALUE	A functional interpretation that translates formulas into existential-universal statements involving witnessing functionals and counterexample functionals.
BACKWARD-COMPUTATION.ALGORITHM	shortlex-unrank/1.0
BACKWARD-COMPUTATION.MATCHES-VALUE	true
ALPHABET	!"#\$%&'()*+,-./0123456789:;<=>?@ABCDEFGHIJKLMNOPQRSTUVWXYZ[\]^_`abcdefghijklmnopqrstuvwxyz{ }~
METRICS.ALPHABET-SYMBOLS	95
METRICS.VALUE-SYMBOLS	155
METRICS.SEQUENTIAL-ID-DIGITS	307
METRICS.NATIVE-INTEGER	false
INSTRUCTION.LANGUAGE	GPIL/1.0
INSTRUCTION.SOURCE-LINE	60
INSTRUCTION.OPERATION	literal
INSTRUCTION.SOURCE	event "Dialectica interpretation" : literal("A functional interpretation that translates formulas into existential-universal statements involving witnessing functionals and counterexample functionals.")

Negative translation

Complete event record 61 of 210

TIME-STAMP	2026-07-18T11:25:00.044Z
LABEL	Negative translation
VALUE	A systematic embedding of classical reasoning into intuitionistic or minimal logic by reformulating formulas with negation while preserving classical derivability.
ID	gpil_0061_681ba6f50939ea4b
SEQUENTIAL-STRING ID	8374475139872311070267048889984880754802079907995190674435131547902937244617581893921534728149692216583664786258559168615160654251848595170436341609056237912158869948365214806313128085159500526974053902058029693469660277653662209151962520308108632288073805427358670436386532102111480076620485782708047624298113776849000815
BACKWARD-COMPUTED VALUE	A systematic embedding of classical reasoning into intuitionistic or minimal logic by reformulating formulas with negation while preserving classical derivability.
BACKWARD-COMPUTATION.ALGORITHM	shortlex-unrank/1.0
BACKWARD-COMPUTATION.MATCHES-VALUE	true
ALPHABET	!"#\$%&'()*+,-./0123456789:;<=>?@ABCDEFGHIJKLMNOPQRSTUVWXYZ[\]^_`abcdefghijklmnopqrstuvwxyz{ }~
METRICS.ALPHABET-SYMBOLS	95
METRICS.VALUE-SYMBOLS	163
METRICS.SEQUENTIAL-ID-DIGITS	322
METRICS.NATIVE-INTEGER	false
INSTRUCTION.LANGUAGE	GPIL/1.0
INSTRUCTION.SOURCE-LINE	61
INSTRUCTION.OPERATION	literal
INSTRUCTION.SOURCE	event "Negative translation" : literal("A systematic embedding of classical reasoning into intuitionistic or minimal logic by reformulating formulas with negation while preserving classical derivability.")

Double-negation translation

Complete event record 62 of 210

TIME-STAMP	2026-07-18T11:25:00.044Z
LABEL	Double-negation translation
VALUE	A negative translation that inserts double negations in a disciplined way so that classical derivations receive intuitionistically acceptable counterparts.
ID	gpil_0062_9eac995b7b0e9973
SEQUENTIAL-STRING ID	1262296514696348692637279467327007227246785460447085229573370508611598735476723192253009710464480480837738462307444426765539280028666777803701846188834412305265808337144656402378151903637499251719235395862980713733469230801388472445117116385790000827067805581283435808301205060998565055190804330627819972245
BACKWARD-COMPUTED VALUE	A negative translation that inserts double negations in a disciplined way so that classical derivations receive intuitionistically acceptable counterparts.
BACKWARD-COMPUTATION.ALGORITHM	shortlex-unrank/1.0
BACKWARD-COMPUTATION.MATCHES-VALUE	true
ALPHABET	!"#\$%&'()*+,-./0123456789:;<=>?@ABCDEFGHIJKLMNOPQRSTUVWXYZ[\]^_`abcdefghijklmnopqrstuvwxyz{ }~
METRICS.ALPHABET-SYMBOLS	95
METRICS.VALUE-SYMBOLS	155
METRICS.SEQUENTIAL-ID-DIGITS	307
METRICS.NATIVE-INTEGER	false
INSTRUCTION.LANGUAGE	GPIL/1.0
INSTRUCTION.SOURCE-LINE	62
INSTRUCTION.OPERATION	literal
INSTRUCTION.SOURCE	event "Double-negation translation" : literal("A negative translation that inserts double negations in a disciplined way so that classical derivations receive intuitionistically acceptable counterparts.")

Friedman translation

Complete event record 63 of 210

TIME-STAMP	2026-07-18T11:25:00.044Z
LABEL	Friedman translation
VALUE	Also called the A-translation, a transformation parameterized by a formula A that replaces atomic content in a way useful for conservation and admissibility results.
ID	gpil_0063_590fc7139a1cbb04
SEQUENTIAL-STRING ID	77356912057933765382325986098446553068330293905827638493322978864606786447894829612676503259744211731293768294116488310131419885549513406486069248822932925572951811154170365175346973733267058217320277584510892361203987459282780593285201644917584448352487107349638149470568117038869202890771595858526252547462520487819469487370
BACKWARD-COMPUTED VALUE	Also called the A-translation, a transformation parameterized by a formula A that replaces atomic content in a way useful for conservation and admissibility results.
BACKWARD-COMPUTATION.ALGORITHM	shortlex-unrank/1.0
BACKWARD-COMPUTATION.MATCHES-VALUE	true
ALPHABET	!"#\$%&'()*+,-./0123456789:;<=>?@ABCDEFGHIJKLMNOPQRSTUVWXYZ[\]^_`abcdefghijklmnopqrstuvwxyz{ }~
METRICS.ALPHABET-SYMBOLS	95
METRICS.VALUE-SYMBOLS	165
METRICS.SEQUENTIAL-ID-DIGITS	326
METRICS.NATIVE-INTEGER	false
INSTRUCTION.LANGUAGE	GPIL/1.0
INSTRUCTION.SOURCE-LINE	63
INSTRUCTION.OPERATION	literal
INSTRUCTION.SOURCE	event "Friedman translation" : literal("Also called the A-translation, a transformation parameterized by a formula A that replaces atomic content in a way useful for conservation and admissibility results.")

Gentzen's LJ

Complete event record 64 of 210

TIME-STAMP	2026-07-18T11:25:00.044Z
LABEL	Gentzen's LJ
VALUE	Gentzen's single-conclusion sequent calculus for intuitionistic logic. Its asymmetric conclusion discipline distinguishes it from the classical calculus LK.
ID	gpil_0064_89bcc126d84c5b97
SEQUENTIAL-STRING ID	143628478085455131392553746830477341455853866422016912287136739455905621915862906803340230886875413421749670145365102575977377106260548464816850290378745624730669422975285281890972790934225000640391473433298351593101212973943760655879460984051066651803139692439808714009034892639313376376695444911977744835820
BACKWARD-COMPUTED VALUE	Gentzen's single-conclusion sequent calculus for intuitionistic logic. Its asymmetric conclusion discipline distinguishes it from the classical calculus LK.
BACKWARD-COMPUTATION.ALGORITHM	shortlex-unrank/1.0
BACKWARD-COMPUTATION.MATCHES-VALUE	true
ALPHABET	!"#\$%&'()*+,-./0123456789:;<=>?@ABCDEFGHIJKLMNOPQRSTUVWXYZ[\]^_`abcdefghijklmnopqrstuvwxyz{ }~
METRICS.ALPHABET-SYMBOLS	95
METRICS.VALUE-SYMBOLS	156
METRICS.SEQUENTIAL-ID-DIGITS	309
METRICS.NATIVE-INTEGER	false
INSTRUCTION.LANGUAGE	GPIL/1.0
INSTRUCTION.SOURCE-LINE	64
INSTRUCTION.OPERATION	literal
INSTRUCTION.SOURCE	event "Gentzen's LJ" : literal("Gentzen's single-conclusion sequent calculus for intuitionistic logic. Its asymmetric conclusion discipline distinguishes it from the classical calculus LK.")

Hypothetical judgment

Complete event record 65 of 210

TIME-STAMP	2026-07-18T11:25:00.044Z
LABEL	Hypothetical judgment
VALUE	A judgment made under one or more assumptions, expressing that a construction is available whenever constructions satisfying the declared context are supplied.
ID	gpil_0065_09deb92cdb6f4192
SEQUENTIAL-STRING ID	102813556849229037291679399231807996622842969943567974680189835786661560107845400792703738669922002072435124506371611323101512934218090405004127508533407204038860703900861048686185890359874352164576955565584849738947371064335616816476641670640505856963355417843749164715996472108571948650639003070767077571635154070
BACKWARD-COMPUTED VALUE	A judgment made under one or more assumptions, expressing that a construction is available whenever constructions satisfying the declared context are supplied.
BACKWARD-COMPUTATION.ALGORITHM	shortlex-unrank/1.0
BACKWARD-COMPUTATION.MATCHES-VALUE	true
ALPHABET	!"#\$%&'()*+,-./0123456789:;<=>?@ABCDEFGHIJKLMNOPQRSTUVWXYZ[\]^_`abcdefghijklmnopqrstuvwxyz{ }~
METRICS.ALPHABET-SYMBOLS	95
METRICS.VALUE-SYMBOLS	159
METRICS.SEQUENTIAL-ID-DIGITS	315
METRICS.NATIVE-INTEGER	false
INSTRUCTION.LANGUAGE	GPIL/1.0
INSTRUCTION.SOURCE-LINE	65
INSTRUCTION.OPERATION	literal
INSTRUCTION.SOURCE	event "Hypothetical judgment" : literal("A judgment made under one or more assumptions, expressing that a construction is available whenever constructions satisfying the declared context are supplied.")

Meaning explanation

Complete event record 66 of 210

TIME-STAMP	2026-07-18T11:25:00.044Z
LABEL	Meaning explanation
VALUE	An account that explains a proposition or type through its canonical verifications and equality conditions, then justifies formal rules from those meanings.
ID	gpil_0066_4273aada4d2d3e1e
SEQUENTIAL-STRING ID	122781895442285202334851163456612348551370566722155145199991310900493150467793856945574469094492476000271942199072324243941142189619614519610449882942585983615992386843945229749976547352479428101635793266312226513222334255947051799317652391703975849120724639189499718582799359010353103016532347143265210739795
BACKWARD-COMPUTED VALUE	An account that explains a proposition or type through its canonical verifications and equality conditions, then justifies formal rules from those meanings.
BACKWARD-COMPUTATION.ALGORITHM	shortlex-unrank/1.0
BACKWARD-COMPUTATION.MATCHES-VALUE	true
ALPHABET	!"#\$%&'()*+,-./0123456789:;<=>?@ABCDEFGHIJKLMNOPQRSTUVWXYZ[\]^_`abcdefghijklmnopqrstuvwxyz{ }~
METRICS.ALPHABET-SYMBOLS	95
METRICS.VALUE-SYMBOLS	156
METRICS.SEQUENTIAL-ID-DIGITS	309
METRICS.NATIVE-INTEGER	false
INSTRUCTION.LANGUAGE	GPIL/1.0
INSTRUCTION.SOURCE-LINE	66
INSTRUCTION.OPERATION	literal
INSTRUCTION.SOURCE	event "Meaning explanation" : literal("An account that explains a proposition or type through its canonical verifications and equality conditions, then justifies formal rules from those meanings.")

Verification

Complete event record 67 of 210

TIME-STAMP	2026-07-18T11:25:00.044Z
LABEL	Verification
VALUE	Direct canonical evidence for a proposition, distinguished in meaning-theoretic accounts from a derivation that may first need to normalize before revealing that evidence.
ID	gpil_0067_eff6a229195f102f
SEQUENTIAL-STRING ID	61711969847580773119095854663460505364936145824493793356455163337102169608392844726428578448827569432635415716214244715246677857264356294663084810175206838389323054853776872761090898565499795188760871217891093384658990600067836640576585174290000642598599752024338190817096758330912978691467687968653542537973194656012800354371440198806115
BACKWARD-COMPUTED VALUE	Direct canonical evidence for a proposition, distinguished in meaning-theoretic accounts from a derivation that may first need to normalize before revealing that evidence.
BACKWARD-COMPUTATION.ALGORITHM	shortlex-unrank/1.0
BACKWARD-COMPUTATION.MATCHES-VALUE	true
ALPHABET	!"#\$%&'()*+,-./0123456789:;<=>?@ABCDEFGHIJKLMNOPQRSTUVWXYZ[\]^_`abcdefghijklmnopqrstuvwxyz{ }~
METRICS.ALPHABET-SYMBOLS	95
METRICS.VALUE-SYMBOLS	171
METRICS.SEQUENTIAL-ID-DIGITS	338
METRICS.NATIVE-INTEGER	false
INSTRUCTION.LANGUAGE	GPIL/1.0
INSTRUCTION.SOURCE-LINE	67
INSTRUCTION.OPERATION	literal
INSTRUCTION.SOURCE	event "Verification" : literal("Direct canonical evidence for a proposition, distinguished in meaning-theoretic accounts from a derivation that may first need to normalize before revealing that evidence.")

Type former

Complete event record 68 of 210

TIME-STAMP	2026-07-18T11:25:00.044Z
LABEL	Type former
VALUE	An operation that constructs a new type from given types or terms and is specified by formation, introduction, elimination, and computation rules.
ID	gpil_0068_b9ee203843e022b6
SEQUENTIAL-STRING ID	2050682787692629054905296042218915989983002375646297189039825625502676489060954479600825595342738858303885421907772106957209680672118563065732255552706222326221313610309690076586958380849844662150299334069699213036659643371141293586726407992179432561221648574234454761378513060637618745745
BACKWARD-COMPUTED VALUE	An operation that constructs a new type from given types or terms and is specified by formation, introduction, elimination, and computation rules.
BACKWARD-COMPUTATION.ALGORITHM	shortlex-unrank/1.0
BACKWARD-COMPUTATION.MATCHES-VALUE	true
ALPHABET	!"#\$%&'()*+,-./0123456789:;<=>?@ABCDEFGHIJKLMNopQRSTUVWXYZ[\]^_`abcdefghijklmnopqrstuvwxyz{ }~
METRICS.ALPHABET-SYMBOLS	95
METRICS.VALUE-SYMBOLS	146
METRICS.SEQUENTIAL-ID-DIGITS	289
METRICS.NATIVE-INTEGER	false
INSTRUCTION.LANGUAGE	GPIL/1.0
INSTRUCTION.SOURCE-LINE	68
INSTRUCTION.OPERATION	literal
INSTRUCTION.SOURCE	event "Type former" : literal("An operation that constructs a new type from given types or terms and is specified by formation, introduction, elimination, and computation rules.")

Function type

Complete event record 69 of 210

TIME-STAMP	2026-07-18T11:25:00.044Z
LABEL	Function type
VALUE	The type A to B whose inhabitants transform any inhabitant of A into an inhabitant of B, giving the computational reading of implication.
ID	gpil_0069_63bdb8fe1440fe26
SEQUENTIAL-STRING ID	5023428950069517612918155629285251366645877993202386940506820209982618864417312746295779413572832357712136374118283970185747927421469777262294568132749549227948023327638698379738732961761644952121957465060714904190811452945352936943854839342133267253139100356466222162770
BACKWARD-COMPUTED VALUE	The type A to B whose inhabitants transform any inhabitant of A into an inhabitant of B, giving the computational reading of implication.
BACKWARD-COMPUTATION.ALGORITHM	shortlex-unrank/1.0
BACKWARD-COMPUTATION.MATCHES-VALUE	true
ALPHABET	!"#\$%&'()*+,-./0123456789:;<=>?@ABCDEFGHIJKLMNOPQRSTUVWXYZ[\]^_`abcdefghijklmnopqrstuvwxyz{ }~
METRICS.ALPHABET-SYMBOLS	95
METRICS.VALUE-SYMBOLS	137
METRICS.SEQUENTIAL-ID-DIGITS	271
METRICS.NATIVE-INTEGER	false
INSTRUCTION.LANGUAGE	GPIL/1.0
INSTRUCTION.SOURCE-LINE	69
INSTRUCTION.OPERATION	literal
INSTRUCTION.SOURCE	event "Function type" : literal("The type A to B whose inhabitants transform any inhabitant of A into an inhabitant of B, giving the computational reading of implication.")

Product type

Complete event record 70 of 210

TIME-STAMP	2026-07-18T11:25:00.044Z
LABEL	Product type
VALUE	A type whose inhabitants are pairs containing one component of each constituent type, giving a computational interpretation of conjunction.
ID	gpil_0070_504e731509dc0f75
SEQUENTIAL-STRING ID	28680880814027165407650706191005386893763461398130285617740685165122637426669236637582888082778797724831159467411710610612121771243573514020876575211231864038973311512653963518175942546575658261722274066479346460010643079404109700004726212306240583200477116248205366674047145
BACKWARD-COMPUTED VALUE	A type whose inhabitants are pairs containing one component of each constituent type, giving a computational interpretation of conjunction.
BACKWARD-COMPUTATION.ALGORITHM	shortlex-unrank/1.0
BACKWARD-COMPUTATION.MATCHES-VALUE	true
ALPHABET	!"#\$%&'()*+,-./0123456789:;<=>?@ABCDEFGHIJKLMNOPQRSTUVWXYZ[\]^_`abcdefghijklmnopqrstuvwxyz{ }~
METRICS.ALPHABET-SYMBOLS	95
METRICS.VALUE-SYMBOLS	139
METRICS.SEQUENTIAL-ID-DIGITS	275
METRICS.NATIVE-INTEGER	false
INSTRUCTION.LANGUAGE	GPIL/1.0
INSTRUCTION.SOURCE-LINE	70
INSTRUCTION.OPERATION	literal
INSTRUCTION.SOURCE	event "Product type" : literal("A type whose inhabitants are pairs containing one component of each constituent type, giving a computational interpretation of conjunction.")

Sum type

Complete event record 71 of 210

TIME-STAMP	2026-07-18T11:25:00.044Z
LABEL	Sum type
VALUE	A type whose inhabitants are tagged as coming from the left or right constituent, giving a computational interpretation of disjunction.
ID	gpil_0071_2fe4794f8b4cd0e5
SEQUENTIAL-STRING ID	352125975878357783106645359578338249629925631609705703519674375016796019425308408790522985199947028399577025923123145222401464593345370777103724783109803214528617106882843669561561567265561181699676088643648058343312183216411442176017589687615176459369606900541234645
BACKWARD-COMPUTED VALUE	A type whose inhabitants are tagged as coming from the left or right constituent, giving a computational interpretation of disjunction.
BACKWARD-COMPUTATION.ALGORITHM	shortlex-unrank/1.0
BACKWARD-COMPUTATION.MATCHES-VALUE	true
ALPHABET	!"#\$%&'()*+,-./0123456789:;<=>?@ABCDEFGHIJKLMNPQRSTUVWXYZ[\]^_`abcdefghijklmnopqrstuvwxyz{ }~
METRICS.ALPHABET-SYMBOLS	95
METRICS.VALUE-SYMBOLS	135
METRICS.SEQUENTIAL-ID-DIGITS	267
METRICS.NATIVE-INTEGER	false
INSTRUCTION.LANGUAGE	GPIL/1.0
INSTRUCTION.SOURCE-LINE	71
INSTRUCTION.OPERATION	literal
INSTRUCTION.SOURCE	event "Sum type" : literal("A type whose inhabitants are tagged as coming from the left or right constituent, giving a computational interpretation of disjunction.")

Unit type

Complete event record 72 of 210

TIME-STAMP	2026-07-18T11:25:00.044Z
LABEL	Unit type
VALUE	A type with a single canonical inhabitant, commonly used as the type-theoretic interpretation of a proposition that is trivially true.
ID	gpil_0072_1cc1c7b4e91ed3ee
SEQUENTIAL-STRING ID	3706589219772187207862236843785406097748252390686255513237302910784985939839033446720858399828696502341047792196049794777027378318341428485266381747566573372000612769737190681676991015030929669866271539245882340201410887154901188406001326302223773709185013133135565
BACKWARD-COMPUTED VALUE	A type with a single canonical inhabitant, commonly used as the type-theoretic interpretation of a proposition that is trivially true.
BACKWARD-COMPUTATION.ALGORITHM	shortlex-unrank/1.0
BACKWARD-COMPUTATION.MATCHES-VALUE	true
ALPHABET	!"#\$%&'()*+,-./0123456789:;<=>?@ABCDEFGHIJKLMNOPQRSTUVWXYZ[\]^_`abcdefghijklmnopqrstuvwxyz{ }~
METRICS.ALPHABET-SYMBOLS	95
METRICS.VALUE-SYMBOLS	134
METRICS.SEQUENTIAL-ID-DIGITS	265
METRICS.NATIVE-INTEGER	false
INSTRUCTION.LANGUAGE	GPIL/1.0
INSTRUCTION.SOURCE-LINE	72
INSTRUCTION.OPERATION	literal
INSTRUCTION.SOURCE	event "Unit type" : literal("A type with a single canonical inhabitant, commonly used as the type-theoretic interpretation of a proposition that is trivially true.")

Empty type

Complete event record 73 of 210

TIME-STAMP	2026-07-18T11:25:00.044Z
LABEL	Empty type
VALUE	A type with no canonical inhabitants. It interprets falsity, and its eliminator expresses that any conclusion follows from impossible evidence.
ID	gpil_0073_ac0e4f079611b6ba
SEQUENTIAL-STRING ID	2336075667853021400313398895798803602371631922316303927821602073397069625581994069637638937901693478107651452217124285778692544104032967081501754862656040356884327506129245366123540242244935943938418522778123845387242068593243552430327882364192936857085960604276798359334853333571740
BACKWARD-COMPUTED VALUE	A type with no canonical inhabitants. It interprets falsity, and its eliminator expresses that any conclusion follows from impossible evidence.
BACKWARD-COMPUTATION.ALGORITHM	shortlex-unrank/1.0
BACKWARD-COMPUTATION.MATCHES-VALUE	true
ALPHABET	!"#\$%&'()*+,-./0123456789:;<=>?@ABCDEFGHIJKLMNOPQRSTUVWXYZ[\]^_`abcdefghijklmnopqrstuvwxyz{ }~
METRICS.ALPHABET-SYMBOLS	95
METRICS.VALUE-SYMBOLS	143
METRICS.SEQUENTIAL-ID-DIGITS	283
METRICS.NATIVE-INTEGER	false
INSTRUCTION.LANGUAGE	GPIL/1.0
INSTRUCTION.SOURCE-LINE	73
INSTRUCTION.OPERATION	literal
INSTRUCTION.SOURCE	event "Empty type" : literal("A type with no canonical inhabitants. It interprets falsity, and its eliminator expresses that any conclusion follows from impossible evidence.")

Identity type

Complete event record 74 of 210

TIME-STAMP	2026-07-18T11:25:00.044Z
LABEL	Identity type
VALUE	A type expressing that two terms of another type are identical, with reflexivity as its basic constructor and substitution as its central use.
ID	gpil_0074_680aa44f6feefafa
SEQUENTIAL-STRING ID	24590270187926356376005500625926496557553087304115331392772810791418782548566662057674194698735123066708038054776857810115494204871529625630200243795059594637390896121432857510803330053881335857196479216099465332956524591369667489515104000861419514694720026108100324934149993927765
BACKWARD-COMPUTED VALUE	A type expressing that two terms of another type are identical, with reflexivity as its basic constructor and substitution as its central use.
BACKWARD-COMPUTATION.ALGORITHM	shortlex-unrank/1.0
BACKWARD-COMPUTATION.MATCHES-VALUE	true
ALPHABET	!"#\$%&'()*+,-./0123456789:;<=>?@ABCDEFGHIJKLMNOPQRSTUVWXYZ[\]^_`abcdefghijklmnopqrstuvwxyz{ }~
METRICS.ALPHABET-SYMBOLS	95
METRICS.VALUE-SYMBOLS	142
METRICS.SEQUENTIAL-ID-DIGITS	281
METRICS.NATIVE-INTEGER	false
INSTRUCTION.LANGUAGE	GPIL/1.0
INSTRUCTION.SOURCE-LINE	74
INSTRUCTION.OPERATION	literal
INSTRUCTION.SOURCE	event "Identity type" : literal("A type expressing that two terms of another type are identical, with reflexivity as its basic constructor and substitution as its central use.")

Natural-number type

Complete event record 75 of 210

TIME-STAMP	2026-07-18T11:25:00.044Z
LABEL	Natural-number type
VALUE	An inductive type generated by zero and successor, equipped with recursion and induction principles that give numerical constructions their computational behavior.
ID	gpil_0075_29a09038e2d3c006
SEQUENTIAL-STRING ID	8574320017427579280704707646700828155001449140984726930002952341200847596396044201915565276097647780473968160975052273140903533916053345461641353551542235118512591703536796719890896514107523597401196328450722589867195681333382738352917456021545785999327087299889236249952101798893999371741801990252951545638923064822564400
BACKWARD-COMPUTED VALUE	An inductive type generated by zero and successor, equipped with recursion and induction principles that give numerical constructions their computational behavior.
BACKWARD-COMPUTATION.ALGORITHM	shortlex-unrank/1.0
BACKWARD-COMPUTATION.MATCHES-VALUE	true
ALPHABET	!"#\$%&'()*+,-./0123456789:;<=>?@ABCDEFGHIJKLMNopQRSTUVWXYZ[\]^_`abcdefghijklmnopqrstuvwxyz{ }~
METRICS.ALPHABET-SYMBOLS	95
METRICS.VALUE-SYMBOLS	163
METRICS.SEQUENTIAL-ID-DIGITS	322
METRICS.NATIVE-INTEGER	false
INSTRUCTION.LANGUAGE	GPIL/1.0
INSTRUCTION.SOURCE-LINE	75
INSTRUCTION.OPERATION	literal
INSTRUCTION.SOURCE	event "Natural-number type" : literal("An inductive type generated by zero and successor, equipped with recursion and induction principles that give numerical constructions their computational behavior.")

Inductive type

Complete event record 76 of 210

TIME-STAMP	2026-07-18T11:25:00.044Z
LABEL	Inductive type
VALUE	The least type generated by specified constructors, together with elimination and computation rules that support definitions and reasoning by structural induction.
ID	gpil_0076_9e3a3bc22e6db580
SEQUENTIAL-STRING ID	13237744689823451722902817809236610885240168475801355850037389154856542004751769129240173922851147330524934804412549625811996372958380225270106964926987111785721356347867961746890198812645493788111690920584066510917722169823333916741943788398842228105698265530499265432808353909169346515615734610884725569141726418831500270
BACKWARD-COMPUTED VALUE	The least type generated by specified constructors, together with elimination and computation rules that support definitions and reasoning by structural induction.
BACKWARD-COMPUTATION.ALGORITHM	shortlex-unrank/1.0
BACKWARD-COMPUTATION.MATCHES-VALUE	true
ALPHABET	!"#\$%&'()*+,-./0123456789:;<=>?@ABCDEFGHIJKLMNOPQRSTUVWXYZ[\]^_`abcdefghijklmnopqrstuvwxyz{ }~
METRICS.ALPHABET-SYMBOLS	95
METRICS.VALUE-SYMBOLS	163
METRICS.SEQUENTIAL-ID-DIGITS	323
METRICS.NATIVE-INTEGER	false
INSTRUCTION.LANGUAGE	GPIL/1.0
INSTRUCTION.SOURCE-LINE	76
INSTRUCTION.OPERATION	literal
INSTRUCTION.SOURCE	event "Inductive type" : literal("The least type generated by specified constructors, together with elimination and computation rules that support definitions and reasoning by structural induction.")

W-type

Complete event record 77 of 210

TIME-STAMP	2026-07-18T11:25:00.044Z
LABEL	W-type
VALUE	A type of well-founded trees parameterized by shapes and branching positions, providing a general type-theoretic representation of inductive generation.
ID	gpil_0077_da719f7be3089ef9
SEQUENTIAL-STRING ID	1472310310736384382918915001378159871371605502172904432188580525888396231412478486377463856581981102042683541151791211495188043625247847691581981160522293330094952472017646961642871764771685784738683977291242658981219306710099022340844426815623669378788942674029751007294696882278463595637243751225270
BACKWARD-COMPUTED VALUE	A type of well-founded trees parameterized by shapes and branching positions, providing a general type-theoretic representation of inductive generation.
BACKWARD-COMPUTATION.ALGORITHM	shortlex-unrank/1.0
BACKWARD-COMPUTATION.MATCHES-VALUE	true
ALPHABET	!"#\$%&'()*+,-./0123456789:;<=>?@ABCDEFGHIJKLMNOPQRSTUVWXYZ[\]^_`abcdefghijklmnopqrstuvwxyz{ }~
METRICS.ALPHABET-SYMBOLS	95
METRICS.VALUE-SYMBOLS	152
METRICS.SEQUENTIAL-ID-DIGITS	301
METRICS.NATIVE-INTEGER	false
INSTRUCTION.LANGUAGE	GPIL/1.0
INSTRUCTION.SOURCE-LINE	77
INSTRUCTION.OPERATION	literal
INSTRUCTION.SOURCE	event "W-type" : literal("A type of well-founded trees parameterized by shapes and branching positions, providing a general type-theoretic representation of inductive generation.")

Universe

Complete event record 78 of 210

TIME-STAMP	2026-07-18T11:25:00.044Z
LABEL	Universe
VALUE	A type whose elements serve as codes for smaller types, usually paired with a decoding operation and arranged in levels to control self-reference and predicativity.
ID	gpil_0078_c2aaa5bc3b97efb0
SEQUENTIAL-STRING ID	795577728576119620936299380579257892364471603637755715528173385281660775921699678466966852046172049434628398411092670836820288781600458359106800505336178504746756885630047001430262653001875438488968444318735155352919653414626313562816478083931115117272215913467020685228272692038690884698118638976486005047890695953163632065
BACKWARD-COMPUTED VALUE	A type whose elements serve as codes for smaller types, usually paired with a decoding operation and arranged in levels to control self-reference and predicativity.
BACKWARD-COMPUTATION.ALGORITHM	shortlex-unrank/1.0
BACKWARD-COMPUTATION.MATCHES-VALUE	true
ALPHABET	!"#\$%&'()*+,-./0123456789:;<=>?@ABCDEFGHIJKLMNOPQRSTUVWXYZ[\]^_`abcdefghijklmnopqrstuvwxyz{ }~
METRICS.ALPHABET-SYMBOLS	95
METRICS.VALUE-SYMBOLS	164
METRICS.SEQUENTIAL-ID-DIGITS	324
METRICS.NATIVE-INTEGER	false
INSTRUCTION.LANGUAGE	GPIL/1.0
INSTRUCTION.SOURCE-LINE	78
INSTRUCTION.OPERATION	literal
INSTRUCTION.SOURCE	event "Universe" : literal("A type whose elements serve as codes for smaller types, usually paired with a decoding operation and arranged in levels to control self-reference and predicativity.")

Definitional equality

Complete event record 79 of 210

TIME-STAMP	2026-07-18T11:25:00.044Z
LABEL	Definitional equality
VALUE	Equality recognized directly by computation, unfolding, and the formal conversion rules, requiring no separate inhabitant as evidence.
ID	gpil_0079_449e903da2236b8e
SEQUENTIAL-STRING ID	4235307383360789103728203502604518901000398815364982546101009344587040925534224499450410731156770733352053285922980230915927999854633141775184302640109363399054116028919639028728068682892175927538332627397572889425715014709604935855428408830225431466248496097243615
BACKWARD-COMPUTED VALUE	Equality recognized directly by computation, unfolding, and the formal conversion rules, requiring no separate inhabitant as evidence.
BACKWARD-COMPUTATION.ALGORITHM	shortlex-unrank/1.0
BACKWARD-COMPUTATION.MATCHES-VALUE	true
ALPHABET	!"#\$%&'()*+,-./0123456789:;<=>?@ABCDEFGHIJKLMNOPQRSTUVWXYZ[\]^_`abcdefghijklmnopqrstuvwxyz{ }~
METRICS.ALPHABET-SYMBOLS	95
METRICS.VALUE-SYMBOLS	134
METRICS.SEQUENTIAL-ID-DIGITS	265
METRICS.NATIVE-INTEGER	false
INSTRUCTION.LANGUAGE	GPIL/1.0
INSTRUCTION.SOURCE-LINE	79
INSTRUCTION.OPERATION	literal
INSTRUCTION.SOURCE	event "Definitional equality" : literal("Equality recognized directly by computation, unfolding, and the formal conversion rules, requiring no separate inhabitant as evidence.")

Propositional equality

Complete event record 80 of 210

TIME-STAMP	2026-07-18T11:25:00.044Z
LABEL	Propositional equality
VALUE	Equality expressed as an identity type whose inhabitants are explicit evidence, allowing equality claims that are not settled merely by computation.
ID	gpil_0080_c39d58917ac2bac4
SEQUENTIAL-STRING ID	20654534397283222393636430593463617818840898416530252584847207397589068863789204763708283638811408282166410659044079510080047954650161823211067323886316657516449070961908057680885438890041447558027120986885290294414435431767887531356807364497398366232626377467549809690234538752783743185709645
BACKWARD-COMPUTED VALUE	Equality expressed as an identity type whose inhabitants are explicit evidence, allowing equality claims that are not settled merely by computation.
BACKWARD-COMPUTATION.ALGORITHM	shortlex-unrank/1.0
BACKWARD-COMPUTATION.MATCHES-VALUE	true
ALPHABET	!"#\$%&'()*+,-./0123456789:;<=>?@ABCDEFGHIJKLMNOPQRSTUVWXYZ[\]^_`abcdefghijklmnopqrstuvwxyz{ }~
METRICS.ALPHABET-SYMBOLS	95
METRICS.VALUE-SYMBOLS	148
METRICS.SEQUENTIAL-ID-DIGITS	293
METRICS.NATIVE-INTEGER	false
INSTRUCTION.LANGUAGE	GPIL/1.0
INSTRUCTION.SOURCE-LINE	80
INSTRUCTION.OPERATION	literal
INSTRUCTION.SOURCE	event "Propositional equality" : literal("Equality expressed as an identity type whose inhabitants are explicit evidence, allowing equality claims that are not settled merely by computation.")

Conversion rule

Complete event record 81 of 210

TIME-STAMP	2026-07-18T11:25:00.044Z
LABEL	Conversion rule
VALUE	A rule allowing a term to be used at a definitionally equal type, so computationally interchangeable type expressions have the same inhabitants.
ID	gpil_0081_bb20d976c73a34b2
SEQUENTIAL-STRING ID	221925712057315347789331671241104433619363125674029966678493264014691752748613104611282799076336204067586201288221641291980660721034015548517300596074480236295093975547748987899207796043166332397584196095080144693057908014921155220086237389076019479266319092551434370629733782738655245
BACKWARD-COMPUTED VALUE	A rule allowing a term to be used at a definitionally equal type, so computationally interchangeable type expressions have the same inhabitants.
BACKWARD-COMPUTATION.ALGORITHM	shortlex-unrank/1.0
BACKWARD-COMPUTATION.MATCHES-VALUE	true
ALPHABET	!"#\$%&'()*+,-./0123456789:;<=>?@ABCDEFGHIJKLMNOPQRSTUVWXYZ[\]^_`abcdefghijklmnopqrstuvwxyz{ }~
METRICS.ALPHABET-SYMBOLS	95
METRICS.VALUE-SYMBOLS	144
METRICS.SEQUENTIAL-ID-DIGITS	285
METRICS.NATIVE-INTEGER	false
INSTRUCTION.LANGUAGE	GPIL/1.0
INSTRUCTION.SOURCE-LINE	81
INSTRUCTION.OPERATION	literal
INSTRUCTION.SOURCE	event "Conversion rule" : literal("A rule allowing a term to be used at a definitionally equal type, so computationally interchangeable type expressions have the same inhabitants.")

Context

Complete event record 82 of 210

TIME-STAMP	2026-07-18T11:25:00.044Z
LABEL	Context
VALUE	An ordered collection of variable declarations and assumptions under which a judgment is formed, recording the dependencies available to a construction.
ID	gpil_0082_b3fea04aa396c6a6
SEQUENTIAL-STRING ID	1507440288533446029435509219005380654288252548211514771786613011593592747383987630878287893259695509511391403094506141671186505671765662675528601757316033283179365707538100863961102388586627042754129561776558749393360770757760229124093376971518873976560179181637963175133811441511407430920809804234645
BACKWARD-COMPUTED VALUE	An ordered collection of variable declarations and assumptions under which a judgment is formed, recording the dependencies available to a construction.
BACKWARD-COMPUTATION.ALGORITHM	shortlex-unrank/1.0
BACKWARD-COMPUTATION.MATCHES-VALUE	true
ALPHABET	!"#\$%&'()*+,-./0123456789:;<=>?@ABCDEFGHIJKLMNOPQRSTUVWXYZ[\]^_`abcdefghijklmnopqrstuvwxyz{ }~
METRICS.ALPHABET-SYMBOLS	95
METRICS.VALUE-SYMBOLS	152
METRICS.SEQUENTIAL-ID-DIGITS	301
METRICS.NATIVE-INTEGER	false
INSTRUCTION.LANGUAGE	GPIL/1.0
INSTRUCTION.SOURCE-LINE	82
INSTRUCTION.OPERATION	literal
INSTRUCTION.SOURCE	event "Context" : literal("An ordered collection of variable declarations and assumptions under which a judgment is formed, recording the dependencies available to a construction.")

Substitution

Complete event record 83 of 210

TIME-STAMP	2026-07-18T11:25:00.044Z
LABEL	Substitution
VALUE	The operation of replacing a declared variable by a well-typed term. The substitution property ensures that valid judgments remain valid after replacement.
ID	gpil_0083_568328eda2aae869
SEQUENTIAL-STRING ID	1995377904403667096479593444871208563732517916776638837411222537402886197610676211195909735194781459389481193078776615095802425641290904467326969430770195849064820866464344447074776870186200938444147023273985288565670961553832001984892208668863736457344791850346639888340556755474509784837369640255444573565
BACKWARD-COMPUTED VALUE	The operation of replacing a declared variable by a well-typed term. The substitution property ensures that valid judgments remain valid after replacement.
BACKWARD-COMPUTATION.ALGORITHM	shortlex-unrank/1.0
BACKWARD-COMPUTATION.MATCHES-VALUE	true
ALPHABET	!"#\$%&'()*+,-./0123456789:;<=>?@ABCDEFGHIJKLMNOPQRSTUVWXYZ[\]^_`abcdefghijklmnopqrstuvwxyz{ }~
METRICS.ALPHABET-SYMBOLS	95
METRICS.VALUE-SYMBOLS	155
METRICS.SEQUENTIAL-ID-DIGITS	307
METRICS.NATIVE-INTEGER	false
INSTRUCTION.LANGUAGE	GPIL/1.0
INSTRUCTION.SOURCE-LINE	83
INSTRUCTION.OPERATION	literal
INSTRUCTION.SOURCE	event "Substitution" : literal("The operation of replacing a declared variable by a well-typed term. The substitution property ensures that valid judgments remain valid after replacement.")

Weakening

Complete event record 84 of 210

TIME-STAMP	2026-07-18T11:25:00.044Z
LABEL	Weakening
VALUE	A structural rule permitting an unused declaration to be added to a context, expressing that a construction remains valid when more assumptions are available.
ID	gpil_0084_cc10186eab63db84
SEQUENTIAL-STRING ID	1082279660231997668753075754697443914544497408545983176181598538144160583262236598042515693491954192408902522626914440297919646650319771293936463058783193905507436398393589780207061428126019668249737998567665590918362957218140486533798492207299014068891311254952010218914533651499048658533095340682185878120036965
BACKWARD-COMPUTED VALUE	A structural rule permitting an unused declaration to be added to a context, expressing that a construction remains valid when more assumptions are available.
BACKWARD-COMPUTATION.ALGORITHM	shortlex-unrank/1.0
BACKWARD-COMPUTATION.MATCHES-VALUE	true
ALPHABET	!"#\$%&'()*+,-./0123456789:;<=>?@ABCDEFGHIJKLMNOPQRSTUVWXYZ[\]^_`abcdefghijklmnopqrstuvwxyz{ }~
METRICS.ALPHABET-SYMBOLS	95
METRICS.VALUE-SYMBOLS	158
METRICS.SEQUENTIAL-ID-DIGITS	313
METRICS.NATIVE-INTEGER	false
INSTRUCTION.LANGUAGE	GPIL/1.0
INSTRUCTION.SOURCE-LINE	84
INSTRUCTION.OPERATION	literal
INSTRUCTION.SOURCE	event "Weakening" : literal("A structural rule permitting an unused declaration to be added to a context, expressing that a construction remains valid when more assumptions are available.")

Exchange

Complete event record 85 of 210

TIME-STAMP	2026-07-18T11:25:00.044Z
LABEL	Exchange
VALUE	A structural rule permitting independent context declarations to be reordered, subject to preserving any dependencies between later and earlier variables.
ID	gpil_0085_c6b853e37b31edd7
SEQUENTIAL-STRING ID	13287555009332312290459106418120719325904465540270356135164233409574519104611097865073711570905172179941856775375718216389386655082470373277041286835511814708844859794171641228453084979223751142337242187486659156292828844633279461599562879010807757977232757172104957311830623816582144059316261618864252620
BACKWARD-COMPUTED VALUE	A structural rule permitting independent context declarations to be reordered, subject to preserving any dependencies between later and earlier variables.
BACKWARD-COMPUTATION.ALGORITHM	shortlex-unrank/1.0
BACKWARD-COMPUTATION.MATCHES-VALUE	true
ALPHABET	!"#\$%&'()*+,-./0123456789:;<=>?@ABCDEFGHIJKLMNOPQRSTUVWXYZ[\]^_`abcdefghijklmnopqrstuvwxyz{ }~
METRICS.ALPHABET-SYMBOLS	95
METRICS.VALUE-SYMBOLS	154
METRICS.SEQUENTIAL-ID-DIGITS	305
METRICS.NATIVE-INTEGER	false
INSTRUCTION.LANGUAGE	GPIL/1.0
INSTRUCTION.SOURCE-LINE	85
INSTRUCTION.OPERATION	literal
INSTRUCTION.SOURCE	event "Exchange" : literal("A structural rule permitting independent context declarations to be reordered, subject to preserving any dependencies between later and earlier variables.")

Contraction

Complete event record 86 of 210

TIME-STAMP	2026-07-18T11:25:00.044Z
LABEL	Contraction
VALUE	A structural principle allowing one assumption to be used more than once or duplicate assumptions to be merged, unlike resource-sensitive systems such as linear logic.
ID	gpil_0086_d2f267f8f7ddafe4
SEQUENTIAL-STRING ID	682106117018167321796923946293293797226936311166348338462226158934871912412017063117230182477713664360162289219798964106934030401336294874402201717569845095900771101539016887188530421079683692374946618272463936447562010242011762979287134810690725064131637109357624839910518149304112434463347350488376017828507329549807112900964700
BACKWARD-COMPUTED VALUE	A structural principle allowing one assumption to be used more than once or duplicate assumptions to be merged, unlike resource-sensitive systems such as linear logic.
BACKWARD-COMPUTATION.ALGORITHM	shortlex-unrank/1.0
BACKWARD-COMPUTATION.MATCHES-VALUE	true
ALPHABET	!"#\$%&'()*+,-./0123456789:;<=>?@ABCDEFGHIJKLMNOPQRSTUVWXYZ[\]^_`abcdefghijklmnopqrstuvwxyz{ }~
METRICS.ALPHABET-SYMBOLS	95
METRICS.VALUE-SYMBOLS	167
METRICS.SEQUENTIAL-ID-DIGITS	330
METRICS.NATIVE-INTEGER	false
INSTRUCTION.LANGUAGE	GPIL/1.0
INSTRUCTION.SOURCE-LINE	86
INSTRUCTION.OPERATION	literal
INSTRUCTION.SOURCE	event "Contraction" : literal("A structural principle allowing one assumption to be used more than once or duplicate assumptions to be merged, unlike resource-sensitive systems such as linear logic.")

Formation rule

Complete event record 87 of 210

TIME-STAMP	2026-07-18T11:25:00.044Z
LABEL	Formation rule
VALUE	A rule stating when an expression is a well-formed proposition or type, including the premises required for each of its parameters.
ID	gpil_0087_a0d8359498e90dd1
SEQUENTIAL-STRING ID	4323154467281083176727997353672398828049185651939313300004021735556182262641835587158100585056860893487857634350349066000693769554469852260477816903276379662221584399090603583667301477099327344272530300438358314069104521887094643296900907403328201083631326445
BACKWARD-COMPUTED VALUE	A rule stating when an expression is a well-formed proposition or type, including the premises required for each of its parameters.
BACKWARD-COMPUTATION.ALGORITHM	shortlex-unrank/1.0
BACKWARD-COMPUTATION.MATCHES-VALUE	true
ALPHABET	!"#\$%&'()*+,-./0123456789:;<=>?@ABCDEFGHIJKLMNOPQRSTUVWXYZ[\]^_`abcdefghijklmnopqrstuvwxyz{ }~
METRICS.ALPHABET-SYMBOLS	95
METRICS.VALUE-SYMBOLS	131
METRICS.SEQUENTIAL-ID-DIGITS	259
METRICS.NATIVE-INTEGER	false
INSTRUCTION.LANGUAGE	GPIL/1.0
INSTRUCTION.SOURCE-LINE	87
INSTRUCTION.OPERATION	literal
INSTRUCTION.SOURCE	event "Formation rule" : literal("A rule stating when an expression is a well-formed proposition or type, including the premises required for each of its parameters.")

Computation rule

Complete event record 88 of 210

TIME-STAMP	2026-07-18T11:25:00.044Z
LABEL	Computation rule
VALUE	A rule describing how an eliminator acts on a canonical constructor, exposing the computational content of an introduction followed by elimination.
ID	gpil_0088_81ce6b13ea6c322b
SEQUENTIAL-STRING ID	190273557375140980742037436972578924663330103545326179519726704240680361296509620110657546496584538627956281786369560370914082637766314232875433898891985921407562502065516939772308358491430186833834140191015572831195996805149864900302407972544695205094706725949770798536687701253686266553395
BACKWARD-COMPUTED VALUE	A rule describing how an eliminator acts on a canonical constructor, exposing the computational content of an introduction followed by elimination.
BACKWARD-COMPUTATION.ALGORITHM	shortlex-unrank/1.0
BACKWARD-COMPUTATION.MATCHES-VALUE	true
ALPHABET	!"#\$%&'()*+,-./0123456789:;<=>?@ABCDEFGHIJKLMNopQRSTUVWXYZ[\]^_`abcdefghijklmnopqrstuvwxyz{ }~
METRICS.ALPHABET-SYMBOLS	95
METRICS.VALUE-SYMBOLS	147
METRICS.SEQUENTIAL-ID-DIGITS	291
METRICS.NATIVE-INTEGER	false
INSTRUCTION.LANGUAGE	GPIL/1.0
INSTRUCTION.SOURCE-LINE	88
INSTRUCTION.OPERATION	literal
INSTRUCTION.SOURCE	event "Computation rule" : literal("A rule describing how an eliminator acts on a canonical constructor, exposing the computational content of an introduction followed by elimination.")

Beta reduction

Complete event record 89 of 210

TIME-STAMP	2026-07-18T11:25:00.044Z
LABEL	Beta reduction
VALUE	The reduction of an introduced object immediately consumed by its eliminator, such as applying a lambda abstraction and substituting the argument into its body.
ID	gpil_0089_e494eef012ab9e67
SEQUENTIAL-STRING ID	15439853865529005928612534354388498712622303509011177615272938685889096589214436651234926398331552706281694345285818021584224880441277028014896227913606233318834808681851001227016947912175451009348767792119760020817940680673980763384887015608085456315376252649823640761057690764243046470423303318643289750638616628790
BACKWARD-COMPUTED VALUE	The reduction of an introduced object immediately consumed by its eliminator, such as applying a lambda abstraction and substituting the argument into its body.
BACKWARD-COMPUTATION.ALGORITHM	shortlex-unrank/1.0
BACKWARD-COMPUTATION.MATCHES-VALUE	true
ALPHABET	!"#\$%&'()*+,-./0123456789:;<=>?@ABCDEFGHIJKLMNopQRSTUVWXYZ[\]^_`abcdefghijklmnopqrstuvwxyz{ }~
METRICS.ALPHABET-SYMBOLS	95
METRICS.VALUE-SYMBOLS	160
METRICS.SEQUENTIAL-ID-DIGITS	317
METRICS.NATIVE-INTEGER	false
INSTRUCTION.LANGUAGE	GPIL/1.0
INSTRUCTION.SOURCE-LINE	89
INSTRUCTION.OPERATION	literal
INSTRUCTION.SOURCE	event "Beta reduction" : literal("The reduction of an introduced object immediately consumed by its eliminator, such as applying a lambda abstraction and substituting the argument into its body.")

Eta conversion

Complete event record 90 of 210

TIME-STAMP	2026-07-18T11:25:00.044Z
LABEL	Eta conversion
VALUE	The principle equating an object with its fully expanded introduction-elimination form, expressing that the available observations completely characterize it.
ID	gpil_0090_f2b69cc0a7aa0ddf
SEQUENTIAL-STRING ID	1710787131187069588091041752362410174281881968225243179474854878461941804135274673351660867070744175487044953117809263415892154074497364635016876052416401643452292969694761232124124074075142565356782027358879928103981079346345812340889666150518473507514176679791794312725510067668300635194596288681895475593792690
BACKWARD-COMPUTED VALUE	The principle equating an object with its fully expanded introduction-elimination form, expressing that the available observations completely characterize it.
BACKWARD-COMPUTATION.ALGORITHM	shortlex-unrank/1.0
BACKWARD-COMPUTATION.MATCHES-VALUE	true
ALPHABET	!"#\$%&'()*+,-./0123456789:;<=>?@ABCDEFGHIJKLMNOPQRSTUVWXYZ[\]^_`abcdefghijklmnopqrstuvwxyz{ }~
METRICS.ALPHABET-SYMBOLS	95
METRICS.VALUE-SYMBOLS	158
METRICS.SEQUENTIAL-ID-DIGITS	313
METRICS.NATIVE-INTEGER	false
INSTRUCTION.LANGUAGE	GPIL/1.0
INSTRUCTION.SOURCE-LINE	90
INSTRUCTION.OPERATION	literal
INSTRUCTION.SOURCE	event "Eta conversion" : literal("The principle equating an object with its fully expanded introduction-elimination form, expressing that the available observations completely characterize it.")

Subject reduction

Complete event record 91 of 210

TIME-STAMP	2026-07-18T11:25:00.044Z
LABEL	Subject reduction
VALUE	The property that evaluation preserves typing: if a term has a type and takes a reduction step, the resulting term retains that type.
ID	gpil_0091_464229cb68ae240b
SEQUENTIAL-STRING ID	61674529144465148916871367381544221815282973648152397850135054323578381786804967765283931435784848849026718952158712055865976688157105607030413275328040437219055014842867555789880642272946863274192261100712984786399070864237364030060986322560420513031385312529565
BACKWARD-COMPUTED VALUE	The property that evaluation preserves typing: if a term has a type and takes a reduction step, the resulting term retains that type.
BACKWARD-COMPUTATION.ALGORITHM	shortlex-unrank/1.0
BACKWARD-COMPUTATION.MATCHES-VALUE	true
ALPHABET	!"#\$%&'()*+,-./0123456789:;<=>?@ABCDEFGHIJKLMNopQRSTUVWXYZ[\]^_`abcdefghijklmnopqrstuvwxyz{ }~
METRICS.ALPHABET-SYMBOLS	95
METRICS.VALUE-SYMBOLS	133
METRICS.SEQUENTIAL-ID-DIGITS	263
METRICS.NATIVE-INTEGER	false
INSTRUCTION.LANGUAGE	GPIL/1.0
INSTRUCTION.SOURCE-LINE	91
INSTRUCTION.OPERATION	literal
INSTRUCTION.SOURCE	event "Subject reduction" : literal("The property that evaluation preserves typing: if a term has a type and takes a reduction step, the resulting term retains that type.")

Strong normalization

Complete event record 92 of 210

TIME-STAMP	2026-07-18T11:25:00.044Z
LABEL	Strong normalization
VALUE	The property that every permitted reduction sequence terminates, ruling out infinite computation within the calculus and supporting canonical-form arguments.
ID	gpil_0092_88f85d2d2523eb9c
SEQUENTIAL-STRING ID	18008285591445580795616468882889051981789787144050830257588979655354002626686722449128897144541888162848416215765083782769389988158584770897848589103928013035173123369206138834708988298790439448159776137746434134641590392719185513168367531995237628512167058162299022964398760721234055690159140036521357852479620
BACKWARD-COMPUTED VALUE	The property that every permitted reduction sequence terminates, ruling out infinite computation within the calculus and supporting canonical-form arguments.
BACKWARD-COMPUTATION.ALGORITHM	shortlex-unrank/1.0
BACKWARD-COMPUTATION.MATCHES-VALUE	true
ALPHABET	!"#\$%&'()*+,-./0123456789:;<=>?@ABCDEFGHIJKLMNOPQRSTUVWXYZ[\]^_`abcdefghijklmnopqrstuvwxyz{ }~
METRICS.ALPHABET-SYMBOLS	95
METRICS.VALUE-SYMBOLS	157
METRICS.SEQUENTIAL-ID-DIGITS	311
METRICS.NATIVE-INTEGER	false
INSTRUCTION.LANGUAGE	GPIL/1.0
INSTRUCTION.SOURCE-LINE	92
INSTRUCTION.OPERATION	literal
INSTRUCTION.SOURCE	event "Strong normalization" : literal("The property that every permitted reduction sequence terminates, ruling out infinite computation within the calculus and supporting canonical-form arguments.")

Confluence

Complete event record 93 of 210

TIME-STAMP	2026-07-18T11:25:00.044Z
LABEL	Confluence
VALUE	The property that diverging reduction paths can be joined again, ensuring that evaluation order does not produce incompatible normal forms.
ID	gpil_0093_deaa0ed817d54b7f
SEQUENTIAL-STRING ID	45336446232211550071725144408428297743832788150608388115741814323658511499575974544486342996028750587697493384844302753888234692870628007587872916470528677603978356209500178925722196339340614126081507246742707800858427581185632476257072711756265808019221816213987610203577195
BACKWARD-COMPUTED VALUE	The property that diverging reduction paths can be joined again, ensuring that evaluation order does not produce incompatible normal forms.
BACKWARD-COMPUTATION.ALGORITHM	shortlex-unrank/1.0
BACKWARD-COMPUTATION.MATCHES-VALUE	true
ALPHABET	!"#\$%&'()*+,-./0123456789:;<=>?@ABCDEFGHIJKLMNOPQRSTUVWXYZ[\]^_`abcdefghijklmnopqrstuvwxyz{ }~
METRICS.ALPHABET-SYMBOLS	95
METRICS.VALUE-SYMBOLS	139
METRICS.SEQUENTIAL-ID-DIGITS	275
METRICS.NATIVE-INTEGER	false
INSTRUCTION.LANGUAGE	GPIL/1.0
INSTRUCTION.SOURCE-LINE	93
INSTRUCTION.OPERATION	literal
INSTRUCTION.SOURCE	event "Confluence" : literal("The property that diverging reduction paths can be joined again, ensuring that evaluation order does not produce incompatible normal forms.")

Type checking

Complete event record 94 of 210

TIME-STAMP	2026-07-18T11:25:00.044Z
LABEL	Type checking
VALUE	The algorithmic task of verifying that a term has a stated type in a context, designed to be decidable in many constructive type theories.
ID	gpil_0094_4f0caa6ef97f9e1b
SEQUENTIAL-STRING ID	477225748171479054632682516925528249102099415556474545243023729713293838024185133912842132037421947067380137385261949698857404648451629926253556552355904352531173392855879288794653117879767334113910739448509364529669472491631100793514406073261805350298373059033730287206120
BACKWARD-COMPUTED VALUE	The algorithmic task of verifying that a term has a stated type in a context, designed to be decidable in many constructive type theories.
BACKWARD-COMPUTATION.ALGORITHM	shortlex-unrank/1.0
BACKWARD-COMPUTATION.MATCHES-VALUE	true
ALPHABET	!"#\$%&'()*+,-./0123456789:;<=>?@ABCDEFGHIJKLMNopQRSTUVWXYZ[\]^_`abcdefghijklmnopqrstuvwxyz{ }~
METRICS.ALPHABET-SYMBOLS	95
METRICS.VALUE-SYMBOLS	138
METRICS.SEQUENTIAL-ID-DIGITS	273
METRICS.NATIVE-INTEGER	false
INSTRUCTION.LANGUAGE	GPIL/1.0
INSTRUCTION.SOURCE-LINE	94
INSTRUCTION.OPERATION	literal
INSTRUCTION.SOURCE	event "Type checking" : literal("The algorithmic task of verifying that a term has a stated type in a context, designed to be decidable in many constructive type theories.")

Type inhabitation

Complete event record 95 of 210

TIME-STAMP	2026-07-18T11:25:00.044Z
LABEL	Type inhabitation
VALUE	The question whether any term has a given type. Under propositions-as-types this is the formal search problem corresponding to derivability.
ID	gpil_0095_cbb7467d0f6fd76e
SEQUENTIAL-STRING ID	4306962393073364501701222217787124724854406304169758542980470393031644266534864169609000070955869710025278354323268760443119995657103367582313525400639533230913244954739557453220848475457170166569869018579492047503972137354066036947081037926730877189740498346785447260735719565
BACKWARD-COMPUTED VALUE	The question whether any term has a given type. Under propositions-as-types this is the formal search problem corresponding to derivability.
BACKWARD-COMPUTATION.ALGORITHM	shortlex-unrank/1.0
BACKWARD-COMPUTATION.MATCHES-VALUE	true
ALPHABET	!"#\$%&'()*+,-./0123456789:;<=>?@ABCDEFGHIJKLMNOPQRSTUVWXYZ[\]^_`abcdefghijklmnopqrstuvwxyz{ }~
METRICS.ALPHABET-SYMBOLS	95
METRICS.VALUE-SYMBOLS	140
METRICS.SEQUENTIAL-ID-DIGITS	277
METRICS.NATIVE-INTEGER	false
INSTRUCTION.LANGUAGE	GPIL/1.0
INSTRUCTION.SOURCE-LINE	95
INSTRUCTION.OPERATION	literal
INSTRUCTION.SOURCE	event "Type inhabitation" : literal("The question whether any term has a given type. Under propositions-as-types this is the formal search problem corresponding to derivability.")

Program extraction

Complete event record 96 of 210

TIME-STAMP	2026-07-18T11:25:00.044Z
LABEL	Program extraction
VALUE	The systematic recovery of executable computational content from a constructive derivation, often by reading or erasing annotations from its term.
ID	gpil_0096_212bf977982eb56b
SEQUENTIAL-STRING ID	3166013129852264251062099944868850836484056386385297280494173762499080836398904480573686215106995135088018854318702377004370077363572824678133342474519474537279283100716919965725397467689847657170432850760995620606309934657774595378605636477551218478138852225797945156384897294747964916500
BACKWARD-COMPUTED VALUE	The systematic recovery of executable computational content from a constructive derivation, often by reading or erasing annotations from its term.
BACKWARD-COMPUTATION.ALGORITHM	shortlex-unrank/1.0
BACKWARD-COMPUTATION.MATCHES-VALUE	true
ALPHABET	!"#\$%&'()*+,-./0123456789:;<=>?@ABCDEFGHIJKLMNOPQRSTUVWXYZ[\]^_`abcdefghijklmnopqrstuvwxyz{ }~
METRICS.ALPHABET-SYMBOLS	95
METRICS.VALUE-SYMBOLS	146
METRICS.SEQUENTIAL-ID-DIGITS	289
METRICS.NATIVE-INTEGER	false
INSTRUCTION.LANGUAGE	GPIL/1.0
INSTRUCTION.SOURCE-LINE	96
INSTRUCTION.OPERATION	literal
INSTRUCTION.SOURCE	event "Program extraction" : literal("The systematic recovery of executable computational content from a constructive derivation, often by reading or erasing annotations from its term.")

Logical relation

Complete event record 97 of 210

TIME-STAMP	2026-07-18T11:25:00.044Z
LABEL	Logical relation
VALUE	A relation defined by recursion on types or formulas and used to establish metatheoretic properties such as normalization, adequacy, and representation independence.
ID	gpil_0097_cf46e0b474439b18
SEQUENTIAL-STRING ID	75579339953339101114845337175112430904846655740257324976374160223906300128214926277089742946319225484399220494547207948301446892446333397777014208063110578661201842940370929947254491513601442532114026329814835027325606251000087766510692227882269002466079350087227715758433750517658035179126033524700525923157324114386661540490
BACKWARD-COMPUTED VALUE	A relation defined by recursion on types or formulas and used to establish metatheoretic properties such as normalization, adequacy, and representation independence.
BACKWARD-COMPUTATION.ALGORITHM	shortlex-unrank/1.0
BACKWARD-COMPUTATION.MATCHES-VALUE	true
ALPHABET	!"#\$%&'()*+,-./0123456789:;<=>?@ABCDEFGHIJKLMNOPQRSTUVWXYZ[\]^_`abcdefghijklmnopqrstuvwxyz{ }~
METRICS.ALPHABET-SYMBOLS	95
METRICS.VALUE-SYMBOLS	165
METRICS.SEQUENTIAL-ID-DIGITS	326
METRICS.NATIVE-INTEGER	false
INSTRUCTION.LANGUAGE	GPIL/1.0
INSTRUCTION.SOURCE-LINE	97
INSTRUCTION.OPERATION	literal
INSTRUCTION.SOURCE	event "Logical relation" : literal("A relation defined by recursion on types or formulas and used to establish metatheoretic properties such as normalization, adequacy, and representation independence.")

Reducibility candidate

Complete event record 98 of 210

TIME-STAMP	2026-07-18T11:25:00.044Z
LABEL	Reducibility candidate
VALUE	A collection of well-behaved terms satisfying closure conditions, used in computability arguments that establish termination and normalization.
ID	gpil_0098_a0aeac1b7ffead08
SEQUENTIAL-STRING ID	2335945517447815742742349744508666180111574026358616629590983907048870870014926113539974841911790109140052969708903531701053852300411184878074398729523685634139600093170827262089124760682820841823029630407096881082357873079504371075054196767347812451409055950262842998414825904006520
BACKWARD-COMPUTED VALUE	A collection of well-behaved terms satisfying closure conditions, used in computability arguments that establish termination and normalization.
BACKWARD-COMPUTATION.ALGORITHM	shortlex-unrank/1.0
BACKWARD-COMPUTATION.MATCHES-VALUE	true
ALPHABET	!"#\$%&'()*+,-./0123456789:;<=>?@ABCDEFGHIJKLMNOPQRSTUVWXYZ[\]^_`abcdefghijklmnopqrstuvwxyz{ }~
METRICS.ALPHABET-SYMBOLS	95
METRICS.VALUE-SYMBOLS	143
METRICS.SEQUENTIAL-ID-DIGITS	283
METRICS.NATIVE-INTEGER	false
INSTRUCTION.LANGUAGE	GPIL/1.0
INSTRUCTION.SOURCE-LINE	98
INSTRUCTION.OPERATION	literal
INSTRUCTION.SOURCE	event "Reducibility candidate" : literal("A collection of well-behaved terms satisfying closure conditions, used in computability arguments that establish termination and normalization.")

Normalization by evaluation

Complete event record 99 of 210

TIME-STAMP	2026-07-18T11:25:00.044Z
LABEL	Normalization by evaluation
VALUE	A technique that evaluates syntax in a semantic domain and then reifies the result as a normal form, often yielding an effective equality procedure.
ID	gpil_0099_d88d4db671582a41
SEQUENTIAL-STRING ID	18076095724376118773577261340449739083301725379758423573161319055472279131801462033180135088801547899228250618140897900282459992831530736651415857370506956716485536201538209924890591927429600269987800439674326917440163562857269050228136215022725036984384558010942118478608218793922150580817490
BACKWARD-COMPUTED VALUE	A technique that evaluates syntax in a semantic domain and then reifies the result as a normal form, often yielding an effective equality procedure.
BACKWARD-COMPUTATION.ALGORITHM	shortlex-unrank/1.0
BACKWARD-COMPUTATION.MATCHES-VALUE	true
ALPHABET	!"#\$%&'()*+,-./0123456789:;<=>?@ABCDEFGHIJKLMNOPQRSTUVWXYZ[\]^_`abcdefghijklmnopqrstuvwxyz{ }~
METRICS.ALPHABET-SYMBOLS	95
METRICS.VALUE-SYMBOLS	148
METRICS.SEQUENTIAL-ID-DIGITS	293
METRICS.NATIVE-INTEGER	false
INSTRUCTION.LANGUAGE	GPIL/1.0
INSTRUCTION.SOURCE-LINE	99
INSTRUCTION.OPERATION	literal
INSTRUCTION.SOURCE	event "Normalization by evaluation" : literal("A technique that evaluates syntax in a semantic domain and then reifies the result as a normal form, often yielding an effective equality procedure.")

Sigma type

Complete event record 100 of 210

TIME-STAMP	2026-07-18T11:25:00.044Z
LABEL	Sigma type
VALUE	A dependent pair type whose inhabitants contain a witness x of A and an inhabitant of B(x), providing the type-theoretic reading of existential quantification.
ID	gpil_0100_a012e7de72930951
SEQUENTIAL-STRING ID	102811491645149087996830887048375920993036246064803810089847098429440493851686360627460340864261765636491058151473902241255903308489510208811484694005410523039121542003482905954704617474790079506096499509758232648782696909168898492140001685515394604389399595620439503531779896739152630535130980228252969213405365895
BACKWARD-COMPUTED VALUE	A dependent pair type whose inhabitants contain a witness x of A and an inhabitant of B(x), providing the type-theoretic reading of existential quantification.
BACKWARD-COMPUTATION.ALGORITHM	shortlex-unrank/1.0
BACKWARD-COMPUTATION.MATCHES-VALUE	true
ALPHABET	!"#\$%&'()*+,-./0123456789:;<=>?@ABCDEFGHIJKLMNOPQRSTUVWXYZ[\]^_`abcdefghijklmnopqrstuvwxyz{ }~
METRICS.ALPHABET-SYMBOLS	95
METRICS.VALUE-SYMBOLS	159
METRICS.SEQUENTIAL-ID-DIGITS	315
METRICS.NATIVE-INTEGER	false
INSTRUCTION.LANGUAGE	GPIL/1.0
INSTRUCTION.SOURCE-LINE	100
INSTRUCTION.OPERATION	literal
INSTRUCTION.SOURCE	event "Sigma type" : literal("A dependent pair type whose inhabitants contain a witness x of A and an inhabitant of B(x), providing the type-theoretic reading of existential quantification.")

Pi type

Complete event record 101 of 210

TIME-STAMP	2026-07-18T11:25:00.044Z
LABEL	Pi type
VALUE	A dependent function type whose inhabitants map each x of A to an inhabitant of B(x), providing the type-theoretic reading of universal quantification.
ID	gpil_0101_d803b795a863a771
SEQUENTIAL-STRING ID	15497185012131852413682628416874761759268200106239406044207566971887330897699137660064494429138827114632212298232420141854653974635844645444899341943432599473971985116422988993392251470085979229742781101427859857496904586105610104230634826926262840752519896226679321746062955203914361634679713959645
BACKWARD-COMPUTED VALUE	A dependent function type whose inhabitants map each x of A to an inhabitant of B(x), providing the type-theoretic reading of universal quantification.
BACKWARD-COMPUTATION.ALGORITHM	shortlex-unrank/1.0
BACKWARD-COMPUTATION.MATCHES-VALUE	true
ALPHABET	!"#\$%&'()*+,-./0123456789:;<=>?@ABCDEFGHIJKLMNOPQRSTUVWXYZ[\]^_`abcdefghijklmnopqrstuvwxyz{ }~
METRICS.ALPHABET-SYMBOLS	95
METRICS.VALUE-SYMBOLS	151
METRICS.SEQUENTIAL-ID-DIGITS	299
METRICS.NATIVE-INTEGER	false
INSTRUCTION.LANGUAGE	GPIL/1.0
INSTRUCTION.SOURCE-LINE	101
INSTRUCTION.OPERATION	literal
INSTRUCTION.SOURCE	event "Pi type" : literal("A dependent function type whose inhabitants map each x of A to an inhabitant of B(x), providing the type-theoretic reading of universal quantification.")

GPIL language contract

Complete event record 102 of 210

TIME-STAMP	2026-07-18T11:25:00.044Z
LABEL	GPIL language contract
VALUE	Every nonblank physical GPIL source line compiles to exactly one verified memory event.
ID	gpil_0102_2d38ca4f57267233
SEQUENTIAL-STRING ID	4725454027275723619508708350959300387424259597814216729428080545116162761777119988587737355883831252071255275529282571738267270553508583192402146165698506510674956885488565
BACKWARD-COMPUTED VALUE	Every nonblank physical GPIL source line compiles to exactly one verified memory event.
BACKWARD-COMPUTATION.ALGORITHM	shortlex-unrank/1.0
BACKWARD-COMPUTATION.MATCHES-VALUE	true
ALPHABET	!"#\$%&'()*+,-./0123456789:;<=>?@ABCDEFGHIJKLMNOPQRSTUVWXYZ[\]^_`abcdefghijklmnopqrstuvwxyz{ }~
METRICS.ALPHABET-SYMBOLS	95
METRICS.VALUE-SYMBOLS	87
METRICS.SEQUENTIAL-ID-DIGITS	172
METRICS.NATIVE-INTEGER	false
INSTRUCTION.LANGUAGE	GPIL/1.0
INSTRUCTION.SOURCE-LINE	102
INSTRUCTION.OPERATION	literal
INSTRUCTION.SOURCE	event "GPIL language contract" : literal("Every nonblank physical GPIL source line compiles to exactly one verified memory event.")

GPIL grammar program

Complete event record 103 of 210

TIME-STAMP	2026-07-18T11:25:00.044Z
LABEL	GPIL grammar program
VALUE	program = zero or more blank-line or event-line productions.
ID	gpil_0103_fb85185401ffb0cb
SEQUENTIAL-STRING ID	39708594368514878661237524172979067915489907659723253012969231924057823992736102438742396160908725810493220784742532220
BACKWARD-COMPUTED VALUE	program = zero or more blank-line or event-line productions.
BACKWARD-COMPUTATION.ALGORITHM	shortlex-unrank/1.0
BACKWARD-COMPUTATION.MATCHES-VALUE	true
ALPHABET	!"#\$%&'()*+,-./0123456789:;<=>?@ABCDEFGHIJKLMNOPQRSTUVWXYZ[\]^_`abcdefghijklmnopqrstuvwxyz{ }~
METRICS.ALPHABET-SYMBOLS	95
METRICS.VALUE-SYMBOLS	60
METRICS.SEQUENTIAL-ID-DIGITS	119
METRICS.NATIVE-INTEGER	false
INSTRUCTION.LANGUAGE	GPIL/1.0
INSTRUCTION.SOURCE-LINE	103
INSTRUCTION.OPERATION	literal
INSTRUCTION.SOURCE	event "GPIL grammar program" : literal("program = zero or more blank-line or event-line productions.")

GPIL grammar event line

Complete event record 104 of 210

TIME-STAMP	2026-07-18T11:25:00.044Z
LABEL	GPIL grammar event line
VALUE	event-line = event string : expression, optional alphabet string, optional semicolon, then newline.
ID	gpil_0104_2d0d215c13234314
SEQUENTIAL-STRING ID	4652692260186473364553252629237266640850835098117809622057166382220019161441407474441063279307084103372961247247576378968266659669685490366493404425874379173746811732359948365169231207092415041640
BACKWARD-COMPUTED VALUE	event-line = event string : expression, optional alphabet string, optional semicolon, then newline.
BACKWARD-COMPUTATION.ALGORITHM	shortlex-unrank/1.0
BACKWARD-COMPUTATION.MATCHES-VALUE	true
ALPHABET	!"#\$%&'()*+,-./0123456789:;<=>?@ABCDEFGHIJKLMNOPQRSTUVWXYZ[\]^_`abcdefghijklmnopqrstuvwxyz{ }~
METRICS.ALPHABET-SYMBOLS	95
METRICS.VALUE-SYMBOLS	99
METRICS.SEQUENTIAL-ID-DIGITS	196
METRICS.NATIVE-INTEGER	false
INSTRUCTION.LANGUAGE	GPIL/1.0
INSTRUCTION.SOURCE-LINE	104
INSTRUCTION.OPERATION	literal
INSTRUCTION.SOURCE	event "GPIL grammar event line" : literal("event-line = event string : expression, optional alphabet string, optional semicolon, then newline.")

GPIL grammar expression

Complete event record 105 of 210

TIME-STAMP

2026-07-18T11:25:00.044Z

LABEL

GPIL grammar expression

VALUE

expression = string, signed decimal, backward event reference, or nested built-in call.

ID

gpil_0105_25b479442279af1c

SEQUENTIAL-STRING ID

861305951571471920089981063025195357561854096467608664848034268800740054997002469476442706732262292
7989916298359043215442572595652997308136949570800690917878880379025209630

BACKWARD-COMPUTED VALUE

expression = string, signed decimal, backward event reference, or nested built-in call.

BACKWARD-COMPUTATION.ALGORITHM

shortlex-unrank/1.0

BACKWARD-COMPUTATION.MATCHES-VALUE

true

ALPHABET

!"#\$%&'()*+,-./0123456789:;<=>?@ABCDEFGHIJKLMNOPQRSTUVWXYZ[\]^_`abcdefghijklmnopqrstuvwxyz{|}~

METRICS.ALPHABET-SYMBOLS

95

METRICS.VALUE-SYMBOLS

87

METRICS.SEQUENTIAL-ID-DIGITS

172

METRICS.NATIVE-INTEGER

false

INSTRUCTION.LANGUAGE

GPIL/1.0

INSTRUCTION.SOURCE-LINE

105

INSTRUCTION.OPERATION

literal

INSTRUCTION.SOURCE

event "GPIL grammar expression" : literal("expression = string, signed decimal, backward event reference, or nested built-in call.")

GPIL backward reference

Complete event record 106 of 210

TIME-STAMP	2026-07-18T11:25:00.044Z
LABEL	GPIL backward reference
VALUE	A \$n expression reads earlier event ordinal n; zero, current, and forward references are rejected.
ID	gpil_0106_f722b797e4557dde
SEQUENTIAL-STRING ID	23486122242639393712202861117161324081473807184030013161194730752982641024207727471352741706289993584540319297576144614049330135825829034434994683237691092030146050016602119365443898246345532695
BACKWARD-COMPUTED VALUE	A \$n expression reads earlier event ordinal n; zero, current, and forward references are rejected.
BACKWARD-COMPUTATION.ALGORITHM	shortlex-unrank/1.0
BACKWARD-COMPUTATION.MATCHES-VALUE	true
ALPHABET	!"#\$%&'()*+,-./0123456789:;<=>?@ABCDEFGHIJKLMNOPQRSTUVWXYZ[\]^_`abcdefghijklmnopqrstuvwxyz{ }~
METRICS.ALPHABET-SYMBOLS	95
METRICS.VALUE-SYMBOLS	98
METRICS.SEQUENTIAL-ID-DIGITS	194
METRICS.NATIVE-INTEGER	false
INSTRUCTION.LANGUAGE	GPIL/1.0
INSTRUCTION.SOURCE-LINE	106
INSTRUCTION.OPERATION	literal
INSTRUCTION.SOURCE	event "GPIL backward reference" : literal("A \$n expression reads earlier event ordinal n; zero, current, and forward references are rejected.")

Instruction array 1

Complete event record 107 of 210

TIME-STAMP	2026-07-18T11:25:00.044Z
LABEL	Instruction array 1
VALUE	Construction instructions are add, apply_directional_polarity, containerize, and decontainerize.
ID	gpil_0107_2c930e00fedb9a34
SEQUENTIAL-STRING ID	2819624542350174524257933595425501701142300100910752224501282517365143813216106385573000040392371122862505503355017246437292741818372805367903483382826031223165081800972349010796684155769315
BACKWARD-COMPUTED VALUE	Construction instructions are add, apply_directional_polarity, containerize, and decontainerize.
BACKWARD-COMPUTATION.ALGORITHM	shortlex-unrank/1.0
BACKWARD-COMPUTATION.MATCHES-VALUE	true
ALPHABET	!"#\$%&'()*+,-./0123456789:;<=>?@ABCDEFGHIJKLMNOPQRSTUVWXYZ[\]^_`abcdefghijklmnopqrstuvwxyz{ }~
METRICS.ALPHABET-SYMBOLS	95
METRICS.VALUE-SYMBOLS	96
METRICS.SEQUENTIAL-ID-DIGITS	190
METRICS.NATIVE-INTEGER	false
INSTRUCTION.LANGUAGE	GPIL/1.0
INSTRUCTION.SOURCE-LINE	107
INSTRUCTION.OPERATION	literal
INSTRUCTION.SOURCE	event "Instruction array 1" : literal("Construction instructions are add, apply_directional_polarity, containerize, and decontainerize.")

Built-in add

Complete event record 108 of 210

TIME-STAMP	2026-07-18T11:25:00.044Z
LABEL	Built-in add
VALUE	add(a,b) returns the arbitrary-precision signed decimal sum of a and b.
ID	gpil_0108_5d91846c355f3174
SEQUENTIAL-STRING ID	184076923983058317374108107098871585791956905081847443539591064086471899497982894581089085747659552793006895276690847124540346276977492333030
BACKWARD-COMPUTED VALUE	add(a,b) returns the arbitrary-precision signed decimal sum of a and b.
BACKWARD-COMPUTATION.ALGORITHM	shortlex-unrank/1.0
BACKWARD-COMPUTATION.MATCHES-VALUE	true
ALPHABET	!"#\$%&'()*+,-./0123456789:;<=>?@ABCDEFGHIJKLMNOPQRSTUVWXYZ[\]^_`abcdefghijklmnopqrstuvwxyz{ }~
METRICS.ALPHABET-SYMBOLS	95
METRICS.VALUE-SYMBOLS	71
METRICS.SEQUENTIAL-ID-DIGITS	141
METRICS.NATIVE-INTEGER	false
INSTRUCTION.LANGUAGE	GPIL/1.0
INSTRUCTION.SOURCE-LINE	108
INSTRUCTION.OPERATION	literal
INSTRUCTION.SOURCE	event "Built-in add" : literal("add(a,b) returns the arbitrary-precision signed decimal sum of a and b.")

Built-in apply directional polarity

Complete event record 109 of 210

TIME-STAMP	2026-07-18T11:25:00.044Z
LABEL	Built-in apply directional polarity
VALUE	apply_directional_polarity(n,d) applies positive, negative, forward, backward, or invert direction to integer n.
ID	gpil_0109_2d93b91f4bbcba5f
SEQUENTIAL-STRING ID	225163290054107461578121187463565629987641159843430681335955915384022618113428647671258177486219203407481812866142855828332851597679428790706953104986661802958411323932194084327897316167596406306877045749088901515394528670
BACKWARD-COMPUTED VALUE	apply_directional_polarity(n,d) applies positive, negative, forward, backward, or invert direction to integer n.
BACKWARD-COMPUTATION.ALGORITHM	shortlex-unrank/1.0
BACKWARD-COMPUTATION.MATCHES-VALUE	true
ALPHABET	!"#\$%&'()*+,-./0123456789:;<=>?@ABCDEFGHIJKLMNOPQRSTUVWXYZ[\]^_`abcdefghijklmnopqrstuvwxyz{ }~
METRICS.ALPHABET-SYMBOLS	95
METRICS.VALUE-SYMBOLS	112
METRICS.SEQUENTIAL-ID-DIGITS	222
METRICS.NATIVE-INTEGER	false
INSTRUCTION.LANGUAGE	GPIL/1.0
INSTRUCTION.SOURCE-LINE	109
INSTRUCTION.OPERATION	literal
INSTRUCTION.SOURCE	event "Built-in apply directional polarity" : literal("apply_directional_polarity(n,d) applies positive, negative, forward, backward, or invert direction to integer n.")

Built-in containerize

Complete event record 110 of 210

TIME-STAMP	2026-07-18T11:25:00.044Z
LABEL	Built-in containerize
VALUE	containerize(x) returns GPDB1, the UTF-8 byte length, and payload x as one validated length-prefixed container.
ID	gpil_0110_5639a8a6c65fc5d9
SEQUENTIAL-STRING ID	2440656009946025285870413174700342753017897356026074366361377697675823364735252088472547310647678903475909274357639371601632874048842039489440265414491739305684163928480174273659295491268482340307534015139892155565199775
BACKWARD-COMPUTED VALUE	containerize(x) returns GPDB1, the UTF-8 byte length, and payload x as one validated length-prefixed container.
BACKWARD-COMPUTATION.ALGORITHM	shortlex-unrank/1.0
BACKWARD-COMPUTATION.MATCHES-VALUE	true
ALPHABET	!"#\$%&'()*+,-./0123456789:;<=>?@ABCDEFGHIJKLMNopQRSTUVWXYZ[\]^_`abcdefghijklmnopqrstuvwxyz{ }~
METRICS.ALPHABET-SYMBOLS	95
METRICS.VALUE-SYMBOLS	111
METRICS.SEQUENTIAL-ID-DIGITS	220
METRICS.NATIVE-INTEGER	false
INSTRUCTION.LANGUAGE	GPIL/1.0
INSTRUCTION.SOURCE-LINE	110
INSTRUCTION.OPERATION	literal
INSTRUCTION.SOURCE	event "Built-in containerize" : literal("containerize(x) returns GPDB1, the UTF-8 byte length, and payload x as one validated length-prefixed container.")

Built-in decontainerize

Complete event record 111 of 210

TIME-STAMP	2026-07-18T11:25:00.044Z
LABEL	Built-in decontainerize
VALUE	decontainerize(c) validates the GPDB1 marker and byte length, then returns the exact UTF-8 payload.
ID	gpil_0111_2154651d28279ae8
SEQUENTIAL-STRING ID	4575337159365533583546909639623633047537311254194343350540924355305374651948149567323314289026703416061981884568244892067203422050907432098110058716940470546818709012620941088938910627283237343470
BACKWARD-COMPUTED VALUE	decontainerize(c) validates the GPDB1 marker and byte length, then returns the exact UTF-8 payload.
BACKWARD-COMPUTATION.ALGORITHM	shortlex-unrank/1.0
BACKWARD-COMPUTATION.MATCHES-VALUE	true
ALPHABET	!"#\$%&'()*+,-./0123456789:;<=>?@ABCDEFGHIJKLMNOPQRSTUVWXYZ[\]^_`abcdefghijklmnopqrstuvwxyz{ }~
METRICS.ALPHABET-SYMBOLS	95
METRICS.VALUE-SYMBOLS	99
METRICS.SEQUENTIAL-ID-DIGITS	196
METRICS.NATIVE-INTEGER	false
INSTRUCTION.LANGUAGE	GPIL/1.0
INSTRUCTION.SOURCE-LINE	111
INSTRUCTION.OPERATION	literal
INSTRUCTION.SOURCE	event "Built-in decontainerize" : literal("decontainerize(c) validates the GPDB1 marker and byte length, then returns the exact UTF-8 payload.")

Instruction array 2

Complete event record 112 of 210

TIME-STAMP	2026-07-18T11:25:00.044Z
LABEL	Instruction array 2
VALUE	Container transformation contains transform_container.
ID	gpil_0112_3283399ee15d1488
SEQUENTIAL-STRING ID	24310834730101046457530803862456616883492735034844351399859036400353747053236200103258430770071462381606025
BACKWARD-COMPUTED VALUE	Container transformation contains transform_container.
BACKWARD-COMPUTATION.ALGORITHM	shortlex-unrank/1.0
BACKWARD-COMPUTATION.MATCHES-VALUE	true
ALPHABET	!"#\$%&'()*+,-./0123456789:;<=>?@ABCDEFGHIJKLMNOPQRSTUVWXYZ[\]^_`abcdefghijklmnopqrstuvwxyz{ }~
METRICS.ALPHABET-SYMBOLS	95
METRICS.VALUE-SYMBOLS	54
METRICS.SEQUENTIAL-ID-DIGITS	107
METRICS.NATIVE-INTEGER	false
INSTRUCTION.LANGUAGE	GPIL/1.0
INSTRUCTION.SOURCE-LINE	112
INSTRUCTION.OPERATION	literal
INSTRUCTION.SOURCE	event "Instruction array 2" : literal("Container transformation contains transform_container.")

Built-in transform container

Complete event record 113 of 210

TIME-STAMP	2026-07-18T11:25:00.044Z
LABEL	Built-in transform container
VALUE	transform_container(c,op,...) transforms only a validated payload by reverse, ASCII case, append, prepend, or replace, then rewraps it.
ID	gpil_0113_91ddc3117456f43d
SEQUENTIAL-STRING ID	888915867589022227794819385219118460153213328481688830770012586893644023357002814354149236753263665249801946592227144214170600183442655238069292797147352691585441241578520267652187625158708728764645287711119465409129388986557118307004270905493588543074390312493148315
BACKWARD-COMPUTED VALUE	transform_container(c,op,...) transforms only a validated payload by reverse, ASCII case, append, prepend, or replace, then rewraps it.
BACKWARD-COMPUTATION.ALGORITHM	shortlex-unrank/1.0
BACKWARD-COMPUTATION.MATCHES-VALUE	true
ALPHABET	!"#\$%&'()*+,-./0123456789:;<=>?@ABCDEFGHIJKLMNOPQRSTUVWXYZ[\]^_`abcdefghijklmnopqrstuvwxyz{ }~
METRICS.ALPHABET-SYMBOLS	95
METRICS.VALUE-SYMBOLS	135
METRICS.SEQUENTIAL-ID-DIGITS	267
METRICS.NATIVE-INTEGER	false
INSTRUCTION.LANGUAGE	GPIL/1.0
INSTRUCTION.SOURCE-LINE	113
INSTRUCTION.OPERATION	literal
INSTRUCTION.SOURCE	event "Built-in transform container" : literal("transform_container(c,op,...) transforms only a validated payload by reverse, ASCII case, append, prepend, or replace, then rewraps it.")

Instruction array 3

Complete event record 114 of 210

TIME-STAMP	2026-07-18T11:25:00.044Z
LABEL	Instruction array 3
VALUE	Arithmetic instructions are subtract, multiply, divide, and modulo.
ID	gpil_0114_321a418006e8776e
SEQUENTIAL-STRING ID	1181297062210786971741827310646766155037364824018426797463907345706660474299865919525377520767217252954320329869136005957003989014290
BACKWARD-COMPUTED VALUE	Arithmetic instructions are subtract, multiply, divide, and modulo.
BACKWARD-COMPUTATION.ALGORITHM	shortlex-unrank/1.0
BACKWARD-COMPUTATION.MATCHES-VALUE	true
ALPHABET	!"#\$%&'()*+,-./0123456789:;<=>?@ABCDEFGHIJKLMNOPQRSTUVWXYZ[\]^_`abcdefghijklmnopqrstuvwxyz{ }~
METRICS.ALPHABET-SYMBOLS	95
METRICS.VALUE-SYMBOLS	67
METRICS.SEQUENTIAL-ID-DIGITS	133
METRICS.NATIVE-INTEGER	false
INSTRUCTION.LANGUAGE	GPIL/1.0
INSTRUCTION.SOURCE-LINE	114
INSTRUCTION.OPERATION	literal
INSTRUCTION.SOURCE	event "Instruction array 3" : literal("Arithmetic instructions are subtract, multiply, divide, and modulo.")

Built-in subtract

Complete event record 115 of 210

TIME-STAMP	2026-07-18T11:25:00.044Z
LABEL	Built-in subtract
VALUE	subtract(a,b) returns the arbitrary-precision signed decimal difference a minus b.
ID	gpil_0115_2f5b4942288c63e5
SEQUENTIAL-STRING ID	1332248146334102700830264560365098177442210792792767609905388468525501300179700689870547223400250513377033597651262627104726318521375800768087883856028079817698030
BACKWARD-COMPUTED VALUE	subtract(a,b) returns the arbitrary-precision signed decimal difference a minus b.
BACKWARD-COMPUTATION.ALGORITHM	shortlex-unrank/1.0
BACKWARD-COMPUTATION.MATCHES-VALUE	true
ALPHABET	!"#\$%&'()*+,-./0123456789:;<=>?@ABCDEFGHIJKLMNOPQRSTUVWXYZ[\]^_`abcdefghijklmnopqrstuvwxyz{ }~
METRICS.ALPHABET-SYMBOLS	95
METRICS.VALUE-SYMBOLS	82
METRICS.SEQUENTIAL-ID-DIGITS	163
METRICS.NATIVE-INTEGER	false
INSTRUCTION.LANGUAGE	GPIL/1.0
INSTRUCTION.SOURCE-LINE	115
INSTRUCTION.OPERATION	literal
INSTRUCTION.SOURCE	event "Built-in subtract" : literal("subtract(a,b) returns the arbitrary-precision signed decimal difference a minus b.")

Built-in multiply

Complete event record 116 of 210

TIME-STAMP	2026-07-18T11:25:00.044Z
LABEL	Built-in multiply
VALUE	multiply(a,b) returns the arbitrary-precision signed decimal product.
ID	gpil_0116_0d04e62a1046f1b2
SEQUENTIAL-STRING ID	24118963466452577015328453132796396750247701375824986854123221961802366321106252461758963806947783575045367739336560048171332474319871040
BACKWARD-COMPUTED VALUE	multiply(a,b) returns the arbitrary-precision signed decimal product.
BACKWARD-COMPUTATION.ALGORITHM	shortlex-unrank/1.0
BACKWARD-COMPUTATION.MATCHES-VALUE	true
ALPHABET	!"#\$%&'()*+,-./0123456789:;<=>?@ABCDEFGHIJKLMNOPQRSTUVWXYZ[\]^_`abcdefghijklmnopqrstuvwxyz{ }~
METRICS.ALPHABET-SYMBOLS	95
METRICS.VALUE-SYMBOLS	69
METRICS.SEQUENTIAL-ID-DIGITS	137
METRICS.NATIVE-INTEGER	false
INSTRUCTION.LANGUAGE	GPIL/1.0
INSTRUCTION.SOURCE-LINE	116
INSTRUCTION.OPERATION	literal
INSTRUCTION.SOURCE	event "Built-in multiply" : literal("multiply(a,b) returns the arbitrary-precision signed decimal product.")

Built-in divide

Complete event record 117 of 210

TIME-STAMP	2026-07-18T11:25:00.044Z
LABEL	Built-in divide
VALUE	divide(a,b) rejects zero and returns arbitrary-precision integer division truncated toward zero.
ID	gpil_0117_0546f784518cdbb1
SEQUENTIAL-STRING ID	5339830256212954368514487810794965891919528790763238283811252059143322159756130306816538102764845909021082187752503557379475419021107196825659736252032549502310561058689021542112090099245440
BACKWARD-COMPUTED VALUE	divide(a,b) rejects zero and returns arbitrary-precision integer division truncated toward zero.
BACKWARD-COMPUTATION.ALGORITHM	shortlex-unrank/1.0
BACKWARD-COMPUTATION.MATCHES-VALUE	true
ALPHABET	!"#\$%&'()*+,-./0123456789:;<=>?@ABCDEFGHIJKLMNOPQRSTUVWXYZ[\]^_`abcdefghijklmnopqrstuvwxyz{ }~
METRICS.ALPHABET-SYMBOLS	95
METRICS.VALUE-SYMBOLS	96
METRICS.SEQUENTIAL-ID-DIGITS	190
METRICS.NATIVE-INTEGER	false
INSTRUCTION.LANGUAGE	GPIL/1.0
INSTRUCTION.SOURCE-LINE	117
INSTRUCTION.OPERATION	literal
INSTRUCTION.SOURCE	event "Built-in divide" : literal("divide(a,b) rejects zero and returns arbitrary-precision integer division truncated toward zero.")

Built-in modulo

Complete event record 118 of 210

TIME-STAMP	2026-07-18T11:25:00.044Z
LABEL	Built-in modulo
VALUE	modulo(a,b) rejects zero and returns the remainder with the dividend sign.
ID	gpil_0118_ea385623508f4766
SEQUENTIAL-STRING ID	186476482426761131962375205857616731231018280157564302127065475720389515102511442696268705608202363000423757908737947336434559405558099248611639945
BACKWARD-COMPUTED VALUE	modulo(a,b) rejects zero and returns the remainder with the dividend sign.
BACKWARD-COMPUTATION.ALGORITHM	shortlex-unrank/1.0
BACKWARD-COMPUTATION.MATCHES-VALUE	true
ALPHABET	!"#\$%&'()*+,-./0123456789:;<=>?@ABCDEFGHIJKLMNOPQRSTUVWXYZ[\]^_`abcdefghijklmnopqrstuvwxyz{ }~
METRICS.ALPHABET-SYMBOLS	95
METRICS.VALUE-SYMBOLS	74
METRICS.SEQUENTIAL-ID-DIGITS	147
METRICS.NATIVE-INTEGER	false
INSTRUCTION.LANGUAGE	GPIL/1.0
INSTRUCTION.SOURCE-LINE	118
INSTRUCTION.OPERATION	literal
INSTRUCTION.SOURCE	event "Built-in modulo" : literal("modulo(a,b) rejects zero and returns the remainder with the dividend sign.")

Instruction array 4

Complete event record 119 of 210

TIME-STAMP	2026-07-18T11:25:00.044Z
LABEL	Instruction array 4
VALUE	Text instructions are literal, identity, concat, symbol_length, reverse, and replace.
ID	gpil_0119_2c3b43e2659989c0
SEQUENTIAL-STRING ID	722995307764219535536694407290128691341661504342767742567378191087426199751476763524622736004583272941442039078416921238918665810538884379130230013235095090449197011490
BACKWARD-COMPUTED VALUE	Text instructions are literal, identity, concat, symbol_length, reverse, and replace.
BACKWARD-COMPUTATION.ALGORITHM	shortlex-unrank/1.0
BACKWARD-COMPUTATION.MATCHES-VALUE	true
ALPHABET	!"#\$%&'()*+,-./0123456789:;<=>?@ABCDEFGHIJKLMNOPQRSTUVWXYZ[\]^_`abcdefghijklmnopqrstuvwxyz{ }~
METRICS.ALPHABET-SYMBOLS	95
METRICS.VALUE-SYMBOLS	85
METRICS.SEQUENTIAL-ID-DIGITS	168
METRICS.NATIVE-INTEGER	false
INSTRUCTION.LANGUAGE	GPIL/1.0
INSTRUCTION.SOURCE-LINE	119
INSTRUCTION.OPERATION	literal
INSTRUCTION.SOURCE	event "Instruction array 4" : literal("Text instructions are literal, identity, concat, symbol_length, reverse, and replace.")

Built-in literal

Complete event record 120 of 210

TIME-STAMP	2026-07-18T11:25:00.044Z
LABEL	Built-in literal
VALUE	literal(x) returns x without alteration.
ID	gpil_0120_f4994af88f2f89ab
SEQUENTIAL-STRING ID	10522906578805333196213250375017769560899761253396324558786547558136036128569020
BACKWARD-COMPUTED VALUE	literal(x) returns x without alteration.
BACKWARD-COMPUTATION.ALGORITHM	shortlex-unrank/1.0
BACKWARD-COMPUTATION.MATCHES-VALUE	true
ALPHABET	!"#\$%&'()*+,-./0123456789:;<=>?@ABCDEFGHIJKLMNOPQRSTUVWXYZ[\]^_`abcdefghijklmnopqrstuvwxyz{ }~
METRICS.ALPHABET-SYMBOLS	95
METRICS.VALUE-SYMBOLS	40
METRICS.SEQUENTIAL-ID-DIGITS	80
METRICS.NATIVE-INTEGER	false
INSTRUCTION.LANGUAGE	GPIL/1.0
INSTRUCTION.SOURCE-LINE	120
INSTRUCTION.OPERATION	literal
INSTRUCTION.SOURCE	event "Built-in literal" : literal("literal(x) returns x without alteration.")

Built-in identity

Complete event record 121 of 210

TIME-STAMP	2026-07-18T11:25:00.044Z
LABEL	Built-in identity
VALUE	identity(x) returns x without alteration and makes dataflow intent explicit.
ID	gpil_0121_c1219c63ef1879c6
SEQUENTIAL-STRING ID	1595106131588257599172627441853730298031900264556329626207038440585363695224948602589690173089042174268468319802340551231065151909920295039560490864315
BACKWARD-COMPUTED VALUE	identity(x) returns x without alteration and makes dataflow intent explicit.
BACKWARD-COMPUTATION.ALGORITHM	shortlex-unrank/1.0
BACKWARD-COMPUTATION.MATCHES-VALUE	true
ALPHABET	!"#\$%&'()*+,-./0123456789:;<=>?@ABCDEFGHIJKLMNOPQRSTUVWXYZ[\]^_`abcdefghijklmnopqrstuvwxyz{ }~
METRICS.ALPHABET-SYMBOLS	95
METRICS.VALUE-SYMBOLS	76
METRICS.SEQUENTIAL-ID-DIGITS	151
METRICS.NATIVE-INTEGER	false
INSTRUCTION.LANGUAGE	GPIL/1.0
INSTRUCTION.SOURCE-LINE	121
INSTRUCTION.OPERATION	literal
INSTRUCTION.SOURCE	event "Built-in identity" : literal("identity(x) returns x without alteration and makes dataflow intent explicit.")

Built-in concat

Complete event record 122 of 210

TIME-STAMP	2026-07-18T11:25:00.044Z
LABEL	Built-in concat
VALUE	concat(x,...) concatenates one or more evaluated UTF-8 values in argument order.
ID	gpil_0122_507a8d5bdbea2157
SEQUENTIAL-STRING ID	119694634948921166096233192327000021901699379903068971882532779482845037600657485114516142377579361594683657424659515606860779727661264921413594621018857544150
BACKWARD-COMPUTED VALUE	concat(x,...) concatenates one or more evaluated UTF-8 values in argument order.
BACKWARD-COMPUTATION.ALGORITHM	shortlex-unrank/1.0
BACKWARD-COMPUTATION.MATCHES-VALUE	true
ALPHABET	!"#\$%&'()*+,-./0123456789:;<=>?@ABCDEFGHIJKLMNOPQRSTUVWXYZ[\]^_`abcdefghijklmnopqrstuvwxyz{ }~
METRICS.ALPHABET-SYMBOLS	95
METRICS.VALUE-SYMBOLS	80
METRICS.SEQUENTIAL-ID-DIGITS	159
METRICS.NATIVE-INTEGER	false
INSTRUCTION.LANGUAGE	GPIL/1.0
INSTRUCTION.SOURCE-LINE	122
INSTRUCTION.OPERATION	literal
INSTRUCTION.SOURCE	event "Built-in concat" : literal("concat(x,...) concatenates one or more evaluated UTF-8 values in argument order.")

Built-in symbol length

Complete event record 123 of 210

TIME-STAMP	2026-07-18T11:25:00.044Z
LABEL	Built-in symbol length
VALUE	symbol_length(x) returns the number of Unicode scalar symbols in x as a decimal string.
ID	gpil_0123_cfbef7ea613b4389
SEQUENTIAL-STRING ID	10313939326836480255959890882245045021582416593160792852431800457288685091686176457415887410767986687836609462199742341443123894589175932329300677839968551833553863975848705
BACKWARD-COMPUTED VALUE	symbol_length(x) returns the number of Unicode scalar symbols in x as a decimal string.
BACKWARD-COMPUTATION.ALGORITHM	shortlex-unrank/1.0
BACKWARD-COMPUTATION.MATCHES-VALUE	true
ALPHABET	!"#\$%&'()*+,-./0123456789:;<=>?@ABCDEFGHIJKLMNOPQRSTUVWXYZ[\]^_`abcdefghijklmnopqrstuvwxyz{ }~
METRICS.ALPHABET-SYMBOLS	95
METRICS.VALUE-SYMBOLS	87
METRICS.SEQUENTIAL-ID-DIGITS	173
METRICS.NATIVE-INTEGER	false
INSTRUCTION.LANGUAGE	GPIL/1.0
INSTRUCTION.SOURCE-LINE	123
INSTRUCTION.OPERATION	literal
INSTRUCTION.SOURCE	event "Built-in symbol length" : literal("symbol_length(x) returns the number of Unicode scalar symbols in x as a decimal string.")

Built-in reverse

Complete event record 124 of 210

TIME-STAMP	2026-07-18T11:25:00.044Z
LABEL	Built-in reverse
VALUE	reverse(x) reverses x by Unicode scalar symbol rather than by UTF-8 byte.
ID	gpil_0124_5a729661cf721a43
SEQUENTIAL-STRING ID	2084810481762883242951815579479941073027434296632642631334593607853996860048184763134732358747424370196477185787103805844607902980986197309735665
BACKWARD-COMPUTED VALUE	reverse(x) reverses x by Unicode scalar symbol rather than by UTF-8 byte.
BACKWARD-COMPUTATION.ALGORITHM	shortlex-unrank/1.0
BACKWARD-COMPUTATION.MATCHES-VALUE	true
ALPHABET	!"#\$%&'()*+,-./0123456789:;<=>?@ABCDEFGHIJKLMNOPQRSTUVWXYZ[\]^_`abcdefghijklmnopqrstuvwxyz{ }~
METRICS.ALPHABET-SYMBOLS	95
METRICS.VALUE-SYMBOLS	73
METRICS.SEQUENTIAL-ID-DIGITS	145
METRICS.NATIVE-INTEGER	false
INSTRUCTION.LANGUAGE	GPIL/1.0
INSTRUCTION.SOURCE-LINE	124
INSTRUCTION.OPERATION	literal
INSTRUCTION.SOURCE	event "Built-in reverse" : literal("reverse(x) reverses x by Unicode scalar symbol rather than by UTF-8 byte.")

Built-in replace

Complete event record 125 of 210

TIME-STAMP	2026-07-18T11:25:00.044Z
LABEL	Built-in replace
VALUE	replace(x,search,replacement) replaces every non-overlapping byte-exact search occurrence and rejects an empty search.
ID	gpil_0125_41a1ca6a17f8f9b5
SEQUENTIAL-STRING ID	207312464007591079514071692016286536479580301980303615961638680749391498339667285102224204981641477068792086021524371860887967282454360460607262365106446088791691562777128721487404748700323425488811601023419762744582149611767034789650
BACKWARD-COMPUTED VALUE	replace(x,search,replacement) replaces every non-overlapping byte-exact search occurrence and rejects an empty search.
BACKWARD-COMPUTATION.ALGORITHM	shortlex-unrank/1.0
BACKWARD-COMPUTATION.MATCHES-VALUE	true
ALPHABET	!"#\$%&'()*+,-./0123456789:;<=>?@ABCDEFGHIJKLMNOPQRSTUVWXYZ[\]^_`abcdefghijklmnopqrstuvwxyz{ }~
METRICS.ALPHABET-SYMBOLS	95
METRICS.VALUE-SYMBOLS	118
METRICS.SEQUENTIAL-ID-DIGITS	234
METRICS.NATIVE-INTEGER	false
INSTRUCTION.LANGUAGE	GPIL/1.0
INSTRUCTION.SOURCE-LINE	125
INSTRUCTION.OPERATION	literal
INSTRUCTION.SOURCE	event "Built-in replace" : literal("replace(x,search,replacement) replaces every non-overlapping byte-exact search occurrence and rejects an empty search.")

Instruction array 5

Complete event record 126 of 210

TIME-STAMP	2026-07-18T11:25:00.044Z
LABEL	Instruction array 5
VALUE	Constructive logic instructions are equal, not, and, or, select, and assert_equal.
ID	gpil_0126_fcd009f036ca3de8
SEQUENTIAL-STRING ID	578176995560440154645401424705966824388294365338733846597380808004302186867246648775727415599445438301414874895318010201446545700984555290307471045822325957116605
BACKWARD-COMPUTED VALUE	Constructive logic instructions are equal, not, and, or, select, and assert_equal.
BACKWARD-COMPUTATION.ALGORITHM	shortlex-unrank/1.0
BACKWARD-COMPUTATION.MATCHES-VALUE	true
ALPHABET	!"#\$%&'()*+,-./0123456789:;<=>?@ABCDEFGHIJKLMNOPQRSTUVWXYZ[\]^_`abcdefghijklmnopqrstuvwxyz{ }~
METRICS.ALPHABET-SYMBOLS	95
METRICS.VALUE-SYMBOLS	82
METRICS.SEQUENTIAL-ID-DIGITS	162
METRICS.NATIVE-INTEGER	false
INSTRUCTION.LANGUAGE	GPIL/1.0
INSTRUCTION.SOURCE-LINE	126
INSTRUCTION.OPERATION	literal
INSTRUCTION.SOURCE	event "Instruction array 5" : literal("Constructive logic instructions are equal, not, and, or, select, and assert_equal.")

Built-in equal

Complete event record 127 of 210

TIME-STAMP	2026-07-18T11:25:00.044Z
LABEL	Built-in equal
VALUE	equal(a,b) returns true exactly when a and b are byte-for-byte equal.
ID	gpil_0127_0a739cb9dc6fb651
SEQUENTIAL-STRING ID	216613009149662056267165675522806143515980948339854512253945044694138921238159815252015009522244677959339211593455202582525864088163201210355
BACKWARD-COMPUTED VALUE	equal(a,b) returns true exactly when a and b are byte-for-byte equal.
BACKWARD-COMPUTATION.ALGORITHM	shortlex-unrank/1.0
BACKWARD-COMPUTATION.MATCHES-VALUE	true
ALPHABET	!"#\$%&'()*+,-./0123456789:;<=>?@ABCDEFGHIJKLMNOPQRSTUVWXYZ[\]^_`abcdefghijklmnopqrstuvwxyz{ }~
METRICS.ALPHABET-SYMBOLS	95
METRICS.VALUE-SYMBOLS	69
METRICS.SEQUENTIAL-ID-DIGITS	137
METRICS.NATIVE-INTEGER	false
INSTRUCTION.LANGUAGE	GPIL/1.0
INSTRUCTION.SOURCE-LINE	127
INSTRUCTION.OPERATION	literal
INSTRUCTION.SOURCE	event "Built-in equal" : literal("equal(a,b) returns true exactly when a and b are byte-for-byte equal.")

Built-in not

Complete event record 128 of 210

TIME-STAMP	2026-07-18T11:25:00.044Z
LABEL	Built-in not
VALUE	not(p) negates the canonical Boolean string true or false.
ID	gpil_0128_6ea22099b7566254
SEQUENTIAL-STRING ID	4290706129790405646382190522401992084745989473049239876445886818948471642089809406605483927544541665886740960833265
BACKWARD-COMPUTED VALUE	not(p) negates the canonical Boolean string true or false.
BACKWARD-COMPUTATION.ALGORITHM	shortlex-unrank/1.0
BACKWARD-COMPUTATION.MATCHES-VALUE	true
ALPHABET	!"#\$%&'()*+,-./0123456789:;<=>?@ABCDEFGHIJKLMNOPQRSTUVWXYZ[\]^_`abcdefghijklmnopqrstuvwxyz{ }~
METRICS.ALPHABET-SYMBOLS	95
METRICS.VALUE-SYMBOLS	58
METRICS.SEQUENTIAL-ID-DIGITS	115
METRICS.NATIVE-INTEGER	false
INSTRUCTION.LANGUAGE	GPIL/1.0
INSTRUCTION.SOURCE-LINE	128
INSTRUCTION.OPERATION	literal
INSTRUCTION.SOURCE	event "Built-in not" : literal("not(p) negates the canonical Boolean string true or false.")

Built-in and

Complete event record 129 of 210

TIME-STAMP	2026-07-18T11:25:00.044Z
LABEL	Built-in and
VALUE	and(p,q) returns true exactly when both canonical Boolean arguments are true.
ID	gpil_0129_8045c8a8c23230e4
SEQUENTIAL-STRING ID	135526891783055104677755342654187540073030577588484488461047104289651201386216074105845623192117660496398205559001691675885476926655351655900610840557440
BACKWARD-COMPUTED VALUE	and(p,q) returns true exactly when both canonical Boolean arguments are true.
BACKWARD-COMPUTATION.ALGORITHM	shortlex-unrank/1.0
BACKWARD-COMPUTATION.MATCHES-VALUE	true
ALPHABET	!"#\$%&'()*+,-./0123456789:;<=>?@ABCDEFGHIJKLMNOPQRSTUVWXYZ[\]^_`abcdefghijklmnopqrstuvwxyz{ }~
METRICS.ALPHABET-SYMBOLS	95
METRICS.VALUE-SYMBOLS	77
METRICS.SEQUENTIAL-ID-DIGITS	153
METRICS.NATIVE-INTEGER	false
INSTRUCTION.LANGUAGE	GPIL/1.0
INSTRUCTION.SOURCE-LINE	129
INSTRUCTION.OPERATION	literal
INSTRUCTION.SOURCE	event "Built-in and" : literal("and(p,q) returns true exactly when both canonical Boolean arguments are true.")

Built-in or

Complete event record 130 of 210

TIME-STAMP	2026-07-18T11:25:00.044Z
LABEL	Built-in or
VALUE	or(p,q) returns true when at least one canonical Boolean argument is true.
ID	gpil_0130_17559b888053b2f2
SEQUENTIAL-STRING ID	191265340233213165149859320282436171462646226954722737015500543085553874570114704491372802316521953681979210376462893110536266967731112792365229315
BACKWARD-COMPUTED VALUE	or(p,q) returns true when at least one canonical Boolean argument is true.
BACKWARD-COMPUTATION.ALGORITHM	shortlex-unrank/1.0
BACKWARD-COMPUTATION.MATCHES-VALUE	true
ALPHABET	!"#\$%&'()*+,-./0123456789:;<=>?@ABCDEFGHIJKLMNOPQRSTUVWXYZ[\]^_`abcdefghijklmnopqrstuvwxyz{ }~
METRICS.ALPHABET-SYMBOLS	95
METRICS.VALUE-SYMBOLS	74
METRICS.SEQUENTIAL-ID-DIGITS	147
METRICS.NATIVE-INTEGER	false
INSTRUCTION.LANGUAGE	GPIL/1.0
INSTRUCTION.SOURCE-LINE	130
INSTRUCTION.OPERATION	literal
INSTRUCTION.SOURCE	event "Built-in or" : literal("or(p,q) returns true when at least one canonical Boolean argument is true.")

Built-in select

Complete event record 131 of 210

TIME-STAMP	2026-07-18T11:25:00.044Z
LABEL	Built-in select
VALUE	select(p,a,b) constructively chooses a when p is true and b when p is false.
ID	gpil_0131_86f5a022d72a7dce
SEQUENTIAL-STRING ID	1808784467151074458761425733542013819166643322531402063614130559830848547885616421093716579350455927468065156934154267309166675582113313187589154895765
BACKWARD-COMPUTED VALUE	select(p,a,b) constructively chooses a when p is true and b when p is false.
BACKWARD-COMPUTATION.ALGORITHM	shortlex-unrank/1.0
BACKWARD-COMPUTATION.MATCHES-VALUE	true
ALPHABET	!"#\$%&'()*+,-./0123456789:;<=>?@ABCDEFGHIJKLMNOPQRSTUVWXYZ[\]^_`abcdefghijklmnopqrstuvwxyz{ }~
METRICS.ALPHABET-SYMBOLS	95
METRICS.VALUE-SYMBOLS	76
METRICS.SEQUENTIAL-ID-DIGITS	151
METRICS.NATIVE-INTEGER	false
INSTRUCTION.LANGUAGE	GPIL/1.0
INSTRUCTION.SOURCE-LINE	131
INSTRUCTION.OPERATION	literal
INSTRUCTION.SOURCE	event "Built-in select" : literal("select(p,a,b) constructively chooses a when p is true and b when p is false.")

Built-in assert equal

Complete event record 132 of 210

TIME-STAMP	2026-07-18T11:25:00.044Z
LABEL	Built-in assert equal
VALUE	assert_equal(a,b) returns a only when a and b are exact equals; otherwise compilation stops.
ID	gpil_0132_df75ba4d9691ffbb
SEQUENTIAL-STRING ID	62839490425461207721515428535369489068615988070478040898318452694515622308522307073926774020992101912002221761993807620765062561144851529240244835371963215938814519548201708998479020
BACKWARD-COMPUTED VALUE	assert_equal(a,b) returns a only when a and b are exact equals; otherwise compilation stops.
BACKWARD-COMPUTATION.ALGORITHM	shortlex-unrank/1.0
BACKWARD-COMPUTATION.MATCHES-VALUE	true
ALPHABET	!"#\$%&'()*+,-./0123456789:;<=>?@ABCDEFGHIJKLMNOPQRSTUVWXYZ[\]^_`abcdefghijklmnopqrstuvwxyz{ }~
METRICS.ALPHABET-SYMBOLS	95
METRICS.VALUE-SYMBOLS	92
METRICS.SEQUENTIAL-ID-DIGITS	182
METRICS.NATIVE-INTEGER	false
INSTRUCTION.LANGUAGE	GPIL/1.0
INSTRUCTION.SOURCE-LINE	132
INSTRUCTION.OPERATION	literal
INSTRUCTION.SOURCE	event "Built-in assert equal" : literal("assert_equal(a,b) returns a only when a and b are exact equals; otherwise compilation stops.")

Instruction array 6

Complete event record 133 of 210

TIME-STAMP

2026-07-18T11:25:00.044Z

LABEL

Instruction array 6

VALUE

Semantic instructions contain semantic_triple.

ID

gpil_0133_f2c81fc7cf08ee87

SEQUENTIAL-STRING ID

5245032312272359539436407772259795704048243098964191511237625615315159231260405570561298340

BACKWARD-COMPUTED VALUE

Semantic instructions contain semantic_triple.

BACKWARD-COMPUTATION.ALGORITHM

shortlex-unrank/1.0

BACKWARD-COMPUTATION.MATCHES-VALUE

true

ALPHABET

!"#\$%&'()*+,-./0123456789:;<=>?@ABCDEFGHIJKLMNOPQRSTUVWXYZ[\]^_`abcdefghijklmnopqrstuvwxyz{|}~

METRICS.ALPHABET-SYMBOLS

95

METRICS.VALUE-SYMBOLS

46

METRICS.SEQUENTIAL-ID-DIGITS

91

METRICS.NATIVE-INTEGER

false

INSTRUCTION.LANGUAGE

GPIL/1.0

INSTRUCTION.SOURCE-LINE

133

INSTRUCTION.OPERATION

literal

INSTRUCTION.SOURCE

event "Instruction array 6" : literal("Semantic instructions contain semantic_triple.")

Built-in semantic triple

Complete event record 134 of 210

TIME-STAMP	2026-07-18T11:25:00.044Z
LABEL	Built-in semantic triple
VALUE	semantic_triple(subject,predicate,object) returns canonical text in the form subject <predicate> object.
ID	gpil_0134_81a5c8a0dd46ec05
SEQUENTIAL-STRING ID	43017811537490825309392036267889114210476751854747991000725215838358477950335663092591774516273407677978694155189470239245488926249881069378428270336324630596743959603331471080707990338726188593643289381790
BACKWARD-COMPUTED VALUE	semantic_triple(subject,predicate,object) returns canonical text in the form subject <predicate> object.
BACKWARD-COMPUTATION.ALGORITHM	shortlex-unrank/1.0
BACKWARD-COMPUTATION.MATCHES-VALUE	true
ALPHABET	!"#\$%&'()*+,-./0123456789:;<=>?@ABCDEFGHIJKLMNOPQRSTUVWXYZ[\]^_`abcdefghijklmnopqrstuvwxyz{ }~
METRICS.ALPHABET-SYMBOLS	95
METRICS.VALUE-SYMBOLS	104
METRICS.SEQUENTIAL-ID-DIGITS	206
METRICS.NATIVE-INTEGER	false
INSTRUCTION.LANGUAGE	GPIL/1.0
INSTRUCTION.SOURCE-LINE	134
INSTRUCTION.OPERATION	literal
INSTRUCTION.SOURCE	event "Built-in semantic triple" : literal("semantic_triple(subject,predicate,object) returns canonical text in the form subject <predicate> object.")

Instruction array 7

Complete event record 135 of 210

TIME-STAMP	2026-07-18T11:25:00.044Z
LABEL	Instruction array 7
VALUE	Sequential-ID instructions are sequential_encode, sequential_decode, and sequential_verify.
ID	gpil_0135_ab91eaf9228db70b
SEQUENTIAL-STRING ID	521571973258576773706577715544964764621087056185295833984230647963061251241312888632091789559164871998149616980522031384004232481703398633999694889113788380315874314728434604778215
BACKWARD-COMPUTED VALUE	Sequential-ID instructions are sequential_encode, sequential_decode, and sequential_verify.
BACKWARD-COMPUTATION.ALGORITHM	shortlex-unrank/1.0
BACKWARD-COMPUTATION.MATCHES-VALUE	true
ALPHABET	!"#\$%&'()*+,-./0123456789:;<=>?@ABCDEFGHIJKLMNOPQRSTUVWXYZ[\]^_`abcdefghijklmnopqrstuvwxyz{ }~
METRICS.ALPHABET-SYMBOLS	95
METRICS.VALUE-SYMBOLS	91
METRICS.SEQUENTIAL-ID-DIGITS	180
METRICS.NATIVE-INTEGER	false
INSTRUCTION.LANGUAGE	GPIL/1.0
INSTRUCTION.SOURCE-LINE	135
INSTRUCTION.OPERATION	literal
INSTRUCTION.SOURCE	event "Instruction array 7" : literal("Sequential-ID instructions are sequential_encode, sequential_decode, and sequential_verify.")

Built-in sequential encode

Complete event record 136 of 210

TIME-STAMP	2026-07-18T11:25:00.044Z
LABEL	Built-in sequential encode
VALUE	sequential_encode(x,A) ranks x in shortlex order over ordered alphabet A and returns an arbitrary-precision decimal ID.
ID	gpil_0136_4d44c7c413f83b31
SEQUENTIAL-STRING ID	19929884516990721507248279975815410494974017326427244106958395644625737867125031306939353725387845420670702345180180835936883788324932407312821695520846300211462992410897524366645684213793203974544292097635206856471576846758999506997455
BACKWARD-COMPUTED VALUE	sequential_encode(x,A) ranks x in shortlex order over ordered alphabet A and returns an arbitrary-precision decimal ID.
BACKWARD-COMPUTATION.ALGORITHM	shortlex-unrank/1.0
BACKWARD-COMPUTATION.MATCHES-VALUE	true
ALPHABET	!"#\$%&'()*+,-./0123456789:;<=>?@ABCDEFGHIJKLMNOPQRSTUVWXYZ[\]^_`abcdefghijklmnopqrstuvwxyz{ }~
METRICS.ALPHABET-SYMBOLS	95
METRICS.VALUE-SYMBOLS	119
METRICS.SEQUENTIAL-ID-DIGITS	236
METRICS.NATIVE-INTEGER	false
INSTRUCTION.LANGUAGE	GPIL/1.0
INSTRUCTION.SOURCE-LINE	136
INSTRUCTION.OPERATION	literal
INSTRUCTION.SOURCE	event "Built-in sequential encode" : literal("sequential_encode(x,A) ranks x in shortlex order over ordered alphabet A and returns an arbitrary-precision decimal ID.")

Built-in sequential decode

Complete event record 137 of 210

TIME-STAMP	2026-07-18T11:25:00.044Z
LABEL	Built-in sequential decode
VALUE	sequential_decode(N,A) un-ranks decimal ID N over alphabet A and returns its unique finite string.
ID	gpil_0137_114b230ea301b989
SEQUENTIAL-STRING ID	58520640549590399178909734595413902835257687130742785787872542865845516175549289149863919726973295825494110342623428344517291487136365464837682897358179022290437728247376549342556122604988739330
BACKWARD-COMPUTED VALUE	sequential_decode(N,A) un-ranks decimal ID N over alphabet A and returns its unique finite string.
BACKWARD-COMPUTATION.ALGORITHM	shortlex-unrank/1.0
BACKWARD-COMPUTATION.MATCHES-VALUE	true
ALPHABET	!"#\$%&'()*+,-./0123456789:;<=>?@ABCDEFGHIJKLMNOPQRSTUVWXYZ[\]^_`abcdefghijklmnopqrstuvwxyz{ }~
METRICS.ALPHABET-SYMBOLS	95
METRICS.VALUE-SYMBOLS	98
METRICS.SEQUENTIAL-ID-DIGITS	194
METRICS.NATIVE-INTEGER	false
INSTRUCTION.LANGUAGE	GPIL/1.0
INSTRUCTION.SOURCE-LINE	137
INSTRUCTION.OPERATION	literal
INSTRUCTION.SOURCE	event "Built-in sequential decode" : literal("sequential_decode(N,A) un-ranks decimal ID N over alphabet A and returns its unique finite string.")

Built-in sequential verify

Complete event record 138 of 210

TIME-STAMP	2026-07-18T11:25:00.044Z
LABEL	Built-in sequential verify
VALUE	sequential_verify(x,A) returns true only when decoding the encoding of x is byte-for-byte x.
ID	gpil_0138_48bf84ca7affd960
SEQUENTIAL-STRING ID	79609966176928125704572582739538137087632498363043815359234709828697202974197874017202047763841411675861998882648573426193040035778044940780607496695128112071643239072316356846446245
BACKWARD-COMPUTED VALUE	sequential_verify(x,A) returns true only when decoding the encoding of x is byte-for-byte x.
BACKWARD-COMPUTATION.ALGORITHM	shortlex-unrank/1.0
BACKWARD-COMPUTATION.MATCHES-VALUE	true
ALPHABET	!"#\$%&'()*+,-./0123456789:;<=>?@ABCDEFGHIJKLMNOPQRSTUVWXYZ[\]^_`abcdefghijklmnopqrstuvwxyz{ }~
METRICS.ALPHABET-SYMBOLS	95
METRICS.VALUE-SYMBOLS	92
METRICS.SEQUENTIAL-ID-DIGITS	182
METRICS.NATIVE-INTEGER	false
INSTRUCTION.LANGUAGE	GPIL/1.0
INSTRUCTION.SOURCE-LINE	138
INSTRUCTION.OPERATION	literal
INSTRUCTION.SOURCE	event "Built-in sequential verify" : literal("sequential_verify(x,A) returns true only when decoding the encoding of x is byte-for-byte x.")

Instruction array 8

Complete event record 139 of 210

TIME-STAMP	2026-07-18T11:25:00.044Z
LABEL	Instruction array 8
VALUE	Memory instructions are backward reference, map_coordinate, automatic event capture, and automatic square-map construction.
ID	gpil_0139_a5cd7bfb6b1f2988
SEQUENTIAL-STRING ID	895406802840959487947448463820582018609344384918301606945657065180547004592395585508453420173189133933482647170229257840107365929792735730992227470751191056440736758922030446772579316952029718467489560721004321235966888524604099927554189000270
BACKWARD-COMPUTED VALUE	Memory instructions are backward reference, map_coordinate, automatic event capture, and automatic square-map construction.
BACKWARD-COMPUTATION.ALGORITHM	shortlex-unrank/1.0
BACKWARD-COMPUTATION.MATCHES-VALUE	true
ALPHABET	!"#\$%&'()*+,-./0123456789:;<=>?@ABCDEFGHIJKLMNOPQRSTUVWXYZ[\]^_`abcdefghijklmnopqrstuvwxyz{ }~
METRICS.ALPHABET-SYMBOLS	95
METRICS.VALUE-SYMBOLS	123
METRICS.SEQUENTIAL-ID-DIGITS	243
METRICS.NATIVE-INTEGER	false
INSTRUCTION.LANGUAGE	GPIL/1.0
INSTRUCTION.SOURCE-LINE	139
INSTRUCTION.OPERATION	literal
INSTRUCTION.SOURCE	event "Instruction array 8" : literal("Memory instructions are backward reference, map_coordinate, automatic event capture, and automatic square-map construction.")

Built-in map coordinate

Complete event record 140 of 210

TIME-STAMP	2026-07-18T11:25:00.044Z
LABEL	Built-in map coordinate
VALUE	map_coordinate(i,n) returns row,column for zero-based event index i in the least square map containing n events.
ID	gpil_0140_fbf9693a73313fa
SEQUENTIAL-STRING ID	265042682960734786966570957456407107993307258013706371361624647979223550969222724821537437013435463293957778402456492752800822317039363200618684598727219883542577834857035481581915767463261928965196857318971054121634701495
BACKWARD-COMPUTED VALUE	map_coordinate(i,n) returns row,column for zero-based event index i in the least square map containing n events.
BACKWARD-COMPUTATION.ALGORITHM	shortlex-unrank/1.0
BACKWARD-COMPUTATION.MATCHES-VALUE	true
ALPHABET	!"#\$%&'()*+,-./0123456789:;<=>?@ABCDEFGHIJKLMNPOQRSTUVWXYZ[\]^_`abcdefghijklmnopqrstuvwxyz{ }~
METRICS.ALPHABET-SYMBOLS	95
METRICS.VALUE-SYMBOLS	112
METRICS.SEQUENTIAL-ID-DIGITS	222
METRICS.NATIVE-INTEGER	false
INSTRUCTION.LANGUAGE	GPIL/1.0
INSTRUCTION.SOURCE-LINE	140
INSTRUCTION.OPERATION	literal
INSTRUCTION.SOURCE	event "Built-in map coordinate" : literal("map_coordinate(i,n) returns row,column for zero-based event index i in the least square map containing n events.")

Automatic event capture

Complete event record 141 of 210

TIME-STAMP	2026-07-18T11:25:00.044Z
LABEL	Automatic event capture
VALUE	After evaluation, the compiler constructs metadata, computes an ID, proves the reverse, and emits one immutable event object.
ID	gpil_0141_75ff5df22a24628c
SEQUENTIAL-STRING ID	6008522716227958428908777487535782918484997618999744571705516337688953100753225152979783979003778574666162648953607381738439825397941487521605261881966882551153923040563818298882631393768455481692029982789711039465204147699668268234674542466725540
BACKWARD-COMPUTED VALUE	After evaluation, the compiler constructs metadata, computes an ID, proves the reverse, and emits one immutable event object.
BACKWARD-COMPUTATION.ALGORITHM	shortlex-unrank/1.0
BACKWARD-COMPUTATION.MATCHES-VALUE	true
ALPHABET	!"#\$%&'()*+,-./0123456789:;<=>?@ABCDEFGHIJKLMNOPQRSTUVWXYZ[\]^_`abcdefghijklmnopqrstuvwxyz{ }~
METRICS.ALPHABET-SYMBOLS	95
METRICS.VALUE-SYMBOLS	125
METRICS.SEQUENTIAL-ID-DIGITS	247
METRICS.NATIVE-INTEGER	false
INSTRUCTION.LANGUAGE	GPIL/1.0
INSTRUCTION.SOURCE-LINE	141
INSTRUCTION.OPERATION	literal
INSTRUCTION.SOURCE	event "Automatic event capture" : literal("After evaluation, the compiler constructs metadata, computes an ID, proves the reverse, and emits one immutable event object.")

Automatic square map

Complete event record 142 of 210

TIME-STAMP	2026-07-18T11:25:00.044Z
LABEL	Automatic square map
VALUE	Events are placed row-major in the least side by side square; remaining cells are explicit null vacancies.
ID	gpil_0142_dcae3950821c0bfe
SEQUENTIAL-STRING ID	178316688475064954970076322497189683304614630918132733584190334054620155988830711516235745925395696394495132548543809799213830702750319492296167782344482883312585784035323972341865411344281410488823470599527995
BACKWARD-COMPUTED VALUE	Events are placed row-major in the least side by side square; remaining cells are explicit null vacancies.
BACKWARD-COMPUTATION.ALGORITHM	shortlex-unrank/1.0
BACKWARD-COMPUTATION.MATCHES-VALUE	true
ALPHABET	!"#\$%&'()*+,-./0123456789:;<=>?@ABCDEFGHIJKLMNPOQRSTUVWXYZ[\]^_`abcdefghijklmnopqrstuvwxyz{ }~
METRICS.ALPHABET-SYMBOLS	95
METRICS.VALUE-SYMBOLS	106
METRICS.SEQUENTIAL-ID-DIGITS	210
METRICS.NATIVE-INTEGER	false
INSTRUCTION.LANGUAGE	GPIL/1.0
INSTRUCTION.SOURCE-LINE	142
INSTRUCTION.OPERATION	literal
INSTRUCTION.SOURCE	event "Automatic square map" : literal("Events are placed row-major in the least side by side square; remaining cells are explicit null vacancies.")

Shortlex encode step 1

Complete event record 143 of 210

TIME-STAMP	2026-07-18T11:25:00.044Z
LABEL	Shortlex encode step 1
VALUE	Validate UTF-8, alphabet uniqueness, alphabet size 2 through 1024, and value length at most 4096 symbols.
ID	gpil_0143_fccef111ce0fd34c
SEQUENTIAL-STRING ID	2686187415761686631422868614091921661554713226716766803072115399248588864869008117725581550648172359760582395236891658461855903095807757875250085070843502407250041959254264172374272788527955382859149715381670
BACKWARD-COMPUTED VALUE	Validate UTF-8, alphabet uniqueness, alphabet size 2 through 1024, and value length at most 4096 symbols.
BACKWARD-COMPUTATION.ALGORITHM	shortlex-unrank/1.0
BACKWARD-COMPUTATION.MATCHES-VALUE	true
ALPHABET	!"#\$%&'()*+,-./0123456789:;<=>?@ABCDEFGHIJKLMNOPQRSTUVWXYZ[\]^_`abcdefghijklmnopqrstuvwxyz{ }~
METRICS.ALPHABET-SYMBOLS	95
METRICS.VALUE-SYMBOLS	105
METRICS.SEQUENTIAL-ID-DIGITS	208
METRICS.NATIVE-INTEGER	false
INSTRUCTION.LANGUAGE	GPIL/1.0
INSTRUCTION.SOURCE-LINE	143
INSTRUCTION.OPERATION	literal
INSTRUCTION.SOURCE	event "Shortlex encode step 1" : literal("Validate UTF-8, alphabet uniqueness, alphabet size 2 through 1024, and value length at most 4096 symbols.")

Shortlex encode step 2

Complete event record 144 of 210

TIME-STAMP	2026-07-18T11:25:00.044Z
LABEL	Shortlex encode step 2
VALUE	Return decimal zero for the empty value.
ID	gpil_0144_8c311d4579e94684
SEQUENTIAL-STRING ID	7000038462576856224299471028231811680853334735243860908107937723128893355488815
BACKWARD-COMPUTED VALUE	Return decimal zero for the empty value.
BACKWARD-COMPUTATION.ALGORITHM	shortlex-unrank/1.0
BACKWARD-COMPUTATION.MATCHES-VALUE	true
ALPHABET	!"#\$%&'()*+,-./0123456789:;<=>?@ABCDEFGHIJKLMNOPQRSTUVWXYZ[\]^_`abcdefghijklmnopqrstuvwxyz{ }~
METRICS.ALPHABET-SYMBOLS	95
METRICS.VALUE-SYMBOLS	40
METRICS.SEQUENTIAL-ID-DIGITS	79
METRICS.NATIVE-INTEGER	false
INSTRUCTION.LANGUAGE	GPIL/1.0
INSTRUCTION.SOURCE-LINE	144
INSTRUCTION.OPERATION	literal
INSTRUCTION.SOURCE	event "Shortlex encode step 2" : literal("Return decimal zero for the empty value.")

Shortlex encode step 3

Complete event record 145 of 210

TIME-STAMP	2026-07-18T11:25:00.044Z
LABEL	Shortlex encode step 3
VALUE	Let base b be the number of ordered alphabet symbols.
ID	gpil_0145_dcebf4819002c408
SEQUENTIAL-STRING ID	317675038799200370328085047866630520132585077548652146203388761053066236542624333748691780592921234912920
BACKWARD-COMPUTED VALUE	Let base b be the number of ordered alphabet symbols.
BACKWARD-COMPUTATION.ALGORITHM	shortlex-unrank/1.0
BACKWARD-COMPUTATION.MATCHES-VALUE	true
ALPHABET	!"#\$%&'()*+,-./0123456789:;<=>?@ABCDEFGHIJKLMNOPQRSTUVWXYZ[\]^_`abcdefghijklmnopqrstuvwxyz{ }~
METRICS.ALPHABET-SYMBOLS	95
METRICS.VALUE-SYMBOLS	53
METRICS.SEQUENTIAL-ID-DIGITS	105
METRICS.NATIVE-INTEGER	false
INSTRUCTION.LANGUAGE	GPIL/1.0
INSTRUCTION.SOURCE-LINE	145
INSTRUCTION.OPERATION	literal
INSTRUCTION.SOURCE	event "Shortlex encode step 3" : literal("Let base b be the number of ordered alphabet symbols.")

Shortlex encode step 4

Complete event record 146 of 210

TIME-STAMP	2026-07-18T11:25:00.044Z
LABEL	Shortlex encode step 4
VALUE	Compute the shorter-string count S as the sum of b to powers 1 through n minus 1.
ID	gpil_0146_6efef6a550e52e73
SEQUENTIAL-STRING ID	6086054762079506427410196734171958471815486395528099833317898230014303944010229009592340820321273044263663438303885425029083013133409465200342532154992516912125
BACKWARD-COMPUTED VALUE	Compute the shorter-string count S as the sum of b to powers 1 through n minus 1.
BACKWARD-COMPUTATION.ALGORITHM	shortlex-unrank/1.0
BACKWARD-COMPUTATION.MATCHES-VALUE	true
ALPHABET	!"#\$%&'()*+,-./0123456789:;<=>?@ABCDEFGHIJKLMNOPQRSTUVWXYZ[\]^_`abcdefghijklmnopqrstuvwxyz{ }~
METRICS.ALPHABET-SYMBOLS	95
METRICS.VALUE-SYMBOLS	81
METRICS.SEQUENTIAL-ID-DIGITS	160
METRICS.NATIVE-INTEGER	false
INSTRUCTION.LANGUAGE	GPIL/1.0
INSTRUCTION.SOURCE-LINE	146
INSTRUCTION.OPERATION	literal
INSTRUCTION.SOURCE	event "Shortlex encode step 4" : literal("Compute the shorter-string count S as the sum of b to powers 1 through n minus 1.")

Shortlex encode step 5

Complete event record 147 of 210

TIME-STAMP	2026-07-18T11:25:00.044Z
LABEL	Shortlex encode step 5
VALUE	Map each value symbol to its zero-based alphabet digit and reject any absent symbol.
ID	gpil_0147_65706a75d42549e1
SEQUENTIAL-STRING ID	6613184765791747751731949771020019044625043550122546768288642596581782498368374818424883952831592459620522126577643984928138590320796145876905001268723849453184462830
BACKWARD-COMPUTED VALUE	Map each value symbol to its zero-based alphabet digit and reject any absent symbol.
BACKWARD-COMPUTATION.ALGORITHM	shortlex-unrank/1.0
BACKWARD-COMPUTATION.MATCHES-VALUE	true
ALPHABET	!"#\$%&'()*+,-./0123456789:;<=>?@ABCDEFGHIJKLMNOPQRSTUVWXYZ[\]^_`abcdefghijklmnopqrstuvwxyz{ }~
METRICS.ALPHABET-SYMBOLS	95
METRICS.VALUE-SYMBOLS	84
METRICS.SEQUENTIAL-ID-DIGITS	166
METRICS.NATIVE-INTEGER	false
INSTRUCTION.LANGUAGE	GPIL/1.0
INSTRUCTION.SOURCE-LINE	147
INSTRUCTION.OPERATION	literal
INSTRUCTION.SOURCE	event "Shortlex encode step 5" : literal("Map each value symbol to its zero-based alphabet digit and reject any absent symbol.")

Shortlex encode step 6

Complete event record 148 of 210

TIME-STAMP	2026-07-18T11:25:00.044Z
LABEL	Shortlex encode step 6
VALUE	Fold digits left with rank = rank times b plus digit.
ID	gpil_0148_77cf5052e94ff5e1
SEQUENTIAL-STRING ID	276734701493597491869880226045537094590936710059537449351951563524576028153207947201332822220691518693815
BACKWARD-COMPUTED VALUE	Fold digits left with rank = rank times b plus digit.
BACKWARD-COMPUTATION.ALGORITHM	shortlex-unrank/1.0
BACKWARD-COMPUTATION.MATCHES-VALUE	true
ALPHABET	!"#\$%&'()*+,-./0123456789:;<=>?@ABCDEFGHIJKLMNOPQRSTUVWXYZ[\]^_`abcdefghijklmnopqrstuvwxyz{ }~
METRICS.ALPHABET-SYMBOLS	95
METRICS.VALUE-SYMBOLS	53
METRICS.SEQUENTIAL-ID-DIGITS	105
METRICS.NATIVE-INTEGER	false
INSTRUCTION.LANGUAGE	GPIL/1.0
INSTRUCTION.SOURCE-LINE	148
INSTRUCTION.OPERATION	literal
INSTRUCTION.SOURCE	event "Shortlex encode step 6" : literal("Fold digits left with rank = rank times b plus digit.")

Shortlex encode step 7

Complete event record 149 of 210

TIME-STAMP	2026-07-18T11:25:00.044Z
LABEL	Shortlex encode step 7
VALUE	Return sequential ID = S plus rank plus one as a normalized decimal string.
ID	gpil_0149_4726a3922530af5b
SEQUENTIAL-STRING ID	11625900759081779808670303563234136354783015180644799623121828368199807295674268244985083071367758541913324400723129602422986573095957782989952411205
BACKWARD-COMPUTED VALUE	Return sequential ID = S plus rank plus one as a normalized decimal string.
BACKWARD-COMPUTATION.ALGORITHM	shortlex-unrank/1.0
BACKWARD-COMPUTATION.MATCHES-VALUE	true
ALPHABET	!"#\$%&'()*+,-./0123456789:;<=>?@ABCDEFGHIJKLMNOPQRSTUVWXYZ[\]^_`abcdefghijklmnopqrstuvwxyz{ }~
METRICS.ALPHABET-SYMBOLS	95
METRICS.VALUE-SYMBOLS	75
METRICS.SEQUENTIAL-ID-DIGITS	149
METRICS.NATIVE-INTEGER	false
INSTRUCTION.LANGUAGE	GPIL/1.0
INSTRUCTION.SOURCE-LINE	149
INSTRUCTION.OPERATION	literal
INSTRUCTION.SOURCE	event "Shortlex encode step 7" : literal("Return sequential ID = S plus rank plus one as a normalized decimal string.")

Shortlex decode step 1

Complete event record 150 of 210

TIME-STAMP	2026-07-18T11:25:00.044Z
LABEL	Shortlex decode step 1
VALUE	Normalize the non-negative decimal ID and return empty for zero.
ID	gpil_0150_139496f951063cbd
SEQUENTIAL-STRING ID	1890086694683975913368698420698430582767649574896041256493354495799923115538805034803063301136564334966848417945058633503526690
BACKWARD-COMPUTED VALUE	Normalize the non-negative decimal ID and return empty for zero.
BACKWARD-COMPUTATION.ALGORITHM	shortlex-unrank/1.0
BACKWARD-COMPUTATION.MATCHES-VALUE	true
ALPHABET	!"#\$%&'()*+,-./0123456789:;<=>?@ABCDEFGHIJKLMNopQRSTUVWXYZ[\]^_`abcdefghijklmnopqrstuvwxyz{ }~
METRICS.ALPHABET-SYMBOLS	95
METRICS.VALUE-SYMBOLS	64
METRICS.SEQUENTIAL-ID-DIGITS	127
METRICS.NATIVE-INTEGER	false
INSTRUCTION.LANGUAGE	GPIL/1.0
INSTRUCTION.SOURCE-LINE	150
INSTRUCTION.OPERATION	literal
INSTRUCTION.SOURCE	event "Shortlex decode step 1" : literal("Normalize the non-negative decimal ID and return empty for zero.")

Shortlex decode step 2

Complete event record 151 of 210

TIME-STAMP	2026-07-18T11:25:00.044Z
LABEL	Shortlex decode step 2
VALUE	Set remaining to ID, length to one, and current bucket size to b.
ID	gpil_0151_6c476e7131eaaed6
SEQUENTIAL-STRING ID	197925810000574567580423660608786833844264955444295401728963360778909328349790985111404869706393683566477833219799157914429171030
BACKWARD-COMPUTED VALUE	Set remaining to ID, length to one, and current bucket size to b.
BACKWARD-COMPUTATION.ALGORITHM	shortlex-unrank/1.0
BACKWARD-COMPUTATION.MATCHES-VALUE	true
ALPHABET	!"#\$%&'()*+,-./0123456789:;<=>?@ABCDEFGHIJKLMNOPQRSTUVWXYZ[\]^_`abcdefghijklmnopqrstuvwxyz{ }~
METRICS.ALPHABET-SYMBOLS	95
METRICS.VALUE-SYMBOLS	65
METRICS.SEQUENTIAL-ID-DIGITS	129
METRICS.NATIVE-INTEGER	false
INSTRUCTION.LANGUAGE	GPIL/1.0
INSTRUCTION.SOURCE-LINE	151
INSTRUCTION.OPERATION	literal
INSTRUCTION.SOURCE	event "Shortlex decode step 2" : literal("Set remaining to ID, length to one, and current bucket size to b.")

Shortlex decode step 3

Complete event record 152 of 210

TIME-STAMP	2026-07-18T11:25:00.044Z
LABEL	Shortlex decode step 3
VALUE	While remaining exceeds the bucket, subtract the bucket, increment length, and multiply the bucket by b.
ID	gpil_0152_cf947d741e443379
SEQUENTIAL-STRING ID	28820504368438257084088707041437695663906898597331177290159601782111307530234457836903573471870492266962861106732575100954021452418855813119784291853656604320164866041968918375242771299799267453320193039780
BACKWARD-COMPUTED VALUE	While remaining exceeds the bucket, subtract the bucket, increment length, and multiply the bucket by b.
BACKWARD-COMPUTATION.ALGORITHM	shortlex-unrank/1.0
BACKWARD-COMPUTATION.MATCHES-VALUE	true
ALPHABET	!"#\$%&'()*+,-./0123456789:;<=>?@ABCDEFGHIJKLMNOPQRSTUVWXYZ[\]^_`abcdefghijklmnopqrstuvwxyz{ }~
METRICS.ALPHABET-SYMBOLS	95
METRICS.VALUE-SYMBOLS	104
METRICS.SEQUENTIAL-ID-DIGITS	206
METRICS.NATIVE-INTEGER	false
INSTRUCTION.LANGUAGE	GPIL/1.0
INSTRUCTION.SOURCE-LINE	152
INSTRUCTION.OPERATION	literal
INSTRUCTION.SOURCE	event "Shortlex decode step 3" : literal("While remaining exceeds the bucket, subtract the bucket, increment length, and multiply the bucket by b.")

Shortlex decode step 4

Complete event record 153 of 210

TIME-STAMP	2026-07-18T11:25:00.044Z
LABEL	Shortlex decode step 4
VALUE	Set the zero-based within-bucket offset to remaining minus one.
ID	gpil_0153_b63a97f50f0904ce
SEQUENTIAL-STRING ID	21930837683559521407072688049515820043707544728655569607329044966566020971400790761128454406895805597207043527475469677857140
BACKWARD-COMPUTED VALUE	Set the zero-based within-bucket offset to remaining minus one.
BACKWARD-COMPUTATION.ALGORITHM	shortlex-unrank/1.0
BACKWARD-COMPUTATION.MATCHES-VALUE	true
ALPHABET	!"#\$%&'()*+,-./0123456789:;<=>?@ABCDEFGHIJKLMNOPQRSTUVWXYZ[\]^_`abcdefghijklmnopqrstuvwxyz{ }~
METRICS.ALPHABET-SYMBOLS	95
METRICS.VALUE-SYMBOLS	63
METRICS.SEQUENTIAL-ID-DIGITS	125
METRICS.NATIVE-INTEGER	false
INSTRUCTION.LANGUAGE	GPIL/1.0
INSTRUCTION.SOURCE-LINE	153
INSTRUCTION.OPERATION	literal
INSTRUCTION.SOURCE	event "Shortlex decode step 4" : literal("Set the zero-based within-bucket offset to remaining minus one.")

Shortlex decode step 5

Complete event record 154 of 210

TIME-STAMP	2026-07-18T11:25:00.044Z
LABEL	Shortlex decode step 5
VALUE	Allocate length result positions initialized to the first alphabet symbol.
ID	gpil_0154_18171680591d0ff4
SEQUENTIAL-STRING ID	82345786141596997145111395688455536632141273807899814143983048614436591017081748193357384286130570601243092122363077268611129644868539217928603455
BACKWARD-COMPUTED VALUE	Allocate length result positions initialized to the first alphabet symbol.
BACKWARD-COMPUTATION.ALGORITHM	shortlex-unrank/1.0
BACKWARD-COMPUTATION.MATCHES-VALUE	true
ALPHABET	!"#\$%&'()*+,-./0123456789:;<=>?@ABCDEFGHIJKLMNOPQRSTUVWXYZ[\]^_`abcdefghijklmnopqrstuvwxyz{ }~
METRICS.ALPHABET-SYMBOLS	95
METRICS.VALUE-SYMBOLS	74
METRICS.SEQUENTIAL-ID-DIGITS	146
METRICS.NATIVE-INTEGER	false
INSTRUCTION.LANGUAGE	GPIL/1.0
INSTRUCTION.SOURCE-LINE	154
INSTRUCTION.OPERATION	literal
INSTRUCTION.SOURCE	event "Shortlex decode step 5" : literal("Allocate length result positions initialized to the first alphabet symbol.")

Shortlex decode step 6

Complete event record 155 of 210

TIME-STAMP	2026-07-18T11:25:00.044Z
LABEL	Shortlex decode step 6
VALUE	From last position to first, divide offset by b and use each remainder as the alphabet digit.
ID	gpil_0155_57d2950ef4727617
SEQUENTIAL-STRING ID	3559226114454373071349122434276050151433996114156354628398204286672179187266997739274519948877848397913713777652973505504775369247068575872365205597381520427141102407828298364806506315
BACKWARD-COMPUTED VALUE	From last position to first, divide offset by b and use each remainder as the alphabet digit.
BACKWARD-COMPUTATION.ALGORITHM	shortlex-unrank/1.0
BACKWARD-COMPUTATION.MATCHES-VALUE	true
ALPHABET	!"#\$%&'()*+,-./0123456789:;<=>?@ABCDEFGHIJKLMNOPQRSTUVWXYZ[\]^_`abcdefghijklmnopqrstuvwxyz{ }~
METRICS.ALPHABET-SYMBOLS	95
METRICS.VALUE-SYMBOLS	93
METRICS.SEQUENTIAL-ID-DIGITS	184
METRICS.NATIVE-INTEGER	false
INSTRUCTION.LANGUAGE	GPIL/1.0
INSTRUCTION.SOURCE-LINE	155
INSTRUCTION.OPERATION	literal
INSTRUCTION.SOURCE	event "Shortlex decode step 6" : literal("From last position to first, divide offset by b and use each remainder as the alphabet digit.")

Shortlex decode step 7

Complete event record 156 of 210

TIME-STAMP	2026-07-18T11:25:00.044Z
LABEL	Shortlex decode step 7
VALUE	Join the symbols and require exact byte equality with the original value before event emission.
ID	gpil_0156_aab9c2bb43ac631d
SEQUENTIAL-STRING ID	35317701001476380573075867640702515872146248848664658293985933015939002976955183666936635807296051426815871749163896598352509226203346822141056998775367367573233748004597587365203835593395
BACKWARD-COMPUTED VALUE	Join the symbols and require exact byte equality with the original value before event emission.
BACKWARD-COMPUTATION.ALGORITHM	shortlex-unrank/1.0
BACKWARD-COMPUTATION.MATCHES-VALUE	true
ALPHABET	!"#\$%&'()*+,-./0123456789:;<=>?@ABCDEFGHIJKLMNOPQRSTUVWXYZ[\]^_`abcdefghijklmnopqrstuvwxyz{ }~
METRICS.ALPHABET-SYMBOLS	95
METRICS.VALUE-SYMBOLS	95
METRICS.SEQUENTIAL-ID-DIGITS	188
METRICS.NATIVE-INTEGER	false
INSTRUCTION.LANGUAGE	GPIL/1.0
INSTRUCTION.SOURCE-LINE	156
INSTRUCTION.OPERATION	literal
INSTRUCTION.SOURCE	event "Shortlex decode step 7" : literal("Join the symbols and require exact byte equality with the original value before event emission.")

Bijection proof

Complete event record 157 of 210

TIME-STAMP	2026-07-18T11:25:00.044Z
LABEL	Bijection proof
VALUE	Disjoint length buckets plus fixed-width base-b positional ranking make encode and decode mutual inverses.
ID	gpil_0157_109b1c1789db9e8f
SEQUENTIAL-STRING ID	173115680677001230174839975099509560460617016315907975617971044812217824093618124873815623108197999771413844172961952836362633658420249254731126459888627683517380163121528881663930022196508704246920943790686120
BACKWARD-COMPUTED VALUE	Disjoint length buckets plus fixed-width base-b positional ranking make encode and decode mutual inverses.
BACKWARD-COMPUTATION.ALGORITHM	shortlex-unrank/1.0
BACKWARD-COMPUTATION.MATCHES-VALUE	true
ALPHABET	!"#\$%&'()*+,-./0123456789:;<=>?@ABCDEFGHIJKLMNOPQRSTUVWXYZ[\]^_`abcdefghijklmnopqrstuvwxyz{ }~
METRICS.ALPHABET-SYMBOLS	95
METRICS.VALUE-SYMBOLS	106
METRICS.SEQUENTIAL-ID-DIGITS	210
METRICS.NATIVE-INTEGER	false
INSTRUCTION.LANGUAGE	GPIL/1.0
INSTRUCTION.SOURCE-LINE	157
INSTRUCTION.OPERATION	literal
INSTRUCTION.SOURCE	event "Bijection proof" : literal("Disjoint length buckets plus fixed-width base-b positional ranking make encode and decode mutual inverses.")

Compiler UTF-8 component

Complete event record 158 of 210

TIME-STAMP	2026-07-18T11:25:00.044Z
LABEL	Compiler UTF-8 component
VALUE	utf8Symbols validates leading bytes, continuation bytes, scalar range, surrogate exclusion, and overlong encodings.
ID	gpil_0158_723e0fa27f77009f
SEQUENTIAL-STRING ID	250913449689825039828009398919222010648158754185562582513102628750867595985734537405394118840946019352425588800695745189054837134619709537622812724269894368430308219746778720102409208220226859511550835970694140528208928240364795
BACKWARD-COMPUTED VALUE	utf8Symbols validates leading bytes, continuation bytes, scalar range, surrogate exclusion, and overlong encodings.
BACKWARD-COMPUTATION.ALGORITHM	shortlex-unrank/1.0
BACKWARD-COMPUTATION.MATCHES-VALUE	true
ALPHABET	!"#\$%&'()*+,-./0123456789:;<=>?@ABCDEFGHIJKLMNOPQRSTUVWXYZ[\]^_`abcdefghijklmnopqrstuvwxyz{ }~
METRICS.ALPHABET-SYMBOLS	95
METRICS.VALUE-SYMBOLS	115
METRICS.SEQUENTIAL-ID-DIGITS	228
METRICS.NATIVE-INTEGER	false
INSTRUCTION.LANGUAGE	GPIL/1.0
INSTRUCTION.SOURCE-LINE	158
INSTRUCTION.OPERATION	literal
INSTRUCTION.SOURCE	event "Compiler UTF-8 component" : literal("utf8Symbols validates leading bytes, continuation bytes, scalar range, surrogate exclusion, and overlong encodings.")

Compiler big-natural normalize

Complete event record 159 of 210

TIME-STAMP	2026-07-18T11:25:00.044Z
LABEL	Compiler big-natural normalize
VALUE	natural::normalize validates decimal digits and removes insignificant leading zeros.
ID	gpil_0159_a477e2dcad80de18
SEQUENTIAL-STRING ID	11286018392290117809862071096836857171218656041238587147733536294097868328316729171181457458276542509644088362996017280990648904614572508887593395767486797994072620245
BACKWARD-COMPUTED VALUE	natural::normalize validates decimal digits and removes insignificant leading zeros.
BACKWARD-COMPUTATION.ALGORITHM	shortlex-unrank/1.0
BACKWARD-COMPUTATION.MATCHES-VALUE	true
ALPHABET	!"#\$%&'()*+,-./0123456789:;<=>?@ABCDEFGHIJKLMNOPQRSTUVWXYZ[\]^_`abcdefghijklmnopqrstuvwxyz{ }~
METRICS.ALPHABET-SYMBOLS	95
METRICS.VALUE-SYMBOLS	84
METRICS.SEQUENTIAL-ID-DIGITS	167
METRICS.NATIVE-INTEGER	false
INSTRUCTION.LANGUAGE	GPIL/1.0
INSTRUCTION.SOURCE-LINE	159
INSTRUCTION.OPERATION	literal
INSTRUCTION.SOURCE	event "Compiler big-natural normalize" : literal("natural::normalize validates decimal digits and removes insignificant leading zeros.")

Compiler big-natural compare

Complete event record 160 of 210

TIME-STAMP	2026-07-18T11:25:00.044Z
LABEL	Compiler big-natural compare
VALUE	natural::compare orders normalized decimals by length and then lexicographically.
ID	gpil_0160_66e7e160cca1d876
SEQUENTIAL-STRING ID	13163456354909016250054605864728506640607146060167030699290787633769616503023578403332607272769955639182516946738009386566784213897411326765807606718218005359490
BACKWARD-COMPUTED VALUE	natural::compare orders normalized decimals by length and then lexicographically.
BACKWARD-COMPUTATION.ALGORITHM	shortlex-unrank/1.0
BACKWARD-COMPUTATION.MATCHES-VALUE	true
ALPHABET	!"#\$%&'()*+,-./0123456789:;<=>?@ABCDEFGHIJKLMNOPQRSTUVWXYZ[\]^_`abcdefghijklmnopqrstuvwxyz{ }~
METRICS.ALPHABET-SYMBOLS	95
METRICS.VALUE-SYMBOLS	81
METRICS.SEQUENTIAL-ID-DIGITS	161
METRICS.NATIVE-INTEGER	false
INSTRUCTION.LANGUAGE	GPIL/1.0
INSTRUCTION.SOURCE-LINE	160
INSTRUCTION.OPERATION	literal
INSTRUCTION.SOURCE	event "Compiler big-natural compare" : literal("natural::compare orders normalized decimals by length and then lexicographically.")

Compiler big-natural add

Complete event record 161 of 210

TIME-STAMP	2026-07-18T11:25:00.044Z
LABEL	Compiler big-natural add
VALUE	natural::add performs right-to-left decimal addition with carry.
ID	gpil_0161_a4965e1185cdb5b5
SEQUENTIAL-STRING ID	3148245910596233572792660881735541970867749181542192327732131704004723270175741341092206442552781666853507783328605703938870390
BACKWARD-COMPUTED VALUE	natural::add performs right-to-left decimal addition with carry.
BACKWARD-COMPUTATION.ALGORITHM	shortlex-unrank/1.0
BACKWARD-COMPUTATION.MATCHES-VALUE	true
ALPHABET	!"#\$%&'()*+,-./0123456789:;<=>?@ABCDEFGHIJKLMNOPQRSTUVWXYZ[\]^_`abcdefghijklmnopqrstuvwxyz{ }~
METRICS.ALPHABET-SYMBOLS	95
METRICS.VALUE-SYMBOLS	64
METRICS.SEQUENTIAL-ID-DIGITS	127
METRICS.NATIVE-INTEGER	false
INSTRUCTION.LANGUAGE	GPIL/1.0
INSTRUCTION.SOURCE-LINE	161
INSTRUCTION.OPERATION	literal
INSTRUCTION.SOURCE	event "Compiler big-natural add" : literal("natural::add performs right-to-left decimal addition with carry.")

Compiler big-natural subtract

Complete event record 162 of 210

TIME-STAMP	2026-07-18T11:25:00.044Z
LABEL	Compiler big-natural subtract
VALUE	natural::subtract performs checked non-negative decimal subtraction with borrow.
ID	gpil_0162_d9b68af691f67a88
SEQUENTIAL-STRING ID	138562698472726486886984830652033130697855971354366399798501243487661458793361392564295474572028853830897802193965250433455433350653444853668790955859246450625
BACKWARD-COMPUTED VALUE	natural::subtract performs checked non-negative decimal subtraction with borrow.
BACKWARD-COMPUTATION.ALGORITHM	shortlex-unrank/1.0
BACKWARD-COMPUTATION.MATCHES-VALUE	true
ALPHABET	!"#\$%&'()*+,-./0123456789:;<=>?@ABCDEFGHIJKLMNOPQRSTUVWXYZ[\]^_`abcdefghijklmnopqrstuvwxyz{ }~
METRICS.ALPHABET-SYMBOLS	95
METRICS.VALUE-SYMBOLS	80
METRICS.SEQUENTIAL-ID-DIGITS	159
METRICS.NATIVE-INTEGER	false
INSTRUCTION.LANGUAGE	GPIL/1.0
INSTRUCTION.SOURCE-LINE	162
INSTRUCTION.OPERATION	literal
INSTRUCTION.SOURCE	event "Compiler big-natural subtract" : literal("natural::subtract performs checked non-negative decimal subtraction with borrow.")

Compiler big-natural multiply

Complete event record 163 of 210

TIME-STAMP	2026-07-18T11:25:00.044Z
LABEL	Compiler big-natural multiply
VALUE	natural::multiply and multiplySmall perform arbitrary-precision decimal multiplication.
ID	gpil_0163_7b01ef4871f62d2d
SEQUENTIAL-STRING ID	9676350019089739757049905466296720060430665568724600789963884737761517496220594741985128669727598450333913456397475048242667148940582496684996896063956353210771705392084645
BACKWARD-COMPUTED VALUE	natural::multiply and multiplySmall perform arbitrary-precision decimal multiplication.
BACKWARD-COMPUTATION.ALGORITHM	shortlex-unrank/1.0
BACKWARD-COMPUTATION.MATCHES-VALUE	true
ALPHABET	!"#\$%&'()*+,-./0123456789:;<=>?@ABCDEFGHIJKLMNOPQRSTUVWXYZ[\]^_`abcdefghijklmnopqrstuvwxyz{ }~
METRICS.ALPHABET-SYMBOLS	95
METRICS.VALUE-SYMBOLS	87
METRICS.SEQUENTIAL-ID-DIGITS	172
METRICS.NATIVE-INTEGER	false
INSTRUCTION.LANGUAGE	GPIL/1.0
INSTRUCTION.SOURCE-LINE	163
INSTRUCTION.OPERATION	literal
INSTRUCTION.SOURCE	event "Compiler big-natural multiply" : literal("natural::multiply and multiplySmall perform arbitrary-precision decimal multiplication.")

Compiler big-natural division

Complete event record 164 of 210

TIME-STAMP	2026-07-18T11:25:00.044Z
LABEL	Compiler big-natural division
VALUE	natural::divide and divideSmall perform long division and return exact quotient-remainder pairs.
ID	gpil_0164_aacbc8cbbf7cf75e
SEQUENTIAL-STRING ID	6098513887820021138350530112896092343915407451841096551581647645479443839867950961496063957308317689077154803574969515222204440135186198082875736450615430035181193255394086839025267769940945
BACKWARD-COMPUTED VALUE	natural::divide and divideSmall perform long division and return exact quotient-remainder pairs.
BACKWARD-COMPUTATION.ALGORITHM	shortlex-unrank/1.0
BACKWARD-COMPUTATION.MATCHES-VALUE	true
ALPHABET	!"#\$%&'()*+,-./0123456789:;<=>?@ABCDEFGHIJKLMNOPQRSTUVWXYZ[\]^_`abcdefghijklmnopqrstuvwxyz{ }~
METRICS.ALPHABET-SYMBOLS	95
METRICS.VALUE-SYMBOLS	96
METRICS.SEQUENTIAL-ID-DIGITS	190
METRICS.NATIVE-INTEGER	false
INSTRUCTION.LANGUAGE	GPIL/1.0
INSTRUCTION.SOURCE-LINE	164
INSTRUCTION.OPERATION	literal
INSTRUCTION.SOURCE	event "Compiler big-natural division" : literal("natural::divide and divideSmall perform long division and return exact quotient-remainder pairs.")

Compiler signed integer component

Complete event record 165 of 210

TIME-STAMP

2026-07-18T11:25:00.044Z

LABEL

Compiler signed integer component

VALUE

BigInteger stores a sign and normalized natural magnitude; helpers implement signed add, multiply, divide, and modulo.

ID

gpil_0165_01edfeff4d5ecde8

SEQUENTIAL-STRING ID

88590430239115654441761143284695082032679224271993359723289213724883369281056653842266526169366921154690872066943984692777303174999786480886484165520367868074624092899792462775226178052479763100816471175462733189545363504763998779915

BACKWARD-COMPUTED VALUE

BigInteger stores a sign and normalized natural magnitude; helpers implement signed add, multiply, divide, and modulo.

BACKWARD-COMPUTATION.ALGORITHM

shortlex-unrank/1.0

BACKWARD-COMPUTATION.MATCHES-VALUE

true

ALPHABET

!"#\$%&'()*+,-./0123456789:;<=>?@ABCDEFGHIJKLMNPOQRSTUVWXYZ[\]^_`abcdefghijklmnopqrstuvwxyz{|}~

METRICS.ALPHABET-SYMBOLS

95

METRICS.VALUE-SYMBOLS

118

METRICS.SEQUENTIAL-ID-DIGITS

233

METRICS.NATIVE-INTEGER

false

INSTRUCTION.LANGUAGE

GPIL/1.0

INSTRUCTION.SOURCE-LINE

165

INSTRUCTION.OPERATION

literal

INSTRUCTION.SOURCE

event "Compiler signed integer component" : literal("BigInteger stores a sign and normalized natural magnitude; helpers implement signed add, multiply, divide, and modulo.")

Compiler Shortlex component

Complete event record 166 of 210

TIME-STAMP	2026-07-18T11:25:00.044Z
LABEL	Compiler Shortlex component
VALUE	Shortlex validates alphabets and implements the same rank and un-rank algorithms as PHP SequentialStringId.
ID	gpil_0166_b86f745bf42a3f01
SEQUENTIAL-STRING ID	22969419194227035727831952977998087170054319843530238294514777064820436977386072338489097101087668096552589235868596818513597310824506430087017495098089730667651932738488571955566528158880136338709929252614109745
BACKWARD-COMPUTED VALUE	Shortlex validates alphabets and implements the same rank and un-rank algorithms as PHP SequentialStringId.
BACKWARD-COMPUTATION.ALGORITHM	shortlex-unrank/1.0
BACKWARD-COMPUTATION.MATCHES-VALUE	true
ALPHABET	!"#\$%&'()*+,-./0123456789:;<=>?@ABCDEFGHIJKLMNOPQRSTUVWXYZ[\]^_`abcdefghijklmnopqrstuvwxyz{ }~
METRICS.ALPHABET-SYMBOLS	95
METRICS.VALUE-SYMBOLS	107
METRICS.SEQUENTIAL-ID-DIGITS	212
METRICS.NATIVE-INTEGER	false
INSTRUCTION.LANGUAGE	GPIL/1.0
INSTRUCTION.SOURCE-LINE	166
INSTRUCTION.OPERATION	literal
INSTRUCTION.SOURCE	event "Compiler Shortlex component" : literal("Shortlex validates alphabets and implements the same rank and un-rank algorithms as PHP SequentialStringId.")

Compiler diagnostic component

Complete event record 167 of 210

TIME-STAMP	2026-07-18T11:25:00.044Z
LABEL	Compiler diagnostic component
VALUE	Diagnostic attaches physical line and column to syntax and semantic failures.
ID	gpil_0167_c7d50cbf07dffde1
SEQUENTIAL-STRING ID	76617661315830881313022068305295278952126265505720557441554704821584783590248391319195762538267398991097736887079742573133567170287384666125169301377495
BACKWARD-COMPUTED VALUE	Diagnostic attaches physical line and column to syntax and semantic failures.
BACKWARD-COMPUTATION.ALGORITHM	shortlex-unrank/1.0
BACKWARD-COMPUTATION.MATCHES-VALUE	true
ALPHABET	!"#\$%&'()*+,-./0123456789:;<=>?@ABCDEFGHIJKLMNOPQRSTUVWXYZ[\]^_`abcdefghijklmnopqrstuvwxyz{ }~
METRICS.ALPHABET-SYMBOLS	95
METRICS.VALUE-SYMBOLS	77
METRICS.SEQUENTIAL-ID-DIGITS	152
METRICS.NATIVE-INTEGER	false
INSTRUCTION.LANGUAGE	GPIL/1.0
INSTRUCTION.SOURCE-LINE	167
INSTRUCTION.OPERATION	literal
INSTRUCTION.SOURCE	event "Compiler diagnostic component" : literal("Diagnostic attaches physical line and column to syntax and semantic failures.")

Compiler lexer component

Complete event record 168 of 210

TIME-STAMP	2026-07-18T11:25:00.044Z
LABEL	Compiler lexer component
VALUE	Lexer recognizes identifiers, escaped UTF-8 strings, signed decimals, \$n references, punctuation, and end of line.
ID	gpil_0168_c5733a9a1938011e
SEQUENTIAL-STRING ID	1390362314206340283123829721593247445630663796958701388120979770056176937679379965668015252076294336039095915789209149196861581608808561903127315280543207755464848826254785674067336124599016348045999862004566076715828535407265
BACKWARD-COMPUTED VALUE	Lexer recognizes identifiers, escaped UTF-8 strings, signed decimals, \$n references, punctuation, and end of line.
BACKWARD-COMPUTATION.ALGORITHM	shortlex-unrank/1.0
BACKWARD-COMPUTATION.MATCHES-VALUE	true
ALPHABET	!"#\$%&'()*+,-./0123456789:;<=>?@ABCDEFGHIJKLMNPQRSTUVWXYZ[\]^_`abcdefghijklmnopqrstuvwxyz{ }~
METRICS.ALPHABET-SYMBOLS	95
METRICS.VALUE-SYMBOLS	114
METRICS.SEQUENTIAL-ID-DIGITS	226
METRICS.NATIVE-INTEGER	false
INSTRUCTION.LANGUAGE	GPIL/1.0
INSTRUCTION.SOURCE-LINE	168
INSTRUCTION.OPERATION	literal
INSTRUCTION.SOURCE	event "Compiler lexer component" : literal("Lexer recognizes identifiers, escaped UTF-8 strings, signed decimals, \$n references, punctuation, and end of line.")

Compiler parser component

Complete event record 169 of 210

TIME-STAMP	2026-07-18T11:25:00.044Z
LABEL	Compiler parser component
VALUE	Parser enforces one event statement per line and builds recursive expression trees with an optional event alphabet.
ID	gpil_0169_2d9e1e8522311ce4
SEQUENTIAL-STRING ID	143510152618155286788430749313343325439190318313329440202555580107889264926161596578217282659406595935806925444103217734268083357538538474948647448893027491909318021136772449571017265046158186559000791450543963031447968599003340
BACKWARD-COMPUTED VALUE	Parser enforces one event statement per line and builds recursive expression trees with an optional event alphabet.
BACKWARD-COMPUTATION.ALGORITHM	shortlex-unrank/1.0
BACKWARD-COMPUTATION.MATCHES-VALUE	true
ALPHABET	!"#\$%&'()*+,-./0123456789:;<=>?@ABCDEFGHIJKLMNOPQRSTUVWXYZ[\]^_`abcdefghijklmnopqrstuvwxyz{ }~
METRICS.ALPHABET-SYMBOLS	95
METRICS.VALUE-SYMBOLS	115
METRICS.SEQUENTIAL-ID-DIGITS	228
METRICS.NATIVE-INTEGER	false
INSTRUCTION.LANGUAGE	GPIL/1.0
INSTRUCTION.SOURCE-LINE	169
INSTRUCTION.OPERATION	literal
INSTRUCTION.SOURCE	event "Compiler parser component" : literal("Parser enforces one event statement per line and builds recursive expression trees with an optional event alphabet.")

Compiler evaluator component

Complete event record 170 of 210

TIME-STAMP	2026-07-18T11:25:00.044Z
LABEL	Compiler evaluator component
VALUE	evaluate resolves earlier event values, evaluates nested expressions inside-out, checks arity, and dispatches built-ins.
ID	gpil_0170_70f2d91d6452576a
SEQUENTIAL-STRING ID	1584518489081600072734993524251980742252822606905994224417462924002734092911869802390843564458696226264395914162916688497771507251650068539859648031040302090770253189236797624392393373806140676818598190948637501923931802667809046395491095
BACKWARD-COMPUTED VALUE	evaluate resolves earlier event values, evaluates nested expressions inside-out, checks arity, and dispatches built-ins.
BACKWARD-COMPUTATION.ALGORITHM	shortlex-unrank/1.0
BACKWARD-COMPUTATION.MATCHES-VALUE	true
ALPHABET	!"#\$%&'()*+,-./0123456789:;<=>?@ABCDEFGHIJKLMNOPQRSTUVWXYZ[\]^_`abcdefghijklmnopqrstuvwxyz{ }~
METRICS.ALPHABET-SYMBOLS	95
METRICS.VALUE-SYMBOLS	120
METRICS.SEQUENTIAL-ID-DIGITS	238
METRICS.NATIVE-INTEGER	false
INSTRUCTION.LANGUAGE	GPIL/1.0
INSTRUCTION.SOURCE-LINE	170
INSTRUCTION.OPERATION	literal
INSTRUCTION.SOURCE	event "Compiler evaluator component" : literal("evaluate resolves earlier event values, evaluates nested expressions inside-out, checks arity, and dispatches built-ins.")

Compiler container component

Complete event record 171 of 210

TIME-STAMP	2026-07-18T11:25:00.044Z
LABEL	Compiler container component
VALUE	containerize and decontainerize implement and validate the GPDB1 length-prefixed UTF-8 container format.
ID	gpil_0171_bf8ceaa433189120
SEQUENTIAL-STRING ID	34949529741547320201317251536652500451002054988807567232632808690491570823851567594942318536763984873021336129784164601082874152730100826858319544105621222865048417527836289897518674707255914692511078349615
BACKWARD-COMPUTED VALUE	containerize and decontainerize implement and validate the GPDB1 length-prefixed UTF-8 container format.
BACKWARD-COMPUTATION.ALGORITHM	shortlex-unrank/1.0
BACKWARD-COMPUTATION.MATCHES-VALUE	true
ALPHABET	!"#\$%&'()*+,-./0123456789:;<=>?@ABCDEFGHIJKLMNOPQRSTUVWXYZ[\]^_`abcdefghijklmnopqrstuvwxyz{ }~
METRICS.ALPHABET-SYMBOLS	95
METRICS.VALUE-SYMBOLS	104
METRICS.SEQUENTIAL-ID-DIGITS	206
METRICS.NATIVE-INTEGER	false
INSTRUCTION.LANGUAGE	GPIL/1.0
INSTRUCTION.SOURCE-LINE	171
INSTRUCTION.OPERATION	literal
INSTRUCTION.SOURCE	event "Compiler container component" : literal("containerize and decontainerize implement and validate the GPDB1 length-prefixed UTF-8 container format.")

Compiler event ID component

Complete event record 172 of 210

TIME-STAMP	2026-07-18T11:25:00.044Z
LABEL	Compiler event ID component
VALUE	eventId combines physical line and deterministic FNV-1a source hash into a stable gpil event ID.
ID	gpil_0172_f2429e3b1cbe52b9
SEQUENTIAL-STRING ID	5426671247030850874550870654844228106812881746390363594145099533151000274128076785022057508964973238041035985544370181229509159468486218533682114103268574097031337262836192518880689467499330
BACKWARD-COMPUTED VALUE	eventId combines physical line and deterministic FNV-1a source hash into a stable gpil event ID.
BACKWARD-COMPUTATION.ALGORITHM	shortlex-unrank/1.0
BACKWARD-COMPUTATION.MATCHES-VALUE	true
ALPHABET	!"#\$%&'()*+,-./0123456789:;<=>?@ABCDEFGHIJKLMNOPQRSTUVWXYZ[\]^_`abcdefghijklmnopqrstuvwxyz{ }~
METRICS.ALPHABET-SYMBOLS	95
METRICS.VALUE-SYMBOLS	96
METRICS.SEQUENTIAL-ID-DIGITS	190
METRICS.NATIVE-INTEGER	false
INSTRUCTION.LANGUAGE	GPIL/1.0
INSTRUCTION.SOURCE-LINE	172
INSTRUCTION.OPERATION	literal
INSTRUCTION.SOURCE	event "Compiler event ID component" : literal("eventId combines physical line and deterministic FNV-1a source hash into a stable gpil event ID.")

Compiler timestamp component

Complete event record 173 of 210

TIME-STAMP	2026-07-18T11:25:00.044Z
LABEL	Compiler timestamp component
VALUE	isoTimestamp emits UTC ISO 8601 timestamps with millisecond precision.
ID	gpil_0173_1a041d45ac8359da
SEQUENTIAL-STRING ID	2174556969893077047037206752766826066556654243983093444213808290125073645488793645625220812647806539590310351283388741364952394575232562145
BACKWARD-COMPUTED VALUE	isoTimestamp emits UTC ISO 8601 timestamps with millisecond precision.
BACKWARD-COMPUTATION.ALGORITHM	shortlex-unrank/1.0
BACKWARD-COMPUTATION.MATCHES-VALUE	true
ALPHABET	!"#\$%&'()*+,-./0123456789:;<=>?@ABCDEFGHIJKLMNOPQRSTUVWXYZ[\]^_`abcdefghijklmnopqrstuvwxyz{ }~
METRICS.ALPHABET-SYMBOLS	95
METRICS.VALUE-SYMBOLS	70
METRICS.SEQUENTIAL-ID-DIGITS	139
METRICS.NATIVE-INTEGER	false
INSTRUCTION.LANGUAGE	GPIL/1.0
INSTRUCTION.SOURCE-LINE	173
INSTRUCTION.OPERATION	literal
INSTRUCTION.SOURCE	event "Compiler timestamp component" : literal("isoTimestamp emits UTC ISO 8601 timestamps with millisecond precision.")

Compiler JSON component

Complete event record 174 of 210

TIME-STAMP	2026-07-18T11:25:00.044Z
LABEL	Compiler JSON component
VALUE	jsonEscape and writeDatabase serialize exact UTF-8 values, proof data, instruction metadata, and complete map cells.
ID	gpil_0174_d38bcd4e589afdb4
SEQUENTIAL-STRING ID	20816958995840745590321058567061463240635559506494982273117555765156861709663161684965049448343505152832676302180011603509280797319167741458449105225624749772339336947397177031480222737596000339598362545336595169431760444213927045
BACKWARD-COMPUTED VALUE	jsonEscape and writeDatabase serialize exact UTF-8 values, proof data, instruction metadata, and complete map cells.
BACKWARD-COMPUTATION.ALGORITHM	shortlex-unrank/1.0
BACKWARD-COMPUTATION.MATCHES-VALUE	true
ALPHABET	!"#\$%&'()*+,-./0123456789:;<=>?@ABCDEFGHIJKLMNOPQRSTUVWXYZ[\]^_`abcdefghijklmnopqrstuvwxyz{ }~
METRICS.ALPHABET-SYMBOLS	95
METRICS.VALUE-SYMBOLS	116
METRICS.SEQUENTIAL-ID-DIGITS	230
METRICS.NATIVE-INTEGER	false
INSTRUCTION.LANGUAGE	GPIL/1.0
INSTRUCTION.SOURCE-LINE	174
INSTRUCTION.OPERATION	literal
INSTRUCTION.SOURCE	event "Compiler JSON component" : literal("jsonEscape and writeDatabase serialize exact UTF-8 values, proof data, instruction metadata, and complete map cells.")

Compiler pipeline component

Complete event record 175 of 210

TIME-STAMP	2026-07-18T11:25:00.044Z
LABEL	Compiler pipeline component
VALUE	parseProgram, compile, and main coordinate input, transactional compilation, output, exit codes, and summary diagnostics.
ID	gpil_0175_4b66dc5b54d985c9
SEQUENTIAL-STRING ID	173410793521214620096081944890758785197274105086873856883339291717686969747069249094066985724952418464642469243488613309309903832301885083569156691726965979739308737180435514610613050983913185918597168276913869934732136669727220448532654945
BACKWARD-COMPUTED VALUE	parseProgram, compile, and main coordinate input, transactional compilation, output, exit codes, and summary diagnostics.
BACKWARD-COMPUTATION.ALGORITHM	shortlex-unrank/1.0
BACKWARD-COMPUTATION.MATCHES-VALUE	true
ALPHABET	!"#\$%&'()*+,-./0123456789:;<=>?@ABCDEFGHIJKLMNopQRSTUVWXYZ[\]^_`abcdefghijklmnopqrstuvwxyz{ }~
METRICS.ALPHABET-SYMBOLS	95
METRICS.VALUE-SYMBOLS	121
METRICS.SEQUENTIAL-ID-DIGITS	240
METRICS.NATIVE-INTEGER	false
INSTRUCTION.LANGUAGE	GPIL/1.0
INSTRUCTION.SOURCE-LINE	175
INSTRUCTION.OPERATION	literal
INSTRUCTION.SOURCE	event "Compiler pipeline component" : literal("parseProgram, compile, and main coordinate input, transactional compilation, output, exit codes, and summary diagnostics.")

Compiler self-test component

Complete event record 176 of 210

TIME-STAMP	2026-07-18T11:25:00.044Z
LABEL	Compiler self-test component
VALUE	selfTest checks known base-2 shortlex ordinals, containers, references, arithmetic, and exact assertions.
ID	gpil_0176_d516d5a1999d6767
SEQUENTIAL-STRING ID	408668701852295429988574961146728210802011181052506596314656134978446049640571816734694779084208597877692900941469281622388610622272548911885446648692293657665568326449198103851105918426015001846104072141595
BACKWARD-COMPUTED VALUE	selfTest checks known base-2 shortlex ordinals, containers, references, arithmetic, and exact assertions.
BACKWARD-COMPUTATION.ALGORITHM	shortlex-unrank/1.0
BACKWARD-COMPUTATION.MATCHES-VALUE	true
ALPHABET	!"#\$%&'()*+,-./0123456789:;<=>?@ABCDEFGHIJKLMNOPQRSTUVWXYZ[\]^_`abcdefghijklmnopqrstuvwxyz{ }~
METRICS.ALPHABET-SYMBOLS	95
METRICS.VALUE-SYMBOLS	105
METRICS.SEQUENTIAL-ID-DIGITS	208
METRICS.NATIVE-INTEGER	false
INSTRUCTION.LANGUAGE	GPIL/1.0
INSTRUCTION.SOURCE-LINE	176
INSTRUCTION.OPERATION	literal
INSTRUCTION.SOURCE	event "Compiler self-test component" : literal("selfTest checks known base-2 shortlex ordinals, containers, references, arithmetic, and exact assertions.")

PHP bootstrap component

Complete event record 177 of 210

TIME-STAMP	2026-07-18T11:25:00.044Z
LABEL	PHP bootstrap component
VALUE	src/bootstrap.php loads BigNatural, SequentialStringId, EventStore, and DatabaseReportPdf in dependency order.
ID	gpil_0177_114e29876749da6c
SEQUENTIAL-STRING ID	31672670545764547124258564513951029525950864008941262019715114612208777376497998245254966926495609326491113947013721226708075251345670329050271509630514349808945942848059840110644245118945891631110318396872349772778525
BACKWARD-COMPUTED VALUE	src/bootstrap.php loads BigNatural, SequentialStringId, EventStore, and DatabaseReportPdf in dependency order.
BACKWARD-COMPUTATION.ALGORITHM	shortlex-unrank/1.0
BACKWARD-COMPUTATION.MATCHES-VALUE	true
ALPHABET	!"#\$%&'()*+,-./0123456789:;<=>?@ABCDEFGHIJKLMNOPQRSTUVWXYZ[\]^_`abcdefghijklmnopqrstuvwxyz{ }~
METRICS.ALPHABET-SYMBOLS	95
METRICS.VALUE-SYMBOLS	110
METRICS.SEQUENTIAL-ID-DIGITS	218
METRICS.NATIVE-INTEGER	false
INSTRUCTION.LANGUAGE	GPIL/1.0
INSTRUCTION.SOURCE-LINE	177
INSTRUCTION.OPERATION	literal
INSTRUCTION.SOURCE	event "PHP bootstrap component" : literal("src/bootstrap.php loads BigNatural, SequentialStringId, EventStore, and DatabaseReportPdf in dependency order.")

PHP BigNatural normalize

Complete event record 178 of 210

TIME-STAMP	2026-07-18T11:25:00.044Z
LABEL	PHP BigNatural normalize
VALUE	BigNatural::normalize validates and canonicalizes non-negative decimal strings.
ID	gpil_0178_47ad40832165aa42
SEQUENTIAL-STRING ID	654886855189253667817366771000153269552241770707144347304138387062325761255978794218634002139130268394665241143665529389436024757044476919239390911934149170
BACKWARD-COMPUTED VALUE	BigNatural::normalize validates and canonicalizes non-negative decimal strings.
BACKWARD-COMPUTATION.ALGORITHM	shortlex-unrank/1.0
BACKWARD-COMPUTATION.MATCHES-VALUE	true
ALPHABET	!"#\$%&'()*+,-./0123456789:;<=>?@ABCDEFGHIJKLMNOPQRSTUVWXYZ[\]^_`abcdefghijklmnopqrstuvwxyz{ }~
METRICS.ALPHABET-SYMBOLS	95
METRICS.VALUE-SYMBOLS	79
METRICS.SEQUENTIAL-ID-DIGITS	156
METRICS.NATIVE-INTEGER	false
INSTRUCTION.LANGUAGE	GPIL/1.0
INSTRUCTION.SOURCE-LINE	178
INSTRUCTION.OPERATION	literal
INSTRUCTION.SOURCE	event "PHP BigNatural normalize" : literal("BigNatural::normalize validates and canonicalizes non-negative decimal strings.")

PHP BigNatural compare

Complete event record 179 of 210

TIME-STAMP	2026-07-18T11:25:00.044Z
LABEL	PHP BigNatural compare
VALUE	BigNatural::compare orders arbitrary-precision decimal strings without native integer conversion.
ID	gpil_0179_ff36ffd73fbcab77
SEQUENTIAL-STRING ID	260130435851245456697735076522517012426015526032014195659848529430828023822861959658939788530664175178884882743849325038646289448968632033723279033163500141931309940849917744427696456571705895
BACKWARD-COMPUTED VALUE	BigNatural::compare orders arbitrary-precision decimal strings without native integer conversion.
BACKWARD-COMPUTATION.ALGORITHM	shortlex-unrank/1.0
BACKWARD-COMPUTATION.MATCHES-VALUE	true
ALPHABET	!"#\$%&'()*+,-./0123456789:;<=>?@ABCDEFGHIJKLMNOPQRSTUVWXYZ[\]^_`abcdefghijklmnopqrstuvwxyz{ }~
METRICS.ALPHABET-SYMBOLS	95
METRICS.VALUE-SYMBOLS	97
METRICS.SEQUENTIAL-ID-DIGITS	192
METRICS.NATIVE-INTEGER	false
INSTRUCTION.LANGUAGE	GPIL/1.0
INSTRUCTION.SOURCE-LINE	179
INSTRUCTION.OPERATION	literal
INSTRUCTION.SOURCE	event "PHP BigNatural compare" : literal("BigNatural::compare orders arbitrary-precision decimal strings without native integer conversion.")

PHP BigNatural add

Complete event record 180 of 210

TIME-STAMP	2026-07-18T11:25:00.044Z
LABEL	PHP BigNatural add
VALUE	BigNatural::add and addSmall implement decimal addition with carry.
ID	gpil_0180_373d7f7dc466ea8b
SEQUENTIAL-STRING ID	1211945275273771157380674801709930041088082977412140854828204409616626115762388338421887858719319129620748237035039877010397542386015
BACKWARD-COMPUTED VALUE	BigNatural::add and addSmall implement decimal addition with carry.
BACKWARD-COMPUTATION.ALGORITHM	shortlex-unrank/1.0
BACKWARD-COMPUTATION.MATCHES-VALUE	true
ALPHABET	!"#\$%&'()*+,-./0123456789:;<=>?@ABCDEFGHIJKLMNOPQRSTUVWXYZ[\]^_`abcdefghijklmnopqrstuvwxyz{ }~
METRICS.ALPHABET-SYMBOLS	95
METRICS.VALUE-SYMBOLS	67
METRICS.SEQUENTIAL-ID-DIGITS	133
METRICS.NATIVE-INTEGER	false
INSTRUCTION.LANGUAGE	GPIL/1.0
INSTRUCTION.SOURCE-LINE	180
INSTRUCTION.OPERATION	literal
INSTRUCTION.SOURCE	event "PHP BigNatural add" : literal("BigNatural::add and addSmall implement decimal addition with carry.")

PHP BigNatural subtract

Complete event record 181 of 210

TIME-STAMP	2026-07-18T11:25:00.044Z
LABEL	PHP BigNatural subtract
VALUE	BigNatural::subtract implements checked decimal subtraction with borrow.
ID	gpil_0181_634a32a5626aaa51
SEQUENTIAL-STRING ID	9377801513000342153184794740231729436567234286732201579713425447075860584677606248936717299568518118074249235321007881168333572049456658560000
BACKWARD-COMPUTED VALUE	BigNatural::subtract implements checked decimal subtraction with borrow.
BACKWARD-COMPUTATION.ALGORITHM	shortlex-unrank/1.0
BACKWARD-COMPUTATION.MATCHES-VALUE	true
ALPHABET	!"#\$%&'()*+,-./0123456789:;<=>?@ABCDEFGHIJKLMNOPQRSTUVWXYZ[\]^_`abcdefghijklmnopqrstuvwxyz{ }~
METRICS.ALPHABET-SYMBOLS	95
METRICS.VALUE-SYMBOLS	72
METRICS.SEQUENTIAL-ID-DIGITS	142
METRICS.NATIVE-INTEGER	false
INSTRUCTION.LANGUAGE	GPIL/1.0
INSTRUCTION.SOURCE-LINE	181
INSTRUCTION.OPERATION	literal
INSTRUCTION.SOURCE	event "PHP BigNatural subtract" : literal("BigNatural::subtract implements checked decimal subtraction with borrow.")

PHP BigNatural multiply

Complete event record 182 of 210

TIME-STAMP	2026-07-18T11:25:00.044Z
LABEL	PHP BigNatural multiply
VALUE	BigNatural::multiplySmall multiplies a decimal string by a bounded native factor.
ID	gpil_0182_91620f7f6c6ed8b1
SEQUENTIAL-STRING ID	5910353868083014352051734821742611758125052619950054887478027899133261360067667062560728024859011662771607675746698059985699301670144090012938933230631096668025
BACKWARD-COMPUTED VALUE	BigNatural::multiplySmall multiplies a decimal string by a bounded native factor.
BACKWARD-COMPUTATION.ALGORITHM	shortlex-unrank/1.0
BACKWARD-COMPUTATION.MATCHES-VALUE	true
ALPHABET	!"#\$%&'()*+,-./0123456789:;<=>?@ABCDEFGHIJKLMNOPQRSTUVWXYZ[\]^_`abcdefghijklmnopqrstuvwxyz{ }~
METRICS.ALPHABET-SYMBOLS	95
METRICS.VALUE-SYMBOLS	81
METRICS.SEQUENTIAL-ID-DIGITS	160
METRICS.NATIVE-INTEGER	false
INSTRUCTION.LANGUAGE	GPIL/1.0
INSTRUCTION.SOURCE-LINE	182
INSTRUCTION.OPERATION	literal
INSTRUCTION.SOURCE	event "PHP BigNatural multiply" : literal("BigNatural::multiplySmall multiplies a decimal string by a bounded native factor.")

PHP BigInteger divide

Complete event record 183 of 210

TIME-STAMP	2026-07-18T11:25:00.044Z
LABEL	PHP BigInteger divide
VALUE	BigInteger::divideSmall returns a normalized decimal quotient and native remainder.
ID	gpil_0183_5d9947b30ca5e072
SEQUENTIAL-STRING ID	53340943659449204527266881595500916370994994502901223620751539768124782443875393386149512212149277324436478529161032502912073994517801001319228277156491908754104150
BACKWARD-COMPUTED VALUE	BigInteger::divideSmall returns a normalized decimal quotient and native remainder.
BACKWARD-COMPUTATION.ALGORITHM	shortlex-unrank/1.0
BACKWARD-COMPUTATION.MATCHES-VALUE	true
ALPHABET	!"#\$%&'()*+,-./0123456789:;<=>?@ABCDEFGHIJKLMNopQRSTUVWXYZ[\]^_`abcdefghijklmnopqrstuvwxyz{ }~
METRICS.ALPHABET-SYMBOLS	95
METRICS.VALUE-SYMBOLS	83
METRICS.SEQUENTIAL-ID-DIGITS	164
METRICS.NATIVE-INTEGER	false
INSTRUCTION.LANGUAGE	GPIL/1.0
INSTRUCTION.SOURCE-LINE	183
INSTRUCTION.OPERATION	literal
INSTRUCTION.SOURCE	event "PHP BigInteger divide" : literal("BigInteger::divideSmall returns a normalized decimal quotient and native remainder.")

PHP SequentialStringId constructor

Complete event record 184 of 210

TIME-STAMP	2026-07-18T11:25:00.044Z
LABEL	PHP SequentialStringId constructor
VALUE	SequentialStringId::__construct validates UTF-8, uniqueness, and the 2 through 1024 symbol alphabet bounds.
ID	gpil_0184_6837e313e038afc8
SEQUENTIAL-STRING ID	22955773505242521891701538127940784505468880105667458176330730153651753132395643127859637616931405598276355309310457667670528800402758006936409314023132834128187538142784131978637449281250690955865206332150163970
BACKWARD-COMPUTED VALUE	SequentialStringId::__construct validates UTF-8, uniqueness, and the 2 through 1024 symbol alphabet bounds.
BACKWARD-COMPUTATION.ALGORITHM	shortlex-unrank/1.0
BACKWARD-COMPUTATION.MATCHES-VALUE	true
ALPHABET	!"#\$%&'()*+,-./0123456789:;<=>?@ABCDEFGHIJKLMNPOQRSTUVWXYZ[\]^_`abcdefghijklmnopqrstuvwxyz{ }~
METRICS.ALPHABET-SYMBOLS	95
METRICS.VALUE-SYMBOLS	107
METRICS.SEQUENTIAL-ID-DIGITS	212
METRICS.NATIVE-INTEGER	false
INSTRUCTION.LANGUAGE	GPIL/1.0
INSTRUCTION.SOURCE-LINE	184
INSTRUCTION.OPERATION	literal
INSTRUCTION.SOURCE	event "PHP SequentialStringId constructor" : literal("SequentialStringId::__construct validates UTF-8, uniqueness, and the 2 through 1024 symbol alphabet bounds.")

PHP SequentialStringId encode

Complete event record 185 of 210

TIME-STAMP	2026-07-18T11:25:00.044Z
LABEL	PHP SequentialStringId encode
VALUE	SequentialStringId::encode performs arbitrary-precision shortlex ranking with a 4096 symbol context limit.
ID	gpil_0185_e084e797e40f3712
SEQUENTIAL-STRING ID	241639721107816019912647769767797731636522396775178897696547602260437520370118446719533722148742404456818615720436156551789056543688895369372976743823020827934761389292331113574747017900732768166120121953969315
BACKWARD-COMPUTED VALUE	SequentialStringId::encode performs arbitrary-precision shortlex ranking with a 4096 symbol context limit.
BACKWARD-COMPUTATION.ALGORITHM	shortlex-unrank/1.0
BACKWARD-COMPUTATION.MATCHES-VALUE	true
ALPHABET	!"#\$%&'()*+,-./0123456789:;<=>?@ABCDEFGHIJKLMNPOQRSTUVWXYZ[\]^_`abcdefghijklmnopqrstuvwxyz{ }~
METRICS.ALPHABET-SYMBOLS	95
METRICS.VALUE-SYMBOLS	106
METRICS.SEQUENTIAL-ID-DIGITS	210
METRICS.NATIVE-INTEGER	false
INSTRUCTION.LANGUAGE	GPIL/1.0
INSTRUCTION.SOURCE-LINE	185
INSTRUCTION.OPERATION	literal
INSTRUCTION.SOURCE	event "PHP SequentialStringId encode" : literal("SequentialStringId::encode performs arbitrary-precision shortlex ranking with a 4096 symbol context limit.")

PHP SequentialStringId decode

Complete event record 186 of 210

TIME-STAMP	2026-07-18T11:25:00.044Z
LABEL	PHP SequentialStringId decode
VALUE	SequentialStringId::decode locates the length bucket and performs repeated base division to reconstruct the value.
ID	gpil_0186_c36d130cbead6cda
SEQUENTIAL-STRING ID	1603087279939160832818814340265408869916271274890128010895883506737759315909328070270830384883065407131875175169517179929284175456039447605891834529722393074453040181487281891947112321251961254762878310507188005050622196113815
BACKWARD-COMPUTED VALUE	SequentialStringId::decode locates the length bucket and performs repeated base division to reconstruct the value.
BACKWARD-COMPUTATION.ALGORITHM	shortlex-unrank/1.0
BACKWARD-COMPUTATION.MATCHES-VALUE	true
ALPHABET	!"#\$%&'()*+,-./0123456789:;<=>?@ABCDEFGHIJKLMNopQRSTUVWXYZ[\]^_`abcdefghijklmnopqrstuvwxyz{ }~
METRICS.ALPHABET-SYMBOLS	95
METRICS.VALUE-SYMBOLS	114
METRICS.SEQUENTIAL-ID-DIGITS	226
METRICS.NATIVE-INTEGER	false
INSTRUCTION.LANGUAGE	GPIL/1.0
INSTRUCTION.SOURCE-LINE	186
INSTRUCTION.OPERATION	literal
INSTRUCTION.SOURCE	event "PHP SequentialStringId decode" : literal("SequentialStringId::decode locates the length bucket and performs repeated base division to reconstruct the value.")

PHP SequentialStringId proof

Complete event record 187 of 210

TIME-STAMP	2026-07-18T11:25:00.044Z
LABEL	PHP SequentialStringId proof
VALUE	SequentialStringId::encodeVerified emits an ID only after decode of that ID exactly equals the original input.
ID	gpil_0187_37c7b7944b466dbe
SEQUENTIAL-STRING ID	19681706309057307206897606252443230115377022043836313334331024212857703210391154200239217103657918989674530363015965711982074946015575525112550840516026667673509411589597550551165719932709458372324510197357325343677865
BACKWARD-COMPUTED VALUE	SequentialStringId::encodeVerified emits an ID only after decode of that ID exactly equals the original input.
BACKWARD-COMPUTATION.ALGORITHM	shortlex-unrank/1.0
BACKWARD-COMPUTATION.MATCHES-VALUE	true
ALPHABET	!"#\$%&'()*+,-./0123456789:;<=>?@ABCDEFGHIJKLMNOPQRSTUVWXYZ[\]^_`abcdefghijklmnopqrstuvwxyz{ }~
METRICS.ALPHABET-SYMBOLS	95
METRICS.VALUE-SYMBOLS	110
METRICS.SEQUENTIAL-ID-DIGITS	218
METRICS.NATIVE-INTEGER	false
INSTRUCTION.LANGUAGE	GPIL/1.0
INSTRUCTION.SOURCE-LINE	187
INSTRUCTION.OPERATION	literal
INSTRUCTION.SOURCE	event "PHP SequentialStringId proof" : literal("SequentialStringId::encodeVerified emits an ID only after decode of that ID exactly equals the original input.")

PHP EventStore constructor

Complete event record 188 of 210

TIME-STAMP	2026-07-18T11:25:00.044Z
LABEL	PHP EventStore constructor
VALUE	EventStore::__construct creates the data directory and initial JSON state when required.
ID	gpil_0188_5b55813c7958b5ab
SEQUENTIAL-STRING ID	448918078151807941535357023023858875175460739518226504818351097543106570950939426056931776990601942009547594377242647006081869845780870190675187672435136613034042867101018570
BACKWARD-COMPUTED VALUE	EventStore::__construct creates the data directory and initial JSON state when required.
BACKWARD-COMPUTATION.ALGORITHM	shortlex-unrank/1.0
BACKWARD-COMPUTATION.MATCHES-VALUE	true
ALPHABET	!"#\$%&'()*+,-./0123456789:;<=>?@ABCDEFGHIJKLMNOPQRSTUVWXYZ[\]^_`abcdefghijklmnopqrstuvwxyz{ }~
METRICS.ALPHABET-SYMBOLS	95
METRICS.VALUE-SYMBOLS	88
METRICS.SEQUENTIAL-ID-DIGITS	174
METRICS.NATIVE-INTEGER	false
INSTRUCTION.LANGUAGE	GPIL/1.0
INSTRUCTION.SOURCE-LINE	188
INSTRUCTION.OPERATION	literal
INSTRUCTION.SOURCE	event "PHP EventStore constructor" : literal("EventStore::__construct creates the data directory and initial JSON state when required.")

PHP EventStore read

Complete event record 189 of 210

TIME-STAMP	2026-07-18T11:25:00.044Z
LABEL	PHP EventStore read
VALUE	EventStore::read uses a shared file lock, decodes JSON, and rebuilds derived square-map metadata.
ID	gpil_0189_bc8f38c985144c9d
SEQUENTIAL-STRING ID	282930353769883015469948280432821195347053165045429140551504481780082708175159468532012999163159568303175191809664553285212974504618261292453167635472105033355319146216773381693398462011309535
BACKWARD-COMPUTED VALUE	EventStore::read uses a shared file lock, decodes JSON, and rebuilds derived square-map metadata.
BACKWARD-COMPUTATION.ALGORITHM	shortlex-unrank/1.0
BACKWARD-COMPUTATION.MATCHES-VALUE	true
ALPHABET	!"#\$%&'()*+,-./0123456789:;<=>?@ABCDEFGHIJKLMNOPQRSTUVWXYZ[\]^_`abcdefghijklmnopqrstuvwxyz{ }~
METRICS.ALPHABET-SYMBOLS	95
METRICS.VALUE-SYMBOLS	97
METRICS.SEQUENTIAL-ID-DIGITS	192
METRICS.NATIVE-INTEGER	false
INSTRUCTION.LANGUAGE	GPIL/1.0
INSTRUCTION.SOURCE-LINE	189
INSTRUCTION.OPERATION	literal
INSTRUCTION.SOURCE	event "PHP EventStore read" : literal("EventStore::read uses a shared file lock, decodes JSON, and rebuilds derived square-map metadata.")

PHP EventStore append

Complete event record 190 of 210

TIME-STAMP	2026-07-18T11:25:00.044Z
LABEL	PHP EventStore append
VALUE	EventStore::append uses an exclusive lock, appends one verified node, rebuilds the map, truncates, writes, and flushes atomically.
ID	gpil_0190_6cdea5bcb8099e3b
SEQUENTIAL-STRING ID	52066515885776280825472427363004304149783732201060766539206630835247586140225550689986320504941688052465389059118978323974327869696248102809475064043278395056335614251139057210197632943325402963086221633546607810960558250699422031368406910150922319774109490
BACKWARD-COMPUTED VALUE	EventStore::append uses an exclusive lock, appends one verified node, rebuilds the map, truncates, writes, and flushes atomically.
BACKWARD-COMPUTATION.ALGORITHM	shortlex-unrank/1.0
BACKWARD-COMPUTATION.MATCHES-VALUE	true
ALPHABET	!"#\$%&'()*+,-./0123456789:;<=>?@ABCDEFGHIJKLMNOPQRSTUVWXYZ[\]^_`abcdefghijklmnopqrstuvwxyz{ }~
METRICS.ALPHABET-SYMBOLS	95
METRICS.VALUE-SYMBOLS	130
METRICS.SEQUENTIAL-ID-DIGITS	257
METRICS.NATIVE-INTEGER	false
INSTRUCTION.LANGUAGE	GPIL/1.0
INSTRUCTION.SOURCE-LINE	190
INSTRUCTION.OPERATION	literal
INSTRUCTION.SOURCE	event "PHP EventStore append" : literal("EventStore::append uses an exclusive lock, appends one verified node, rebuilds the map, truncates, writes, and flushes atomically.")

PHP EventStore square map

Complete event record 191 of 210

TIME-STAMP	2026-07-18T11:25:00.044Z
LABEL	PHP EventStore square map
VALUE	EventStore::buildSquareMap computes the least square, row-major cells, and total, occupied, and vacant metrics.
ID	gpil_0191_9bd274ce2d012355
SEQUENTIAL-STRING ID	1379780543658771742821049947257878211835399421285423669618262393514653962847925257581722744372429818673538235738186618141964938304237167331453216028985737000925917362590207034404342913250827910362839675418259814855141195
BACKWARD-COMPUTED VALUE	EventStore::buildSquareMap computes the least square, row-major cells, and total, occupied, and vacant metrics.
BACKWARD-COMPUTATION.ALGORITHM	shortlex-unrank/1.0
BACKWARD-COMPUTATION.MATCHES-VALUE	true
ALPHABET	!"#\$%&'()*+,-./0123456789:;<=>?@ABCDEFGHIJKLMNPOQRSTUVWXYZ[\]^_`abcdefghijklmnopqrstuvwxyz{ }~
METRICS.ALPHABET-SYMBOLS	95
METRICS.VALUE-SYMBOLS	111
METRICS.SEQUENTIAL-ID-DIGITS	220
METRICS.NATIVE-INTEGER	false
INSTRUCTION.LANGUAGE	GPIL/1.0
INSTRUCTION.SOURCE-LINE	191
INSTRUCTION.OPERATION	literal
INSTRUCTION.SOURCE	event "PHP EventStore square map" : literal("EventStore::buildSquareMap computes the least square, row-major cells, and total, occupied, and vacant metrics.")

PHP EventStore timestamp

Complete event record 192 of 210

TIME-STAMP	2026-07-18T11:25:00.044Z
LABEL	PHP EventStore timestamp
VALUE	EventStore::timestamp returns a UTC ISO 8601 timestamp with millisecond precision.
ID	gpil_0192_71dcc53d0c11a9c3
SEQUENTIAL-STRING ID	610696545393967821715632837331547593821088236406675230156370815193655002175191919579275689688164348029181842922985178207519834886395093784608630713293165320452770
BACKWARD-COMPUTED VALUE	EventStore::timestamp returns a UTC ISO 8601 timestamp with millisecond precision.
BACKWARD-COMPUTATION.ALGORITHM	shortlex-unrank/1.0
BACKWARD-COMPUTATION.MATCHES-VALUE	true
ALPHABET	!"#\$%&'()*+,-./0123456789:;<=>?@ABCDEFGHIJKLMNOPQRSTUVWXYZ[\]^_`abcdefghijklmnopqrstuvwxyz{ }~
METRICS.ALPHABET-SYMBOLS	95
METRICS.VALUE-SYMBOLS	82
METRICS.SEQUENTIAL-ID-DIGITS	162
METRICS.NATIVE-INTEGER	false
INSTRUCTION.LANGUAGE	GPIL/1.0
INSTRUCTION.SOURCE-LINE	192
INSTRUCTION.OPERATION	literal
INSTRUCTION.SOURCE	event "PHP EventStore timestamp" : literal("EventStore::timestamp returns a UTC ISO 8601 timestamp with millisecond precision.")

PHP EventStore decoder

Complete event record 193 of 210

TIME-STAMP	2026-07-18T11:25:00.044Z
LABEL	PHP EventStore decoder
VALUE	EventStore::decodeState rejects invalid JSON structures, preserves top-level manifests, and refreshes derived map data.
ID	gpil_0193_0afad31c8b5ee492
SEQUENTIAL-STRING ID	9153746033583594799543258984755292369254282460840612035009326105832627123541660235253393587891799222763324639646170705666737945529115974900870619144594127562781259086828064777060053169514679097249965100462513575886799055596348982543910
BACKWARD-COMPUTED VALUE	EventStore::decodeState rejects invalid JSON structures, preserves top-level manifests, and refreshes derived map data.
BACKWARD-COMPUTATION.ALGORITHM	shortlex-unrank/1.0
BACKWARD-COMPUTATION.MATCHES-VALUE	true
ALPHABET	!"#\$%&'()*+,-./0123456789:;<=>?@ABCDEFGHIJKLMNOPQRSTUVWXYZ[\]^_`abcdefghijklmnopqrstuvwxyz{ }~
METRICS.ALPHABET-SYMBOLS	95
METRICS.VALUE-SYMBOLS	119
METRICS.SEQUENTIAL-ID-DIGITS	235
METRICS.NATIVE-INTEGER	false
INSTRUCTION.LANGUAGE	GPIL/1.0
INSTRUCTION.SOURCE-LINE	193
INSTRUCTION.OPERATION	literal
INSTRUCTION.SOURCE	event "PHP EventStore decoder" : literal("EventStore::decodeState rejects invalid JSON structures, preserves top-level manifests, and refreshes derived map data.")

PHP API GET

Complete event record 194 of 210

TIME-STAMP	2026-07-18T11:25:00.044Z
LABEL	PHP API GET
VALUE	api.php GET returns database state and compiler limits; export-report returns the current complete PDF snapshot.
ID	gpil_0194_bed61fade5f19578
SEQUENTIAL-STRING ID	225160434175332952864976689997713742007275912070714030649954303058439659947003141467794816944964146837446757083224248173131382538839686149191154554137329723292641247178753117073835145954031775044995919440802220922599777215
BACKWARD-COMPUTED VALUE	api.php GET returns database state and compiler limits; export-report returns the current complete PDF snapshot.
BACKWARD-COMPUTATION.ALGORITHM	shortlex-unrank/1.0
BACKWARD-COMPUTATION.MATCHES-VALUE	true
ALPHABET	!"#\$%&'()*+,-./0123456789:;<=>?@ABCDEFGHIJKLMNOPQRSTUVWXYZ[\]^_`abcdefghijklmnopqrstuvwxyz{ }~
METRICS.ALPHABET-SYMBOLS	95
METRICS.VALUE-SYMBOLS	112
METRICS.SEQUENTIAL-ID-DIGITS	222
METRICS.NATIVE-INTEGER	false
INSTRUCTION.LANGUAGE	GPIL/1.0
INSTRUCTION.SOURCE-LINE	194
INSTRUCTION.OPERATION	literal
INSTRUCTION.SOURCE	event "PHP API GET" : literal("api.php GET returns database state and compiler limits; export-report returns the current complete PDF snapshot.")

PHP API POST

Complete event record 195 of 210

TIME-STAMP	2026-07-18T11:25:00.044Z
LABEL	PHP API POST
VALUE	api.php POST validates JSON create-event input, ranks and proves the value, constructs all required fields, and commits it.
ID	gpil_0195_942fcb54bd2472bd
SEQUENTIAL-STRING ID	1280712757359371751859558578363089163713447882096947987453706592593925259488727808136138005280694061757037383108487491719486621142474786616437504016032647892302837715674826711014059502975551061944312470700555201747816251875659190281370244354565
BACKWARD-COMPUTED VALUE	api.php POST validates JSON create-event input, ranks and proves the value, constructs all required fields, and commits it.
BACKWARD-COMPUTATION.ALGORITHM	shortlex-unrank/1.0
BACKWARD-COMPUTATION.MATCHES-VALUE	true
ALPHABET	!"#\$%&'()*+,-./0123456789:;<=>?@ABCDEFGHIJKLMNOPQRSTUVWXYZ[\]^_`abcdefghijklmnopqrstuvwxyz{ }~
METRICS.ALPHABET-SYMBOLS	95
METRICS.VALUE-SYMBOLS	123
METRICS.SEQUENTIAL-ID-DIGITS	244
METRICS.NATIVE-INTEGER	false
INSTRUCTION.LANGUAGE	GPIL/1.0
INSTRUCTION.SOURCE-LINE	195
INSTRUCTION.OPERATION	literal
INSTRUCTION.SOURCE	event "PHP API POST" : literal("api.php POST validates JSON create-event input, ranks and proves the value, constructs all required fields, and commits it.")

PHP API payload built-ins

Complete event record 196 of 210

TIME-STAMP	2026-07-18T11:25:00.044Z
LABEL	PHP API payload built-ins
VALUE	readPayload uses content type checks, file_get_contents, json_decode, and JSON_THROW_ON_ERROR for strict request parsing.
ID	gpil_0196_a954d9ab1decdd54
SEQUENTIAL-STRING ID	177740976002618500739713323733844488374258569637206709747088266394098444013108314102903963751607783701913653621487126230804366106695506436292529987079824692505725782692703867606914397839425485317610640014746587956982073283596935373563586830
BACKWARD-COMPUTED VALUE	readPayload uses content type checks, file_get_contents, json_decode, and JSON_THROW_ON_ERROR for strict request parsing.
BACKWARD-COMPUTATION.ALGORITHM	shortlex-unrank/1.0
BACKWARD-COMPUTATION.MATCHES-VALUE	true
ALPHABET	!"#\$%&'()*+,-./0123456789:;<=>?@ABCDEFGHIJKLMNOPQRSTUVWXYZ[\]^_`abcdefghijklmnopqrstuvwxyz{ }~
METRICS.ALPHABET-SYMBOLS	95
METRICS.VALUE-SYMBOLS	121
METRICS.SEQUENTIAL-ID-DIGITS	240
METRICS.NATIVE-INTEGER	false
INSTRUCTION.LANGUAGE	GPIL/1.0
INSTRUCTION.SOURCE-LINE	196
INSTRUCTION.OPERATION	literal
INSTRUCTION.SOURCE	event "PHP API payload built-ins" : literal("readPayload uses content type checks, file_get_contents, json_decode, and JSON_THROW_ON_ERROR for strict request parsing.")

PHP API event ID built-ins

Complete event record 197 of 210

TIME-STAMP	2026-07-18T11:25:00.044Z
LABEL	PHP API event ID built-ins
VALUE	createEventId uses microtime, floor, base_convert, random_bytes, bin2hex, and sprintf for collision-resistant runtime IDs.
ID	gpil_0197_0828b32290fb224d
SEQUENTIAL-STRING ID	13888615422540223900094817839931487645849208742862153332323417518396008469578814147676778476289381356945652441102041004885409534242721752938602030401878217982256425591823118803842671080120678441263957435549785989150680913357790714936921067920
BACKWARD-COMPUTED VALUE	createEventId uses microtime, floor, base_convert, random_bytes, bin2hex, and sprintf for collision-resistant runtime IDs.
BACKWARD-COMPUTATION.ALGORITHM	shortlex-unrank/1.0
BACKWARD-COMPUTATION.MATCHES-VALUE	true
ALPHABET	!"#\$%&'()*+,-./0123456789:;<=>?@ABCDEFGHIJKLMNopQRSTUVWXYZ[\]^_`abcdefghijklmnopqrstuvwxyz{ }~
METRICS.ALPHABET-SYMBOLS	95
METRICS.VALUE-SYMBOLS	122
METRICS.SEQUENTIAL-ID-DIGITS	242
METRICS.NATIVE-INTEGER	false
INSTRUCTION.LANGUAGE	GPIL/1.0
INSTRUCTION.SOURCE-LINE	197
INSTRUCTION.OPERATION	literal
INSTRUCTION.SOURCE	event "PHP API event ID built-ins" : literal("createEventId uses microtime, floor, base_convert, random_bytes, bin2hex, and sprintf for collision-resistant runtime IDs.")

PHP API native integer metric

Complete event record 198 of 210

TIME-STAMP	2026-07-18T11:25:00.044Z
LABEL	PHP API native integer metric
VALUE	isNativeInteger compares decimal digit length and lexicographic order with PHP_INT_MAX without lossy conversion.
ID	gpil_0198_7ca186d0f49827c8
SEQUENTIAL-STRING ID	252197685572990714855863947017098416697043382564828874613067204859503570983839599518375009241590852611270259045630353057909215255281191105922978586101386467089857891610753946565317043569704032095098722936585582914579518395
BACKWARD-COMPUTED VALUE	isNativeInteger compares decimal digit length and lexicographic order with PHP_INT_MAX without lossy conversion.
BACKWARD-COMPUTATION.ALGORITHM	shortlex-unrank/1.0
BACKWARD-COMPUTATION.MATCHES-VALUE	true
ALPHABET	!"#\$%&'()*+,-./0123456789:;<=>?@ABCDEFGHIJKLMNopQRSTUVWXYZ[\]^_`abcdefghijklmnopqrstuvwxyz{ }~
METRICS.ALPHABET-SYMBOLS	95
METRICS.VALUE-SYMBOLS	112
METRICS.SEQUENTIAL-ID-DIGITS	222
METRICS.NATIVE-INTEGER	false
INSTRUCTION.LANGUAGE	GPIL/1.0
INSTRUCTION.SOURCE-LINE	198
INSTRUCTION.OPERATION	literal
INSTRUCTION.SOURCE	event "PHP API native integer metric" : literal("isNativeInteger compares decimal digit length and lexicographic order with PHP_INT_MAX without lossy conversion.")

PHP PDF report builder

Complete event record 199 of 210

TIME-STAMP	2026-07-18T11:25:00.044Z
LABEL	PHP PDF report builder
VALUE	DatabaseReportPdf traverses every top-level property, map cell, and recursively flattened field of every event.
ID	gpil_0199_a4afb9c10079d234
SEQUENTIAL-STRING ID	1336554558863395279988678560013985446755403467200790357547230981984141469206293753508939978877212122503382319817831091135527793584874015188886937168458649696869645078474345686557453030174980837484081842009682101426504190
BACKWARD-COMPUTED VALUE	DatabaseReportPdf traverses every top-level property, map cell, and recursively flattened field of every event.
BACKWARD-COMPUTATION.ALGORITHM	shortlex-unrank/1.0
BACKWARD-COMPUTATION.MATCHES-VALUE	true
ALPHABET	!"#\$%&'()*+,-./0123456789:;<=>?@ABCDEFGHIJKLMNOPQRSTUVWXYZ[\]^_`abcdefghijklmnopqrstuvwxyz{ }~
METRICS.ALPHABET-SYMBOLS	95
METRICS.VALUE-SYMBOLS	111
METRICS.SEQUENTIAL-ID-DIGITS	220
METRICS.NATIVE-INTEGER	false
INSTRUCTION.LANGUAGE	GPIL/1.0
INSTRUCTION.SOURCE-LINE	199
INSTRUCTION.OPERATION	literal
INSTRUCTION.SOURCE	event "PHP PDF report builder" : literal("DatabaseReportPdf traverses every top-level property, map cell, and recursively flattened field of every event.")

PHP PDF layout engine

Complete event record 200 of 210

TIME-STAMP	2026-07-18T11:25:00.044Z
LABEL	PHP PDF layout engine
VALUE	Page, title, section, paragraph, field, wrap, drawing, and continuation helpers create consistent paginated A4 output.
ID	gpil_0200_e1e431f70eb04177
SEQUENTIAL-STRING ID	123038957698356199730369964611314518725614975265952996774346565090776180677462894198813895892958554818901277680987117827909476443028232066661426271769268205882764580673965759735020203661544341544107954337351215770993470654427858422240
BACKWARD-COMPUTED VALUE	Page, title, section, paragraph, field, wrap, drawing, and continuation helpers create consistent paginated A4 output.
BACKWARD-COMPUTATION.ALGORITHM	shortlex-unrank/1.0
BACKWARD-COMPUTATION.MATCHES-VALUE	true
ALPHABET	!"#\$%&'()*+,-./0123456789:;<=>?@ABCDEFGHIJKLMNopQRSTUVWXYZ[\]^_`abcdefghijklmnopqrstuvwxyz{ }~
METRICS.ALPHABET-SYMBOLS	95
METRICS.VALUE-SYMBOLS	118
METRICS.SEQUENTIAL-ID-DIGITS	234
METRICS.NATIVE-INTEGER	false
INSTRUCTION.LANGUAGE	GPIL/1.0
INSTRUCTION.SOURCE-LINE	200
INSTRUCTION.OPERATION	literal
INSTRUCTION.SOURCE	event "PHP PDF layout engine" : literal("Page, title, section, paragraph, field, wrap, drawing, and continuation helpers create consistent paginated A4 output.")

PHP PDF Unicode preservation

Complete event record 201 of 210

TIME-STAMP	2026-07-18T11:25:00.044Z
LABEL	PHP PDF Unicode preservation
VALUE	asciiSafe represents non-ASCII scalar values as explicit U+ notation when standard PDF fonts cannot encode them.
ID	gpil_0201_c5efc3b6dc138a5f
SEQUENTIAL-STRING ID	225264771957207982294350123631075604139305303800069705943765909950486055421206917407130101215262026207026841970901014864497985039473829681800803044102752862692500326892682898292203075916599863074184628712053844879892308800
BACKWARD-COMPUTED VALUE	asciiSafe represents non-ASCII scalar values as explicit U+ notation when standard PDF fonts cannot encode them.
BACKWARD-COMPUTATION.ALGORITHM	shortlex-unrank/1.0
BACKWARD-COMPUTATION.MATCHES-VALUE	true
ALPHABET	!"#\$%&'()*+,-./0123456789:;<=>?@ABCDEFGHIJKLMNOPQRSTUVWXYZ[\]^_`abcdefghijklmnopqrstuvwxyz{ }~
METRICS.ALPHABET-SYMBOLS	95
METRICS.VALUE-SYMBOLS	112
METRICS.SEQUENTIAL-ID-DIGITS	222
METRICS.NATIVE-INTEGER	false
INSTRUCTION.LANGUAGE	GPIL/1.0
INSTRUCTION.SOURCE-LINE	201
INSTRUCTION.OPERATION	literal
INSTRUCTION.SOURCE	event "PHP PDF Unicode preservation" : literal("asciiSafe represents non-ASCII scalar values as explicit U+ notation when standard PDF fonts cannot encode them.")

PHP PDF serializer

Complete event record 202 of 210

TIME-STAMP	2026-07-18T11:25:00.044Z
LABEL	PHP PDF serializer
VALUE	compilePdf writes PDF 1.4 objects, content streams, fonts, pages, xref offsets, metadata, page numbers, and EOF trailer.
ID	gpil_0202_6dd1e94084a164b8
SEQUENTIAL-STRING ID	1538219432087053402920186653822712066857397138359387997401243795511114100812378564230078963200235854023738134748462972712285613814662674402314897888405622873835636325260603362624660919222817209681659086341958994487937275665960706060485025
BACKWARD-COMPUTED VALUE	compilePdf writes PDF 1.4 objects, content streams, fonts, pages, xref offsets, metadata, page numbers, and EOF trailer.
BACKWARD-COMPUTATION.ALGORITHM	shortlex-unrank/1.0
BACKWARD-COMPUTATION.MATCHES-VALUE	true
ALPHABET	!"#\$%&'()*+,-./0123456789:;<=>?@ABCDEFGHIJKLMNOPQRSTUVWXYZ[\]^_`abcdefghijklmnopqrstuvwxyz{ }~
METRICS.ALPHABET-SYMBOLS	95
METRICS.VALUE-SYMBOLS	120
METRICS.SEQUENTIAL-ID-DIGITS	238
METRICS.NATIVE-INTEGER	false
INSTRUCTION.LANGUAGE	GPIL/1.0
INSTRUCTION.SOURCE-LINE	202
INSTRUCTION.OPERATION	literal
INSTRUCTION.SOURCE	event "PHP PDF serializer" : literal("compilePdf writes PDF 1.4 objects, content streams, fonts, pages, xref offsets, metadata, page numbers, and EOF trailer.")

PHP web entry component

Complete event record 203 of 210

TIME-STAMP	2026-07-18T11:25:00.044Z
LABEL	PHP web entry component
VALUE	index.php supplies security headers and the Memory Square form, proof output, map modes, search, canvas, and report export control.
ID	gpil_0203_5aa71264999112cf
SEQUENTIAL-STRING ID	9510392601905396715149830651342741256812914652813818026910757523109009642914741697684797221857555515554580061678178396929204059948896134062364294795541114989115488023377099210444420229816009264410466029947985556009670415063481572089072304064037205278647603955
BACKWARD-COMPUTED VALUE	index.php supplies security headers and the Memory Square form, proof output, map modes, search, canvas, and report export control.
BACKWARD-COMPUTATION.ALGORITHM	shortlex-unrank/1.0
BACKWARD-COMPUTATION.MATCHES-VALUE	true
ALPHABET	!"#\$%&'()*+,-./0123456789:;<=>?@ABCDEFGHIJKLMNOPQRSTUVWXYZ[\]^_`abcdefghijklmnopqrstuvwxyz{ }~
METRICS.ALPHABET-SYMBOLS	95
METRICS.VALUE-SYMBOLS	131
METRICS.SEQUENTIAL-ID-DIGITS	259
METRICS.NATIVE-INTEGER	false
INSTRUCTION.LANGUAGE	GPIL/1.0
INSTRUCTION.SOURCE-LINE	203
INSTRUCTION.OPERATION	literal
INSTRUCTION.SOURCE	event "PHP web entry component" : literal("index.php supplies security headers and the Memory Square form, proof output, map modes, search, canvas, and report export control.")

JavaScript API component

Complete event record 204 of 210

TIME-STAMP	2026-07-18T11:25:00.044Z
LABEL	JavaScript API component
VALUE	assets/app.js loads state, posts create-event JSON, displays server errors, and renders every database response.
ID	gpil_0204_0192e56ef0de0069
SEQUENTIAL-STRING ID	225270726970577414860686224681512255409061589525235719194944877929442385472720324425299719248925965342532625231979283560003514797659130999758370628456614511514882249621039828056948242354411801982634606105333277577362403640
BACKWARD-COMPUTED VALUE	assets/app.js loads state, posts create-event JSON, displays server errors, and renders every database response.
BACKWARD-COMPUTATION.ALGORITHM	shortlex-unrank/1.0
BACKWARD-COMPUTATION.MATCHES-VALUE	true
ALPHABET	!"#\$%&'()*+,-./0123456789:;<=>?@ABCDEFGHIJKLMNOPQRSTUVWXYZ[\]^_`abcdefghijklmnopqrstuvwxyz{ }~
METRICS.ALPHABET-SYMBOLS	95
METRICS.VALUE-SYMBOLS	112
METRICS.SEQUENTIAL-ID-DIGITS	222
METRICS.NATIVE-INTEGER	false
INSTRUCTION.LANGUAGE	GPIL/1.0
INSTRUCTION.SOURCE-LINE	204
INSTRUCTION.OPERATION	literal
INSTRUCTION.SOURCE	event "JavaScript API component" : literal("assets/app.js loads state, posts create-event JSON, displays server errors, and renders every database response.")

JavaScript map component

Complete event record 205 of 210

TIME-STAMP	2026-07-18T11:25:00.044Z
LABEL	JavaScript map component
VALUE	InfiniteMemoryMap builds node positions from cells and implements high-DPI resize, fit, center, pan, zoom, hit testing, and drawing.
ID	gpil_0205_c0c26c7bbfb8fe63
SEQUENTIAL-STRING ID	517174405012220600569950512019417090541424283300371416716840940028110741666266216258991768702637822994926148718124555249898969218111855736417535699084165833698917213724596445637919792213299703143376608945205095068407598395181745298954677274852453383033442520580
BACKWARD-COMPUTED VALUE	InfiniteMemoryMap builds node positions from cells and implements high-DPI resize, fit, center, pan, zoom, hit testing, and drawing.
BACKWARD-COMPUTATION.ALGORITHM	shortlex-unrank/1.0
BACKWARD-COMPUTATION.MATCHES-VALUE	true
ALPHABET	!"#\$%&'()*+,-./0123456789:;<=>?@ABCDEFGHIJKLMNPOQRSTUVWXYZ[\]^_`abcdefghijklmnopqrstuvwxyz{ }~
METRICS.ALPHABET-SYMBOLS	95
METRICS.VALUE-SYMBOLS	132
METRICS.SEQUENTIAL-ID-DIGITS	261
METRICS.NATIVE-INTEGER	false
INSTRUCTION.LANGUAGE	GPIL/1.0
INSTRUCTION.SOURCE-LINE	205
INSTRUCTION.OPERATION	literal
INSTRUCTION.SOURCE	event "JavaScript map component" : literal("InfiniteMemoryMap builds node positions from cells and implements high-DPI resize, fit, center, pan, zoom, hit testing, and drawing.")

JavaScript search component

Complete event record 206 of 210

TIME-STAMP
2026-07-18T11:25:00.044Z

LABEL
JavaScript search component

VALUE
renderSearch matches label, value, event ID, sequential ID, and timestamp and opens complete node details.

ID
gpil_0206_88c98d6acaf115a4

SEQUENTIAL-STRING ID
383655079608929055951982990068942937802287579323106442194946219497108665548711113749602282949631794090613740255748460417911422349305706327920114984050870565340297543725409312400518789418120392109041638351343045

BACKWARD-COMPUTED VALUE
renderSearch matches label, value, event ID, sequential ID, and timestamp and opens complete node details.

BACKWARD-COMPUTATION.ALGORITHM
shortlex-unrank/1.0

BACKWARD-COMPUTATION.MATCHES-VALUE
true

ALPHABET
!"#\$%&'()*+,-./0123456789:;<=>?@ABCDEFGHIJKLMNOPQRSTUVWXYZ[\]^_`abcdefghijklmnopqrstuvwxyz{|}~

METRICS.ALPHABET-SYMBOLS
95

METRICS.VALUE-SYMBOLS
106

METRICS.SEQUENTIAL-ID-DIGITS
210

METRICS.NATIVE-INTEGER
false

INSTRUCTION.LANGUAGE
GPIL/1.0

INSTRUCTION.SOURCE-LINE
206

INSTRUCTION.OPERATION
literal

INSTRUCTION.SOURCE
event "JavaScript search component" : literal("renderSearch matches label, value, event ID, sequential ID, and timestamp and opens complete node details.")

Template compiler guidance

Complete event record 207 of 210

TIME-STAMP	2026-07-18T11:25:00.044Z
LABEL	Template compiler guidance
VALUE	template/compiler.html contributes character-map loading, base conversion, encode, decode, frequency, preview, and C syntax display patterns.
ID	gpil_0207_a106158e664ec465
SEQUENTIAL-STRING ID	652403802688024200109907431986621030757723605079090382875273868584282148618302476393853866218208579550115861157102030442235487370413958767756882323409381236235215523861020979136080874371872370000235262890295076292369349358118923112289354499019184841858028554605851377877712301845
BACKWARD-COMPUTED VALUE	template/compiler.html contributes character-map loading, base conversion, encode, decode, frequency, preview, and C syntax display patterns.
BACKWARD-COMPUTATION.ALGORITHM	shortlex-unrank/1.0
BACKWARD-COMPUTATION.MATCHES-VALUE	true
ALPHABET	!"#\$%&'()*+,-./0123456789:;<=>?@ABCDEFGHIJKLMNPOQRSTUVWXYZ[\]^_`abcdefghijklmnopqrstuvwxyz{ }~
METRICS.ALPHABET-SYMBOLS	95
METRICS.VALUE-SYMBOLS	141
METRICS.SEQUENTIAL-ID-DIGITS	279
METRICS.NATIVE-INTEGER	false
INSTRUCTION.LANGUAGE	GPIL/1.0
INSTRUCTION.SOURCE-LINE	207
INSTRUCTION.OPERATION	literal
INSTRUCTION.SOURCE	event "Template compiler guidance" : literal("template/compiler.html contributes character-map loading, base conversion, encode, decode, frequency, preview, and C syntax display patterns.")

System source inventory

Complete event record 208 of 210

TIME-STAMP	2026-07-18T11:25:00.044Z
LABEL	System source inventory
VALUE	Constructive artifacts are compiler source and binary, docs, examples, PHP classes, api.php, index.php, assets, tests, and tools.
ID	gpil_0208_034bb52bca993c59
SEQUENTIAL-STRING ID	518883972928560548645865136410585961642230640925050690494956802941567013434506709603219131870732392054720755047482769325108692912423881850544952950876974199114849032125642073423168464877457982187666948261378124283473932616135909144200147309428795419233545
BACKWARD-COMPUTED VALUE	Constructive artifacts are compiler source and binary, docs, examples, PHP classes, api.php, index.php, assets, tests, and tools.
BACKWARD-COMPUTATION.ALGORITHM	shortlex-unrank/1.0
BACKWARD-COMPUTATION.MATCHES-VALUE	true
ALPHABET	!"#\$%&'()*+,-./0123456789:;<=>?@ABCDEFGHIJKLMNopQRSTUVWXYZ[\]^_`abcdefghijklmnopqrstuvwxyz{ }~
METRICS.ALPHABET-SYMBOLS	95
METRICS.VALUE-SYMBOLS	129
METRICS.SEQUENTIAL-ID-DIGITS	255
METRICS.NATIVE-INTEGER	false
INSTRUCTION.LANGUAGE	GPIL/1.0
INSTRUCTION.SOURCE-LINE	208
INSTRUCTION.OPERATION	literal
INSTRUCTION.SOURCE	event "System source inventory" : literal("Constructive artifacts are compiler source and binary, docs, examples, PHP classes, api.php, index.php, assets, tests, and tools.")

Database schema

Complete event record 209 of 210

TIME-STAMP	2026-07-18T11:25:00.044Z
LABEL	Database schema
VALUE	memory-event-node/2.0 stores full reversible event evidence and a complete least-square row-major memory map.
ID	gpil_0209_c6f41424a4458207
SEQUENTIAL-STRING ID	309296889445705712133973531830667091437399203404804237767333237183620071709793392805412331487570774632852285791760329782526480900626980124103548497926431634234145303143458905121355739928424539313924444929595212679235
BACKWARD-COMPUTED VALUE	memory-event-node/2.0 stores full reversible event evidence and a complete least-square row-major memory map.
BACKWARD-COMPUTATION.ALGORITHM	shortlex-unrank/1.0
BACKWARD-COMPUTATION.MATCHES-VALUE	true
ALPHABET	!"#\$%&'()*+,-./0123456789:;<=>?@ABCDEFGHIJKLMNopQRSTUVWXYZ[\]^_`abcdefghijklmnopqrstuvwxyz{ }~
METRICS.ALPHABET-SYMBOLS	95
METRICS.VALUE-SYMBOLS	109
METRICS.SEQUENTIAL-ID-DIGITS	216
METRICS.NATIVE-INTEGER	false
INSTRUCTION.LANGUAGE	GPIL/1.0
INSTRUCTION.SOURCE-LINE	209
INSTRUCTION.OPERATION	literal
INSTRUCTION.SOURCE	event "Database schema" : literal("memory-event-node/2.0 stores full reversible event evidence and a complete least-square row-major memory map.")

Report completeness

Complete event record 210 of 210

TIME-STAMP	2026-07-18T11:25:00.044Z
LABEL	Report completeness
VALUE	report.pdf includes schema, update time, map dimensions and occupancy, every map cell, every event field, and all compiler and PHP component events.
ID	gpil_0210_55fb754872de5788
SEQUENTIAL-STRING ID	44497292995756040753301690542920324946541309907250982749371154376181459353694389949103814717447271016514811493465590083065525427514094228221260880887953471902386214154729891224711518956717100748142783684375371065058434976123050567383066407412783394107439268525216976415811107359430989605795245
BACKWARD-COMPUTED VALUE	report.pdf includes schema, update time, map dimensions and occupancy, every map cell, every event field, and all compiler and PHP component events.
BACKWARD-COMPUTATION.ALGORITHM	shortlex-unrank/1.0
BACKWARD-COMPUTATION.MATCHES-VALUE	true
ALPHABET	!"#\$%&'()*+,-./0123456789:;<=>?@ABCDEFGHIJKLMNopQRSTUVWXYZ[\]^_`abcdefghijklmnopqrstuvwxyz{ }~
METRICS.ALPHABET-SYMBOLS	95
METRICS.VALUE-SYMBOLS	148
METRICS.SEQUENTIAL-ID-DIGITS	293
METRICS.NATIVE-INTEGER	false
INSTRUCTION.LANGUAGE	GPIL/1.0
INSTRUCTION.SOURCE-LINE	210
INSTRUCTION.OPERATION	literal
INSTRUCTION.SOURCE	event "Report completeness" : literal("report.pdf includes schema, update time, map dimensions and occupancy, every map cell, every event field, and all compiler and PHP component events.")